Chapter 11
Light, Eyes and Vision

Abstract The structure and function of the eye are overviewed. This is followed by developing the basics of imaging of light with lenses, and these are used to show how sources are imaged onto the eye retina. Several optical characteristics of the eye and vision are discussed, including accommodation, visual acuity and depth of field, along with modes of imperfect vision and how to correct them with eyeglasses and contact lenses. The relation of the image formed on your retina to what you actually perceive is then analyzed in terms of its brightness and color.

Our eyes image a source onto light-sensitive retinas. Cells in the retina convert the imaged light into electrical signals. This information is carried to the brain by neurons via the optic nerve. The visual cortex in the brain processes this information, and we somehow perceive a visual image. Each of these four steps in the vision process is important and contains interesting physics. The physics of the first step is the optics of imaging the source onto the retina and the second is the quantum physics of the absorption of light by the retina. The physics of the conduction of electrical signals in nerves is step three and we will discuss this in Chap. 12. Physical processes and processing are also important in the fourth step, within the brain [22–24, 63, 65]. (By the way, David et al. shared the Nobel Prize in Physiology or Medicine in 1981 for their discoveries concerning information processing in the visual system.) In this chapter, we will focus on the physics of imaging and briefly consider the absorption of light by retinal cells. Feedback and control (Chap. 13) are also important in vision. For example, the body controls the focal length of the eye lens (*accommodation*) to enable the imaging of both near and far objects and changes the diameter of the pupil to adjust the amount of light entering the eye.

For general references on vision see [3, 6, 10, 13, 19, 30, 39, 47, 48, 51, 63]. For general references on optics and the optics of vision see [1, 3, 6, 8, 21, 27, 28, 40, 42, 52, 54, 66].

11.1 Structure of the Eye

Figure 11.1 shows the structure of the human eye. Light enters the cornea and passes through the aqueous humor in the anterior chamber, the pupil in the iris, the lens

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(commonly called the *crystalline lens* although it is not a crystal), and the vitreous humor, and images on the retina, which contains *rod* and *cone* photoreceptor cells. The three pairs of muscles that control eye motion were described in Fig. 1.14 and Tables 1.3 and 1.4. The eye has an average radius of 12 mm.

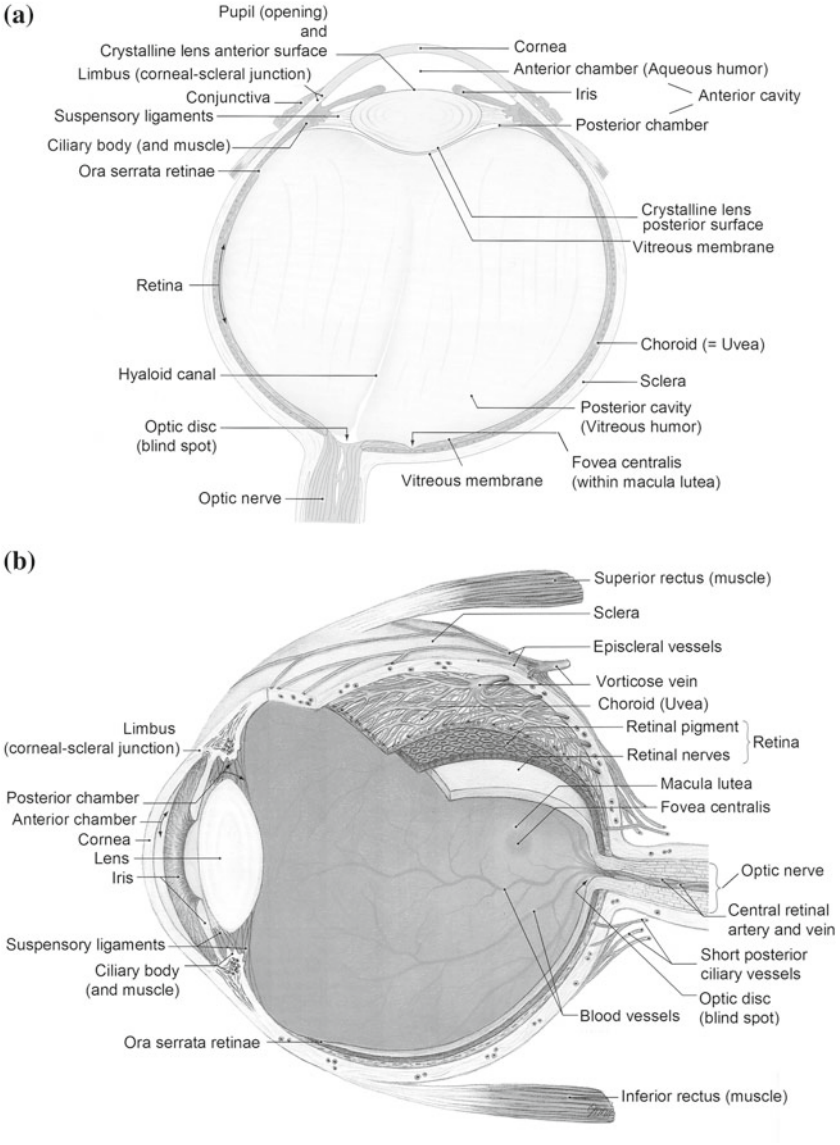


Fig. 11.1 Structure of the eye: **a** sagittal section, **b** three-dimensional perspective (From [59])

The adult cornea is 0.52 mm thick in the center and 0.65 mm thick in the periphery, and is about 12.6 mm in diameter horizontally and 11.7 mm vertically. It is composed of several layers, from anterior to posterior: the outer epithelium, the basement membrane, anterior limiting lamina (or the Bowman's layer), corneal stroma, posterior limiting lamina (or the Descemet's membrane), and the epithelium. The corneal stroma constitutes 90% of the corneal thickness and is composed of 50 layers, each with similarly oriented collagen fibers, with the fibers always parallel to the cornea surface. The cornea is transparent because it is uniform in structure, avascular (i.e., it has no blood vessels) except in the extreme periphery, and relatively dehydrated. (More on this below.) It is covered by a 7–10 μm thick layer of tears, which, among other things, smoothes over optical irregularities on the anterior surface of the cornea and supplies the cornea with oxygen. The average radius of curvature of the anterior surface of the cornea is about 7.8 mm in the central region, with a variation among people of about ± 0.4 mm, and is flatter in the periphery.

The aqueous humor fills the anterior chamber (with a volume of 0.3 cm^3) bounded by the cornea, iris, and the anterior surface of the crystalline lens, and the posterior chamber (0.2 cm^3) on the periphery of the lens. It has many fewer proteins (0.1 g/L) than blood plasma (60–70 g/L). The pupil in the iris is usually slightly nasal and inferior to the center of the iris, and can vary roughly from 1.5 to 10 mm in diameter. The diameter of the pupil is controlled by an opposing pair of smooth muscles: the sphincter pupillae (which is a ring of muscles that encircle the pupil) contracts it and the dilator pupillae (which has the form of a thin disc) widens it.

The crystalline lens is suspended from the ciliary body by zonular fibers and rests on the posterior surface of the iris. It is composed of about 66% water and 33% protein. This crystalline lens is about 4 mm thick and 9 mm in diameter. It continues to grow during life, with new layers growing on older layers, forming a layered structure like an onion (Fig. 11.1). At 30 years of age, the lens has a mass of 170 mg, which increases by about 1.2 mg per year; similarly the lens width is about 4 mm and increases by about 0.02 mm per year. The crystalline lens is avascular and almost completely transparent. Still, it is slightly birefractive (i.e., it has slightly different refractive indices for different polarizations of light), becomes more yellow with age, and can become opaque (and this forms a cataract). Aphakia describes the condition when the crystalline lens is absent. The vitreous humor is about 99% water, with the remaining 1% composed of collagen (0.5 g/L proteins) and hyaluronic acid; the latter gives it its gelatinous, viscous physical characteristics. This humor accounts for about 5 cm^3 of the 7–8 cm^3 volume of the eye.

As we will see below, the formation of an image on the retina is determined by the indices of refraction of each eye component that the light passes through and by the shapes of the surfaces of these elements. The cornea and crystalline lens are the actual focusing elements in the eye. The cornea performs about two-thirds of the focusing and the crystalline lens the remaining one-third. The shape and consequently the focal length of the crystalline lens are adjustable and do the fine-tuning of imaging

for accommodation. The measured refractive index of the tears and the vitreous humor is about 1.336 and that of the aqueous humor is a bit higher, 1.3374. The refractive indices of the cornea, about 1.3771, and the crystalline lens are higher. At the center of the crystalline lens (which is called the nuclear region), the index is about 1.40–1.41 and it decreases to 1.385–1.388 in the direction towards the "poles" and to 1.375 in the direction toward the "equator"; it is 1.360 in the capsule, which is the elastic membrane that encloses the crystalline lens. There is still some uncertainty in these values; optical models of the eye use values close to these cited numbers.

Only about 50% of visible light (400–700 nm) incident on the eye actually reaches the retina as direct light. Then light must pass through the (transparent) ganglion and other retinal neurons before reaching and forming an image on the backward-facing photoreceptors on the retina (Fig. 11.2). The *fovea* or fovea centralis is the central region of the retina, and the region of sharpest vision because it has the highest density of cone cells on the retina (Fig. 11.3). The *visual axis* of the eye is the line from the point you are focusing on, through the center of pupil, to the fovea. The *optical axis* of the eye is a line that passes through the centers of the cornea, pupil and crystalline lens to the retina. They are not parallel. The optic nerve leaves the eyeball at a *blind spot* (*optic disk*), a region with no rods or cones (Fig. 11.3); it is 13–18° away from the fovea in the "nasal" direction.

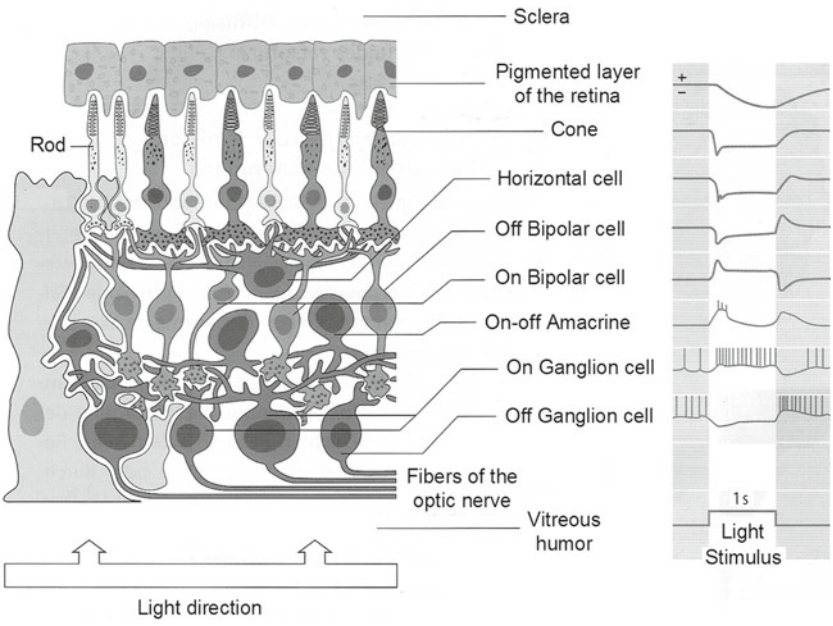


Fig. 11.2 Schematic of the retina in the eye, with the arrangement of rods and cones and other neurons, along with electrical excitation by the shown light stimulus (From [50])

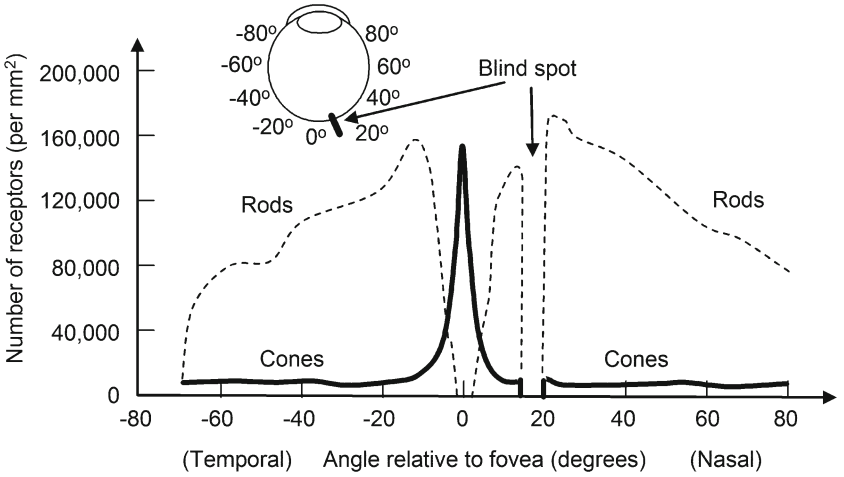


Fig. 11.3 Distribution of rods and cones on the retina, and the location of the blind spot (Based on [13, 63])

We are usually not aware of the blind spot when we use both eyes because the part of the image that forms on the blind spot in one eye is located in a functional region in the other eye and the brain fuses the images of the two eyes. It is easy to prove the existence of the blind spot. Close your right eye and use your left eye to look at the × in Fig. 11.4. When you move the book about 10 cm from your left eye, you will find one position where the central dot disappears because of the blind spot in your left eye.

There are about 120 million rod cells per retina (Fig. 11.5). They have high sensitivity, low spatial acuity, and are relatively more numerous in the periphery of the

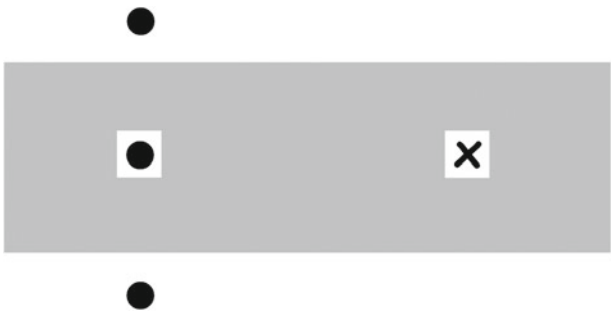


Fig. 11.4 Fixate on the × using your *left* eye, with your *right* eye closed. Keep the book about 10 cm from your *left* eye, and then move it back and forth until you do not see the central dot. This dot is then on the blind spot. The dots above and below it are still visible, but fuzzy because of the lower visual acuity outside the fovea. (Also see Fig. 11.24) (Based on [63])

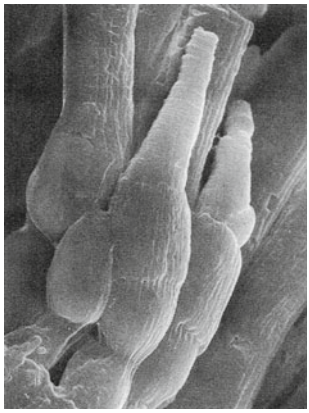


Fig. 11.5 Scanning electron micrograph of rod and cone outer segments, with the cone seen (with its tapered end) in the *center* and the end and beginning of two rods (which are longer than the cones) seen beneath and to the *left* of it (Reprinted from [29]. Used with permission of Elsevier)

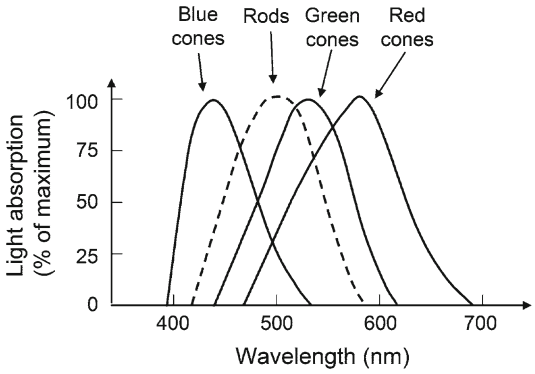


Fig. 11.6 Relative spectral sensitivity of rods and cones. The absolute sensitivity of rods is 1,000× larger those of the cones (Based on [5, 17, 34])

retina. The sensitivity of rods peaks near 500 nm (Fig. 11.6). Vision using only rods results in various shades of gray. Night vision and peripheral vision are mostly due to rods. Rods are about 2 μm in diameter. Far from the fovea the rods become more widely spaced and many (in some cases several hundred) rods are connected to the same nerve fiber. Both factors decrease visual acuity in the outer portions of the retina.

There are about 6.5 million cone cells per retina. They have low sensitivity—about 1,000× lower than rods—high spatial acuity, and are concentrated in the fovea. There are three types of cone cells, with spectral sensitivities peaking near 445 nm (blue or S cones—S for short wavelength peak sensitivity), 535 nm (green or M cones—M for middle wavelengths), and 570 nm (red or L cones—L for long wavelengths) (Fig. 11.6). The overall spectral sensitivity due to the rods and cones of humans closely matches the spectrum of solar light reaching land. Sharp vision and color vision are due to cones, and consequently damage to the fovea leads to visual images that are fuzzy. Cones are about 1.0–1.5 μm in diameter and are about 2.0–2.5 μm

apart in the fovea. There are only about 1 million nerve fibers in the eye, so there are some cones (as well as rods) connected to the same nerve cells. We will not delve into the cellular structure of the rods and cones, but will focus on two physical aspects of these sensors: the absorption of light and acuity of vision.

The absorption of light by the rods and cones is a fundamental quantum-mechanical process in which one photon (or quantum) of light is absorbed by the pigment rhodopsin. Quantum mechanics is the physics of small-scale objects, and has features that are distinct from the physics of larger-scale objects, which is the classical physics we have been using throughout this book. One feature of quantum physics is the quantization of energy levels in molecules, which means that a molecule can have only distinct energies. Consequently, a molecule can absorb light only at those specific energies (or frequencies) corresponding to the differences of its energy levels. Moreover, in quantum mechanics, light acts like light packets, called photons. The energy of a photon is

$$E = h\nu = \frac{hc}{\lambda}, \tag{11.1}$$

where h is Planck's constant (6.626×10^{-34} J–s, as in (6.37)), ν (or f) is the frequency of the light, c is the speed of light (3.0×10^8 m/s), and λ is the wavelength of light. The last two parts of this equation reflect the relationship between frequency, wavelength, and propagation speed for these electromagnetic waves,

$$c = \lambda\nu, \tag{11.2}$$

as in (10.3). Absorption occurs when the photons have energy in ranges that can be absorbed by the photosensitive molecules in these cells.

Rhodopsin consists of a chromophore (i.e., the part of the molecule responsible for its color) covalently attached to the protein opsin. The chromophore is retinal, which is a derivative of vitamin A, and the absorption of a single photon of light isomerizes it (i.e., changes its molecular conformation) from 11-*cis* retinal to all-*trans* retinal. This isomerization triggers a change in the conformation of rhodopsin that starts a sequence of sensory transduction processes (Fig. 11.7). Proteins themselves have absorption bands in the ultraviolet, and cannot absorb in the visible. The absorption of free 11-*cis* retinal is in the near ultraviolet, 360–380 nm; however, the binding of the retinal to the protein red shifts the absorption by about 200 nm to the visible. Differences in the opsin proteins in the rods and the three cones cause the different wavelength responses for these four types of photoreceptor cells. (By the way, Ragnar Granit, Haldan Keffer Hartline, and George Wald were awarded the Nobel Prize in Physiology or Medicine in 1967 for their discoveries concerning the primary physiological and chemical visual processes in the eye.)

The pressure in the eyeball maintains its shape. It is normally about 15 mmHg (ranging from 10 to 20 mmHg), and is determined by the rates of formation of the aqueous humor (about 1% of the total volume is produced per minute) and drainage of the aqueous humor through the canal of Schlemm. If the exit of the aqueous

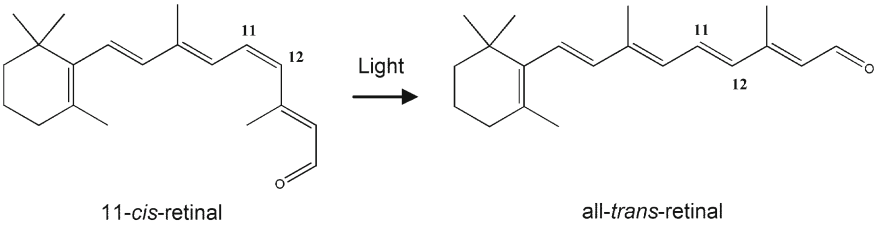


Fig. 11.7 The chromophore 11-*cis* retinal is photoisomerized by light to all-*trans* retinal (11-*trans* retinal)

humor is impaired, the eyeball pressure increases, leading to glaucoma and possible blindness (as is addressed below). Intraocular pressure (IOP) is measured by the amount of force needed to flatten to a given area (or the area flattened by a given force) by using a tonometer. (This is explored in Problem 11.53.)

11.2 Focusing and Imaging with Lenses

11.2.1 Image Formation

Figure 11.8 shows how an object or source is imaged by a *convex (converging or positive) lens*. By convention in optics the object is placed a positive distance d_1 to the left of the lens and optical rays propagate from the left to the right. The object has a size (or height) y_1. For a convex lens the *focal length* f is positive, hence the name positive lens. The central axis (the z-axis) is known as the *optic axis*. All rays passing through the lens form an image a positive distance d_2 to the right of the lens, where d_2 is given by the lens equation

$$\frac{1}{d_1} + \frac{1}{d_2} = \frac{1}{f}. \tag{11.3}$$

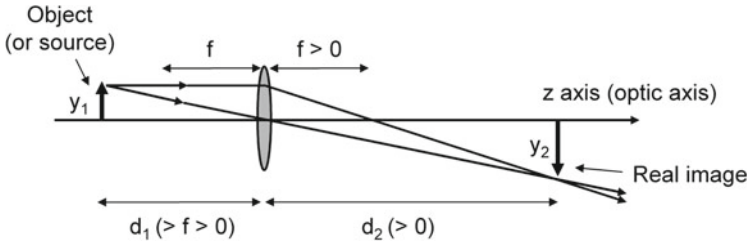


Fig. 11.8 Imaging by a thin, positive lens

A *real image* forms at d_2 when $d_1 > f$, which means that you will see the image at d_2 if you place a screen, such as a piece of paper, there. The image is inverted and its size y_2 is magnified by $M = d_2/d_1$ (the *transverse magnification*). This can be seen from the triangles in Fig. 11.8 that give $y_1/d_1 = y_2/d_2$, so the magnification is

$$M = \frac{d_2}{d_1} = \frac{y_2}{y_1}. \tag{11.4}$$

(Actually, it is magnified when $d_2/d_1 > 1$ and reduced in size (minified) when $d_2/d_1 < 1$.) When $d_1 = \infty$, parallel rays are incident on the lens and an image forms at $d_2 = f$ (Fig. 11.9a). When the object is at the focus and so $d_1 = f$, the image is at ∞ (Fig. 11.9b). When the object is closer to the lens than the focal point and so $d_1 < f$, then $d_2 < 0$ and the image is to the left of the lens (Fig. 11.9c). This is a *virtual image*. Placing a screen there will give no image. However, if the light rays

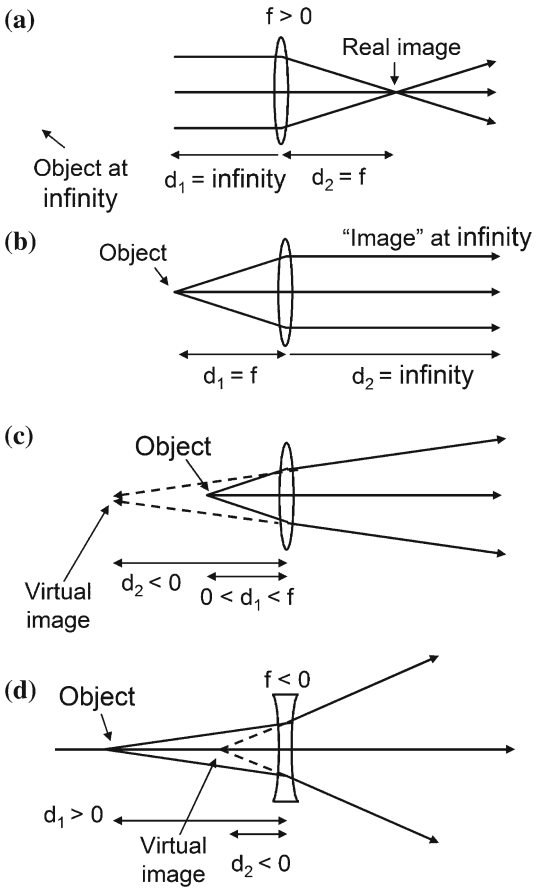


Fig. 11.9 Special cases of imaging with positive lenses (**a**–**c**), and imaging with negative lenses **d**

to the right of the lens are traced backward to the left of the lens, they will seem to emanate from this virtual image.

This same lens equation (11.3) can be used to determine the location of the image for a *concave (diverging or negative) lens*, which has a negative focal length f. Concave lenses produce virtual images (Fig. 11.9d).

Within the eye the cornea and crystalline lens are positive lenses, because they need to form a real image on the retina. Corrective lenses (eyeglass lenses and contact lenses) can have positive or negative focal lengths, depending on the necessary correction. We will explore this later in this chapter. Focal lengths are expressed as distances, in cm or m. We will see that in discussing the eye and corrective lenses it is very common to discuss $1/f$ and use units of *diopters* (D), with 1 D = 1/m.

In a very simple model of the eye imaging system, the eye is treated as a thin lens with a 17 mm focal length in air (*Standard eye* model). For an image at $d_1 \gg f$, (11.3) shows that $d_2 \simeq f$. We will see that two points are resolvable by at best $\Delta y_2 = 2\,\mu\text{m}$ on the fovea. So for a source that is 10 m away, two points are resolvable when separated by at least $\Delta y_1 = (d_1/d_2)\Delta y_2 = (10\,\text{m}/17\,\text{mm})(2\,\mu\text{m}) = 1.2\,\text{mm}$; this corresponds to an angle of 1.2 mm/10 m = 0.12 mrad = 25 s of arc. The $\sim 300\,\mu\text{m}$ foveal diameter corresponds to a lateral separation of $\sim(10\,\text{m}/17\,\text{mm})(300\,\mu\text{m}) = 18\,\text{cm}$ at 10 m or $\sim 18\,\text{cm}/10\,\text{m} = 18$ mrad $\sim 1^\circ$ of arc.

In clinical, ophthalmic optics, the lens equation (11.3) is phrased and used differently. It is expressed in terms of *vergences*, which indicate the angles rays make with the optic axis and signify ray convergence with positive vergence and divergence with negative vergence. U is the *object convergence*, sometimes also known as the *object proximity*. If all optical elements are in a medium with refractive index n, then $U = -n/d_1$. The *image vergence* (or *image proximity*) is $V = n/d_2$. The refractive power of the lens $P = n/f$, as we will see. U, V, and P are expressed in diopters (1 D = 1/m). The lens equation (11.3) expressed in vergences is then:

$$U + P = V. \tag{11.5}$$

This is interpreted as meaning that the propagation of light from an object through the lens to the image increases the ray vergence by an amount equal to the power of the lens (or more generally, the power of the optical interface or optical system).

We have assumed *geometric optics*, which ignores the wave-like features of light due to optical diffraction; this is a good approximation for very short wavelengths and for much of the imaging in the eye. Our analysis also assumes only *paraxial rays*, i.e., all rays are near the optic axis and make small angles to it. (Rays that are farther away from the optic axis—nearer where the maximum amount of light is transmitted—are called *zonal* rays, and those at the margin of the lens are *marginal* rays. We will evaluate the importance of diffraction and of these zonal and marginal rays later in this chapter.)

11.2.2 Scientific Basis for Imaging

We will trace rays by following how they propagate in straight lines in uniform media and how they refract at interfaces by using Snell's Law. Snell's Law of refraction shows that light from medium 1 with index of refraction n_1 impinging at an angle θ_1 (relative to the normal) on a flat interface with medium 2 with refractive index n_2, is refracted to an angle θ_2 given by (Fig. 11.10).

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \tag{11.6}$$

and for small angles ($\theta_1, \theta_2 \ll 1$)

$$n_1\theta_1 = n_2\theta_2. \tag{11.7}$$

Some important indices of refraction are 1.0 for air, 1.33 for water, 1.5–1.6 for different types of glass, and 1.44–1.50 for plastics. Refractive indices actually vary some with wavelength and temperature, but we will ignore those variations at present.

Imaging can occur when the interfaces are curved. Let us consider the refraction of paraxial rays at the interface in Fig. 11.11 from medium 1 to medium 2, which has a spherical radius of curvature R_{12}. (This region can be formed by slicing off a section from a sphere with radius R_{12}, composed of material 2.) As shown here, this radius is defined to be positive (see the Fig. 11.11 inset). Equation (11.6) still applies, but the angle of incidence for a light ray parallel to the optic axis varies with the distance y the ray is displaced from this axis . We see that $\theta_1 = y/R_{12}$. For $y \ll R_{12}$ (so $\theta_1 \ll 1$), (11.7) gives

$$\theta_2 = \frac{n_1}{n_2}\,\theta_1 = \frac{n_1}{n_2}\frac{y}{R_{12}}. \tag{11.8}$$

This refracted ray makes an angle $\theta_1 - \theta_2 = (1 - n_1/n_2)(y/R_{12}) = [(n_2 - n_1)/n_2]$ (y/R_{12}) with the horizontal (Fig. 11.11). Geometry shows that it hits the optic axis a distance $F(y) = y/(\theta_1 - \theta_2)$ after the interface. This distance is independent of y, and so all parallel rays impinging on the curved surface hit the optic axis at this same distance, which is called the focal length f (or f_{12} for this interface)

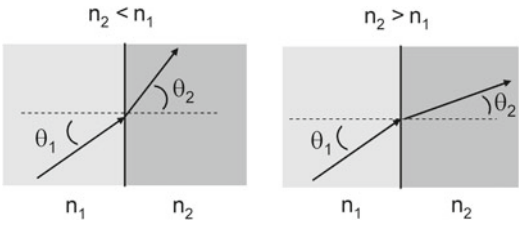


Fig. 11.10 Snell's law

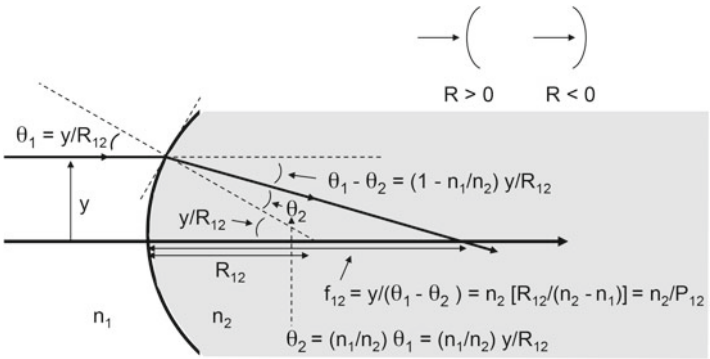


Fig. 11.11 Refraction at a curved interface. The *inset* shows the convention for the radius of curvature. In this figure, $R_{12} > 0$ and $n_2 > n_1$, and $\theta_1 \ll 1$ so $R_{12} \gg y$

$$f_{12} = \frac{y}{[(n_2 - n_1)/n_2](y/R_{12})} = n_2 \frac{R_{12}}{n_2 - n_1} = \frac{n_2}{P_{12}}. \tag{11.9}$$

The last expression has been written in terms of the *refractive power* (or sometimes called the *convergence*) of the interface

$$P_{12} = \frac{n_2 - n_1}{R_{12}}. \tag{11.10}$$

The focal length is defined in terms of the refractive power of the interface and the refractive index of the medium the ray enters. The ratio of the indices of refraction of the two media is important and not their individual values. (For $n_2 = 1$, we see that $f_{12} = 1/P_{12}$.) The units of the refractive power are diopters (1 D = 1/m).

We see that this derivation giving the focal length in (11.9) is similar to that for (11.3), except that the object and image are in regions with different refractive indices here. If n_1 and n_2 were interchanged, the sign of the focal length would change and its magnitude would change to $\mid n_1/P_{12} \mid$. The analog of (11.3) at this refractive interface is

$$\frac{n_1}{d_1} + \frac{n_2}{d_2} = \frac{n_2 - n_1}{R_{12}}. \tag{11.11}$$

For $P_{12} > 0$, when $d_1 = \infty$, a real image occurs at $d_2 = f_{12} > 0$ (for rays traveling from left to right, as is the convention). For rays traveling from right to left and $d_2 = \infty$, a real image occurs at $d_1 = f_{21} > 0$. So in general we see that

$$P_{12} = \frac{n_2 - n_1}{R_{12}} = \frac{n_2}{f_{12}} = \frac{n_1}{f_{21}}. \tag{11.12}$$

(For $P_{12} > 0$, this gives $f_{12} > 0$ and $f_{21} > 0$. Another popular convention is to define f_{21} as the negative of our definition, so the last term in (11.12) would be $-n_1/f_{21}$.)

The magnification, M, is modified from (11.4) to give

$$M = \frac{d_2/n_2}{d_1/n_1} = \frac{y_2}{y_1}. \tag{11.13}$$

Here, the object vergence is $U = -n_1/d_1$ and the image vergence is $V = n_2/d_2$. Then (11.11) can be written as

$$U + P_{12} = V, \tag{11.14}$$

which is similar to (11.5).

What happens when there are two refracting interfaces in succession (Fig. 11.12)? If they are separated by a distance D that is "very small," the same reasoning gives an overall focal length f for this lens

$$f = \frac{n_3}{P_{12} + P_{23}}, \tag{11.15}$$

where P_{12} is given by (11.10) and

$$P_{23} = \frac{n_3 - n_2}{R_{23}}. \tag{11.16}$$

This is called the *thin lens approximation*. Equation (11.15) can be expressed as $f = n_3/P_{\text{total}}$, where

$$P_{\text{total}} = P_{12} + P_{23}, \tag{11.17}$$

so the refractive powers add in this approximation.

For a thin lens of refractive index $n_2 = n$ in air or vacuum (with refractive index $n_1 = n_3 = 1$, so $f_{13} = f_{31}$), (11.15) reduces to

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_{12}} - \frac{1}{R_{23}}\right), \tag{11.18}$$

which is known as the *Lensmaker's equation*. This focal length can also be expressed as

$$f = \frac{1}{P_{12} + P_{23}}, \tag{11.19}$$

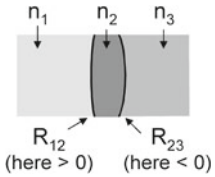


Fig. 11.12 Refraction at two spherical interfaces. As drawn here, the radii of curvature $R_{12} > 0$ and $R_{23} < 0$

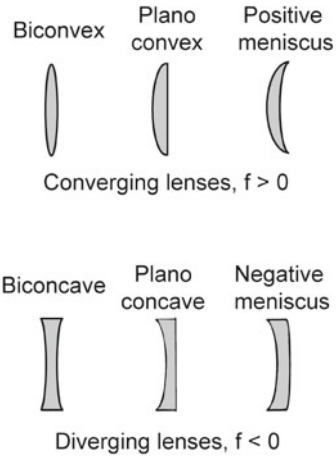


Fig. 11.13 Types of positive and negative lenses

with $P_{12} = (n-1)/R_{12}$ and $P_{23} = -(n-1)/R_{23}$.

Lenses can have a range of shapes even for the same focal length (Fig. 11.13). For positive focal lengths, they can be either biconvex (as shown $R_{12} > 0$ and $R_{23} < 0$), planoconvex (one side flat), or positive meniscus as shown ($R_{23} > R_{12} > 0$). For negative focal lengths, they can be either biconcave (as shown $R_{12} < 0$ and $R_{23} > 0$), planoconcave (one side flat), or negative meniscus (as shown $R_{23} < R_{12} < 0$). The isolated cornea is a negative meniscus lens and the crystalline lens is an asymmetric biconvex lens. We will see below that in the eye the cornea serves as a positive lens.

The length of the eyeball is approximately 24 mm, so the distances from the cornea/crystalline lens to the retina and the focal length of optical rays focused by the cornea/crystalline lens propagating in the vitreous humor and imaging on the retina, are also about 24 mm. The refractive index of the vitreous humor is approximately 1.33. Equation (11.15) shows that a system with the same refractive power has a focal length that is proportional to this refractive index, so in air the focal length would be smaller by a factor of 1.33, or 24 mm/1.33 = 17 mm. That is why a model eyeball can be treated as if the cornea/crystalline lens system had an effective focal length of 17 mm, with the effective lens separated by 17 mm of air from the retina; we will call this the *Standard eye* model. (This differs from the eye models in Table 11.1.)

Moving Lenses

Two lenses have the same *effectivity* (i.e., effectiveness) if both can image the same object to the same image, placed at the appropriate distance between the object and image as given by (11.3). For example, an image forms 4 cm after a lens with focal length 2 cm that is placed 4 cm after an object. If you wanted move the lens to the left by 2 cm, you would have to change its focal lens to 1.5 cm for it to have the image at the same place and so have the same effectivity. (Why?)

Let us consider the special case of parallel rays hitting a converging lens with power P in a medium with refractive index n. They form an image a distance n/P after the lens. If we move the same lens a distance D to the left and want the rays to focus in the same place, we now need them to focus a distance $n/P + D$ after the lens and so the lens will need to have a power P' such that $n/P' = n/P + D$. Therefore we see that

$$P' = \frac{P}{1 + \frac{D}{n}P}. \tag{11.20}$$

This is known as the *effectivity formula*. (This equation is sometimes displayed with a negative sign in the denominator because the lens is being moved a distance D to the right.)

Why is this imaging effectivity important to us? The corrective powers for prescriptions for eyeglasses and contact lenses are different because eyeglasses are placed about 1.2 cm anterior to the cornea, while contact lenses sit right on the cornea, and so are different in two ways. We just addressed the consequences of this different lateral position and will address below those due to refractive power changes that occur because the contact lens is physically on the cornea. In any case, you want both prescriptions to have the same effectivity for objects at infinity, so (11.20) is used with $n = 1$.

11.2.3 Combinations of Lenses or Refractive Surfaces

The eye itself is a combination of four distinct curved imaging interfaces. When imaging on the retina is not perfect, corrective lenses (with two additional interfaces) are chosen so the combined effect of these supplemental lenses and the eye imaging system produces a more perfect image on the retina. To first order we can apply (11.19), including all of the curved interfaces (for the eye or eye + corrective lens), with the ray finally entering a medium j with index of refraction n_j. We see that the focal length is

$$f = \frac{n_j}{\sum_{i=1 \text{ to } j-1} P_{i,i+1}}. \tag{11.21}$$

Merely using (11.21) to determine the imaging properties of these complex systems of curved surfaces or lenses is not sufficient, because it is based on the thin lens approximation, which ignores the propagation of rays from one refracting surface to the next. Because the refractive interfaces are displaced from one another, this usually needs to be corrected.

Two Thin Lenses

Before we explain how to account for the separation of the curved interfaces in a very general way, let us consider the imaging by two separated thin lenses. This approach

is rigorous but it is cumbersome when extended to more than two lenses. It can be applied, for example, to the two refractive surfaces of the cornea or the crystalline lens. We will use it to learn how to correct vision by considering the combined effect of eyeglasses or contact lenses and the eye, with both modeled as simple thin lenses.

Let us consider two thin lenses with focal lengths f_1 and f_2 that are separated by a distance D (Fig. 11.14). An object is placed at a distance d_a to the left of the first lens and the final image is formed at a distance d_b to the right of the second lens. In the *forward propagation approach*, we consider propagation from the left to the right in Fig. 11.14a. The first lens forms an image at a distance d_i to the right of this lens, where

$$\frac{1}{d_a} + \frac{1}{d_i} = \frac{1}{f_1}. \tag{11.22}$$

This intermediate image is at a distance $D - d_i$ to the left of the second lens, and serves as the object for this second lens. Therefore, the second lens forms an image at a distance d_b to its right, where

$$\frac{1}{D - d_i} + \frac{1}{d_b} = \frac{1}{f_2}. \tag{11.23}$$

This distance d_b is determined by inserting d_i from (11.22) into (11.23)

$$D = \frac{d_a f_1}{d_a - f_1} + \frac{d_b f_2}{d_b - f_2}. \tag{11.24}$$

This interrelates all five parameters (D, d_a, d_b, f_1, f_2) and can be used to determine the fifth parameter, such as d_b, when the other four are known.

Sometimes it is simpler to work backward from the final image to the initial source, especially when the location of the final image is fixed (Fig. 11.14b). For example, consider using this *backward propagation approach* when corrective lenses (lens 1) are needed to correct the images of the effective eye lens (lens 2) on the retina. Let us say we know that the eye lens images perfectly on the retina when an object is at a distance z_1 away (to the left of the effective eye lens), but we would like to have clear images of objects at a distance z_2 away. The eye by itself cannot accomplish this, as we will soon see, due to insufficient accommodation or to myopia or hyperopia. The distance from the corrective lens to the eye lens is fixed, $D \sim 1.2$–$1.5\,\text{cm}$ for eyeglass lenses and $D \sim 0\,\text{cm}$ for contact lenses. Therefore this distance z_1 to the left of lens 2 is at a known distance $z_1 - D$ to the left of lens 1. The focal length of corrective lens 1, f_1, is chosen so that it takes an object $d_a = z_2 - D$ to the left of it and forms an image $d_i = D - z_1$ to the right of it. (The eye can then image this intermediate object quite well by itself.) Using (11.3), this condition becomes

$$\frac{1}{z_2 - D} + \frac{1}{D - z_1} = \frac{1}{f_1}, \tag{11.25}$$

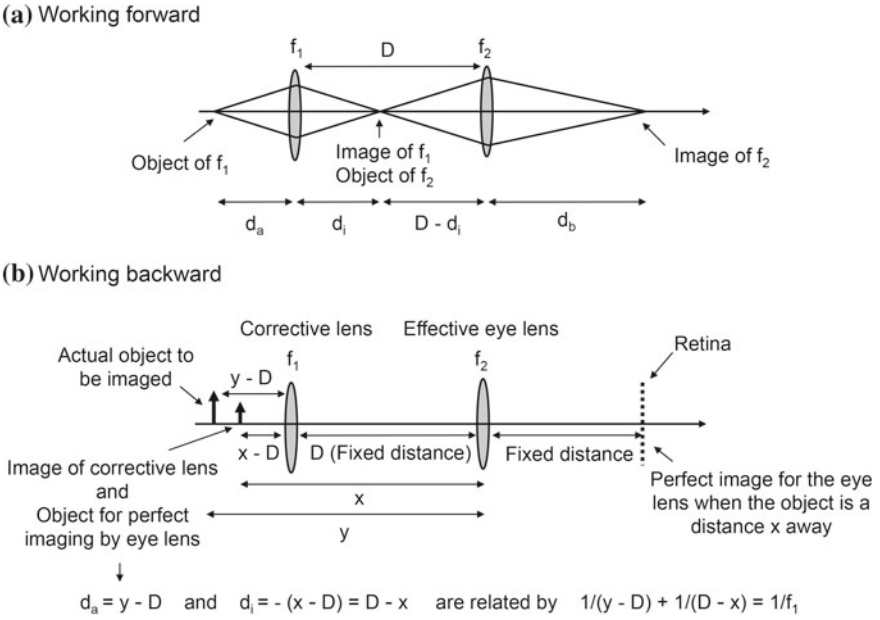


Fig. 11.14 Imaging by two lenses using **a** forward and **b** backward propagation

and so the focal length of the necessary corrective lens is $f_{\text{corrective}} = f_1$. One or both of the terms on the left-hand side of this equation can be negative. We will use this approach below to prescribe corrective lenses.

Of course, you can analyze two lenses with the forward and backward propagation approaches by using the totally equivalent method of vergences.

Complex Optical Systems (Advanced Topic)

We will now see that any system with several refracting surfaces, such as the eye, can be reduced to a simple effective optical system that is similar to a thin lens (Fig. 11.15). The effective thin lens system is analyzed by tracing optical rays with the help of six reference points on the optic axis called *cardinal points*. There are two principal points (P, P′), two focal points (F, F′), and two nodal points (N, N′).

Rays traveling parallel to the optic axis (from left to right) in the first medium (with refractive index n where the object is located) refract in the optical system and travel as straight rays that converge at the second (or image) *principal focus* at point F′ (after the last interface in the last medium with refractive index n') (Fig. 11.15). When these rays are backtracked, they each intersect the initial parallel rays at a surface called the second (or image) *principal plane* **P′**. Similarly, parallel rays traveling from right to left in this last medium intersect in the first medium at the first (or

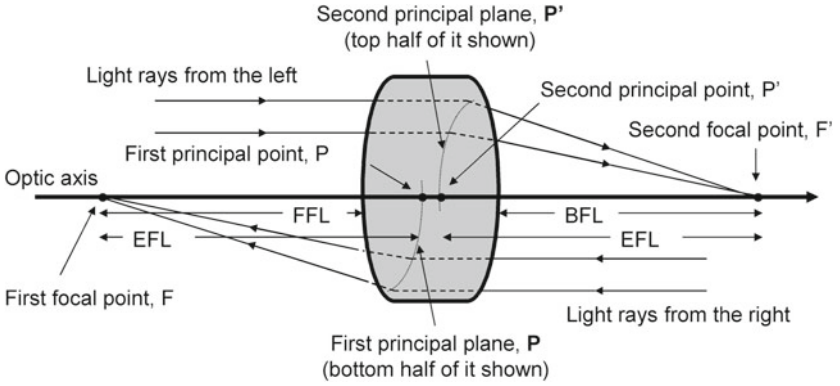


Fig. 11.15 Optical planes and imaging in a thick lens (or any multielement optical system) for rays that are parallel to the axis. The second (or first) focal length is the distance from the second (or first) principal point to the second (or first) focal point. If the source and image media refractive indices, n and n' are the same, both are called the effective focal length (EFL). The back (or front) focal length (BFL or FFL) is measured from the last (first) vertex of the last (first) optical element

object) *principal focus* at point F, and when these rays are backtracked, they each intersect the initial parallel rays at a surface called the first (or object) principal plane **P**. (These "planes" are really curved surfaces that are approximately planar near the optic axis.) The first and second (or object and image) *principal points* P and P′ are the intersections of these planes with the optic axis. We will see that there are also *nodal points* N and N′ on the optic axis, which help in analyzing image formation (Fig. 11.16).

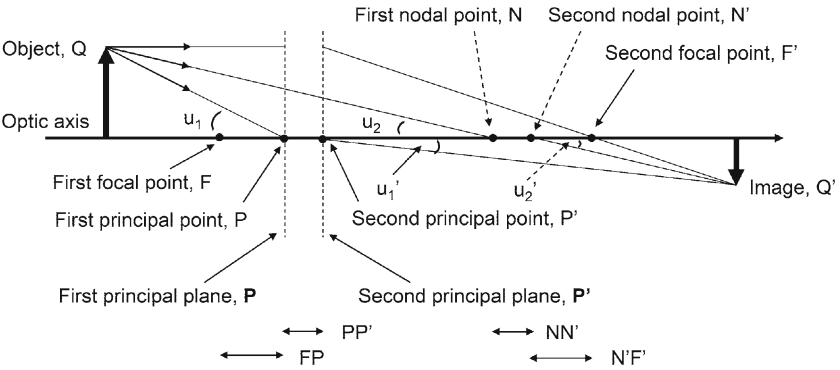


Fig. 11.16 Image formation using cardinal points. The location and size of the image (in a medium with refractive index n') can be determined by tracing rays from the object (in a medium with refractive index n): the object ray parallel to the optic axis and either the ray going through principal point P at angle u_1 and emerging from P′ at angle $u'_1 = (n/n')u_1$ or that going through nodal point N at angle u_2 and emerging from N′ at the same angle $u'_2 = u_2$. This example is similar to that of Schematic eye 2 depicted in the top half of Fig. 11.20, with $n' > n$

We will adopt a sign convention that is convenient for biconvex, positive lenses, so that allfocal lengths will be defined to be positive for such lenses. (This is consistent with (11.12), but is not universal notation.) The first principal focus F is a distance f to the left of P when $f > 0$ and the second principal focus at F′ is a distance f' to the right of P′ when $f' > 0$. We will call distances along the optic axis, say between P and another point A—which is PA, positive when A is to the right of P and negative when it is to the left of it; therefore, AP = −PA. There is symmetry in the positions of the six cardinal points, with the separations PP′ = NN′ and FP = N′F′, as is depicted in Fig. 11.16.

Rays traveling from left to right emanate from the same object point Q and intersect to form an image at Q′. Any ray from Q hits plane **P** and emerges from plane **P**′ as if it were refracted by a thin lens using (11.3), with a "no man's land" between the two principal planes. So, with $d_1 = \text{QP}$ (which is the distance along the optic axis) and $d_2 = \text{P}'\text{Q}'$, the distance from object to image is really

$$\text{QQ}' = \text{QP} + \text{PP}' + \text{P}'\text{Q}' = d_1 + d_2 + \text{PP}'. \tag{11.26}$$

Rays that hit point P, at an angle u, act as if they "emerge" from P′ at an angle u' (Fig. 11.16), where

$$n'u' = nu. \tag{11.27}$$

This can be viewed as Snell's Law in this paraxial limit. Nodal points are also defined in a way that rays on course to hit the first nodal point N, act as if they "emerge" from the second nodal point N′ with no change in angle. Again, these two sets of rays emanating from the same point Q intersect to form an image at Q′, as is seen in Fig. 11.16. The principal and nodal points coincide when the media to the left and right have the same refractive index; this is not the case for imaging by the eye. (For a thin lens in air, P, P′, N, and N′ merge into one point (inside the lens for a biconvex lens), and $f' = f$.)

Two Arbitrary Optical Refractive Systems (Advanced Topic)

Let us now consider two arbitrary optical systems [28] (Fig. 11.17). The first has principal points P_a and P'_a, object and image refractive indices n_a and n'_a, and refracting power P_a. It is followed by the second with principal points P_b and P'_b, with object and image refractive indices n_b and n'_b, and refracting power P_b. The reduced distance between the second principal point of the first system and the first principal point of the second system is defined as

$$\delta = \frac{t}{n'_\text{a}} = \frac{t}{n_\text{b}} = \frac{t}{n_2}, \tag{11.28}$$

with $t = \text{P}'_\text{a}\ P_\text{b}$. We will now call $n_\text{a} = n_1$, $n'_\text{a} = n_\text{b} = n_2$, and $n'_\text{b} = n_3$.

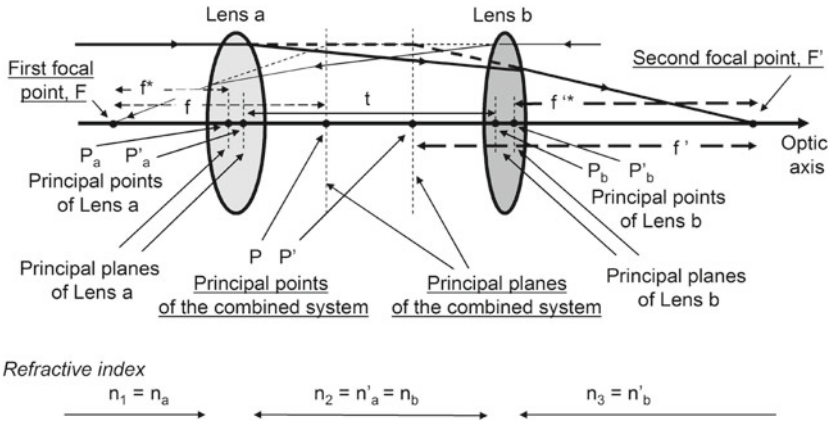


Fig. 11.17 Analyzing the combination of two optical systems, represented here by two "lenses." The notation for the planes and points of the combined systems is *underlined*. The *solid line* ray refracts at the principal planes of the individual systems, while the *dashed ray* refracts at the principal planes of the overall system

Optical analysis that is beyond our scope gives general expressions for combining these two optical systems. The effective refractive power of the entire system can be shown to be

$$P_{\text{eff}} = P_{\text{a}} + P_{\text{b}} - \delta P_{\text{a}} P_{\text{b}} \tag{11.29}$$

or

$$P_{\text{eff}} = P_{\text{a}} + P_{\text{b}} - \frac{t}{n_2} P_{\text{a}} P_{\text{b}} = \frac{n_3}{f'} = \frac{n_1}{f}. \tag{11.30}$$

(This is consistent with (11.12). Using some other sign conventions for focal lengths, this last term is expressed as $-n_1/f$.)

Where are the principal points of the combined system? The position of the second principal point of the overall system P′ relative to the second principal point of the second system is

$$\frac{\text{P}'_{\text{b}}\text{P}'}{n_3} = -\frac{t}{n_2} \frac{P_{\text{a}}}{P_{\text{eff}}}, \tag{11.31}$$

which means that P′ is to the left of P'_b in Fig. 11.17 when $P_{\text{a}}/P_{\text{eff}} > 0$ (such as for a biconvex lens).

Similarly, the position of the first principal point of the overall system P relative to the first principal point of the first system is

$$\frac{\text{P}_{\text{a}}\text{P}}{n_1} = +\frac{t}{n_2} \frac{P_{\text{b}}}{P_{\text{eff}}}, \tag{11.32}$$

which means that P is to the right of P_a in Fig. 11.17 when $P_{\text{b}}/P_{\text{eff}} > 0$ (such as for a biconvex lens).

What are the new focal lengths? The second principal focus of the combined system F′ is a distance

$$\mathrm{P'F'} = f' \tag{11.33}$$

to the right of new second principal point P′. The new first principal focus F is a distance

$$\mathrm{FP} = f \tag{11.34}$$

to the left of new first principal point P. The focal lengths are given by (11.30). If $n = n'$, these focal lengths are the same (and are the effective focal length (EFL), as seen in Fig. 11.15).

Where are these foci relative to the principal points of the original two systems? Using (11.30) and combining (11.31)–(11.34) gives the position of the second focal point relative to the second principal plane of the second system $\mathrm{P'_bF'} = \mathrm{P'_bP'} + \mathrm{P'F'}$:

$$\mathrm{P'_bF'} = f'\left(1 - \frac{t}{n_2}P_\mathrm{a}\right) = f'^* \tag{11.35}$$

and the position of the first focal point relative to the first principal plane of the first system $\mathrm{FP_a} = \mathrm{FP} + \mathrm{PP_a}$:

$$\mathrm{FP_a} = f\left(1 - \frac{t}{n_2}P_\mathrm{b}\right) = f^*. \tag{11.36}$$

The parameters f^* and f'^* are sometimes called the front and back (vertex) focal lengths (FFL and BFL in Fig. 11.15) because for a thick lens these are the distances from the front and back surfaces. In analogy with (11.30), sometimes the front and back vertex powers are defined as

$$P^* = \frac{n_1}{f^*} \tag{11.37}$$

and

$$P'^* = \frac{n_3}{f'^*}. \tag{11.38}$$

Combining these with (11.30), and using (11.35) and (11.36) shows that

$$P^* = \frac{P_\mathrm{b}}{1 - \delta P_\mathrm{b}} + P_\mathrm{a} \tag{11.39}$$

and

$$P'^* = \frac{P_\mathrm{a}}{1 - \delta P_\mathrm{a}} + P_\mathrm{b}, \tag{11.40}$$

with $\delta = t/n_2$. This is an equivalent way of determining the location of the focal points of the combined optical system.

Equations (11.39) and (11.40) look very similar to the effectivity formula, (11.20), and for good reason: they are very closely related. Let us consider the effectivity formula for a thin lens "a" with refractive power P_a that is moved a distance $t = n_2\delta$ to the right, so it would be coincident with thin lens "b" with refractive power P_b. If its effectivity were not to change, its power must change from P_a to $P'_a = P_a/[1 - (t/n)P_a]$, as given by the effectivity formula with $D = -t$. (Remember that D corresponds to a leftward displacement.) The new refractive power at the second lens is then $P'_a + P_b$. This gives (11.40). (Of course, now everything is referred to the principal planes of the optical systems rather than the location of the thin lenses.) Equation (11.39) is similarly obtained.

Because $\mathrm{P'P'_b}$, $\mathrm{F'P'}$, $P_a\mathrm{P}$, and FP are known, the nodal points F and F′ can be found using $\mathrm{PP'} = \mathrm{NN'}$ and $\mathrm{FP} = \mathrm{N'F'}$, with F′ following F.

This approach can be used to evaluate complex optical systems. For example, for the eye it can be applied first to the cornea as a lens (see below), then to the crystalline lens, and then to combine these two optical elements to describe the paraxial optics of the entire eye. If the subject wears corrective lenses, the combined effect of these lenses and the eye can then be evaluated.

Thick Lenses. The focal length of a thick lens of thickness t follows from (11.30) and substituting $P_a = (n_2 - n_1)/R_{12}$ and $P_b = (n_3 - n_2)/R_{23}$ from (11.10)

$$\frac{n_3}{f_{\text{eff}}} = \frac{n_2 - n_1}{R_{12}} + \frac{n_3 - n_2}{R_{23}} - \frac{(n_2 - n_1)(n_3 - n_2)t}{n_2 R_{12} R_{23}} \tag{11.41}$$

and the back focal length is

$$f_{\text{bfl}} = f_{\text{eff}} - \frac{f_{\text{eff}}(n_2 - n_1)t}{n_2 R_{12}}. \tag{11.42}$$

The front focal length can similarly be determined.

For a single lens of refractive index n in air, the distance rays propagate between refractive surfaces affects the focal length. With the refractive surfaces separated by a distance t, the Lensmaker's equation, (11.18), for thin lenses can now be generalized to the *thick lens equation*

$$\frac{1}{f_{\text{eff}}} = (n-1)\left(\frac{1}{R_{12}} - \frac{1}{R_{23}} + \frac{(n-1)t}{nR_{12}R_{23}}\right). \tag{11.43}$$

This effective focal length is the distance from the second principal point. The distance from the second surface to the focus is the (back) focal length f_{bfl}

$$f_{\text{bfl}} = f_{\text{eff}} - \frac{f_{\text{eff}}(n-1)t}{nR_{12}}, \tag{11.44}$$

so for incident parallel rays the image forms a distance f_{bfl} past the second surface of the lens.

11.3 Imaging and Detection by the Eye

11.3.1 Transmission of Light in the Eye

Figure 11.18 shows the percentage of incident light that reaches the aqueous humor, the crystalline lens, the vitreous humor, and the retina. Perhaps it is surprising that only about 50% of visible light (400–700 nm) incident on the eye actually reaches the retina to form an image. These losses are due to reflection of light at the interfaces between the different ocular media, and the absorption and scattering in these ocular media. Much of this loss is due to scattering in the eye. This scattered light does not contribute to the desired image even if it hits the retina. Also note the large transmission through the retina, so there is little loss due to absorption and scattering with the retina before light hits the rods and cones (Fig. 11.2).

The expression for the fraction of light that reflects from a planar interface between two semi-infinite media with refractive indices n_1 and n_2 is actually the same as that for the reflection of acoustic waves between the interface of media. In Figs. 10.3 and (10.33) the acoustic impedances Z_1 and Z_2 are replaced by the respective refractive indices n_1 and n_2 to give

$$R = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 . \tag{11.45}$$

The reflected fraction at the air/cornea interface $[(1.377 - 1)^2/(1.377 + 1)^2] \sim 2\%$ is much larger than that at any other interface in the eye because of the very large difference in refractive indices between air and the cornea. The actual reflected fraction is really a bit larger because the reflected fraction increases for other angles of incidence (and all light rays do not enter the eye at normal incidence) and the ocular interfaces are curved. Still, reflection accounts for little of the transmission losses.

When a penlight is shined into an eye, four images, called *Purkinje images*, are formed from the reflection off the anterior and posterior surfaces of the cornea and crystalline lens. Purkinje image I from the anterior cornea surface is the strongest, as was just shown; it is a virtual, erect image 3.85 mm from the corneal apex. Purkinje image II from the posterior cornea surface is the next strongest; it is a virtual, erect image 3.77 mm from the corneal apex. Purkinje images III and IV from the anterior and posterior crystalline lens surfaces are the weakest; image III is a virtual, erect image 10.50 mm from the corneal apex and image IV is a real, inverted image 3.96 mm from the corneal apex.

Figure 11.18 shows that most of the light from 300 to 400 nm is absorbed by the crystalline lens, mostly by the yellow macular pigment xanthophyll. There are no important sources of absorption in the visible (which is one reason why this light is "visible"). Most of the transmission losses in the visible are due to scattering within the eye components due to inhomogeneities in the index of refraction, such as those caused by cells and submicroscopic particles. The efficiency of scattering

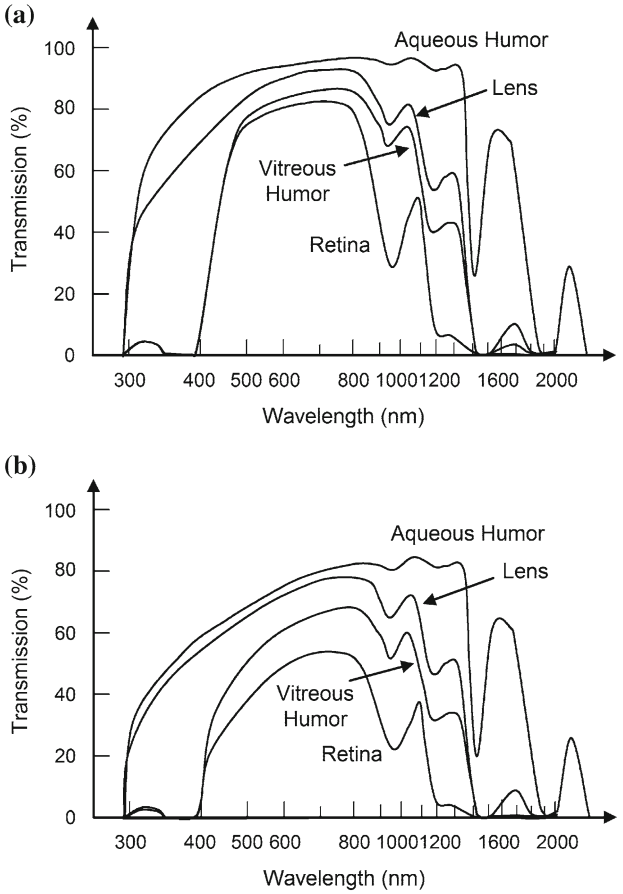


Fig. 11.18 Transmission of near-ultraviolet, visible, and near-infrared light through the eye up to the labeled part of the eye. In **a** only transmission losses due to absorption are included, while in **b** all transmission losses are accounted for, including those due to absorption and scattering (Based on [4, 19])

attributed by changes in the refractive index over distances $\ll\lambda$, is called Rayleigh scattering. Scattering by a concentration N of nonabsorbing particles of diameter D and refractive index n_p in an ambient medium of refractive index n_a (such as that of the crystalline lens, cornea, or the humors) decreases the transmission of light intensity (I in W/m^2) according to Beer's Law (10.18).

$$I(z) = I(z=0)\exp(-\alpha_{\text{light scattering}}z), \tag{11.46}$$

where I is the incident intensity (say in W/m^2). For a light of wavelength λ (as measured in vacuum or air)

$$\alpha_{\text{light scattering}} = \frac{8\pi^4}{3}\frac{ND^6}{\lambda^4}\left(\frac{{n_\text{p}}^2 - n_\text{a}^2}{{n_\text{p}}^2 + 2n_\text{a}^2}\right)^2. \tag{11.47}$$

(The derivation of this is beyond the current scope.) The important point here is that the efficiency of scattering varies as $1/\lambda^4$ or, using (11.2), as ν^4. So blue light scatters more efficiently than red light, and this explains the greater loss intransmission at shorter visible wavelengths in Fig. 11.18b. (This is also one of the reasons why the sky is blue; blue light scatters more efficiently.)

The cornea, white of the eye, skin, tendons and ligaments all largely consist of collagen fibrils in a matrix. The cornea is very transparent while these other tissues are opaque. Why? The refractive index of the fibrils, ~1.47, differs from that of the surrounding media, 1.345, and the strength of the elastic scattering of light depends on this difference. Since the fibrils are densely packed in the cornea and it is relatively thick (~0.05 cm), we would expect the cornea to be opaque also. However, in the cornea the fibrils are partially ordered, and scattering is not incoherent (the sum of the scattered intensity from each fibril) but partially coherent and there is destructive interference (cancellation) of the scattered waves that decrease overall scattering and, thankfully, improves transparency [18, 36, 56].

11.3.2 The Eye as a Compound Lens

Parameters for several progressively simpler *schematic* models for the imaging of paraxial rays in the eye are given in Table 11.1 [3, 6, 8, 9, 15, 28]. In each case the eye is assumed to be fully relaxed (or unaccommodated), with the exception of Schematic eye 2′, which accounts for accommodating Schematic eye 2 from imaging distant objects to imaging near objects. The most complete model is the Gullstrand *exact eye* (Schematic exact eye), which has six refractive surfaces: at the anterior and posterior surfaces of both the cornea and crystalline lens and two within the crystalline lens, so the variation of the refractive index within the lens is included (in a mathematical, but not a totally accurate physical, manner). (By the way, Allvar Gullstrand was the recipient of the 1911 Nobel Prize for Physiology or Medicine for his work in this area, the dioptrics of the eye. Dioptrics is the branch of geometrical optics dealing with the formation of images by refraction, especially by lenses.)

We will call the next simplest model Schematic eye 1 (the classic Emsley model), for which the refractive index within the crystalline lens is uniform, so there are only four refractive surfaces: at the anterior and posterior surfaces of both the cornea and crystalline lens. Figure 11.19 is a diagram of this model, showing the radii or curvature and locations of the four interfaces and the refractive indices of the five media. The given refractive index of the crystalline lens, 1.4200, is an averaged value; it ranges from about 1.406 near the center to about 1.386 far away from the center.

Table 11.1 Dimensions for Schematic (Schem.) and Reduced (Red.) eyes

	Schem. exact eye	Schem. eye 1	Schem. eye 2	Schem. eye 2'	Schem. eye 3	Red. eye
Radii of surfaces						
Anterior cornea	7.70	7.80	7.80	7.80	7.80	5.55
Posterior cornea	6.80	6.50	–	–	–	–
Anterior lens	10.00	10.20	10.00	5.00	11.00	–
First internal lens	7.911	–	–	–	–	–
Second internal lens	–5.76	–	–	–	–	–
Posterior lens	–6.00	–6.00	–6.00	–5.00	–6.476	–
Distance from anterior cornea						
Posterior cornea	0.50	0.55	–	–	–	–
Anterior lens	3.60	3.60	3.60	3.20	3.60	–
First internal lens	4.146					
Second internal lens	6.565					
Posterior lens	7.20	7.60	7.20	7.20	7.30	–
Retina	23.9	24.20	23.89	23.89	24.09	–
First principal point P	1.348	1.59	1.55	1.78	1.51	0
Second principal point P′	1.602	1.91	1.85	2.13	1.82	0
First nodal point N	7.078	7.20	7.06	6.56	7.11	5.55
Second nodal point N′	7.332	7.51	7.36	6.91	7.42	5.55
First focal point F	–15.707	–15.09	–14.98	–12.56	–15.16	–16.67
Second focal point F′	24.387	24.20	23.89	21.25	24.09	22.22[a]
Refractive indices						
Cornea	1.376	1.3771	–	–	–	4/3
Aqueous humor	1.336	1.3374	1.3333	1.3333	1.336	4/3
Crystalline lens–anterior	1.386	1.4200	1.4160	1.4160	1.422	4/3
Crystalline lens–nucleus	1.406					
Crystalline lens–posterior	1.386					
Vitreous humor	1.336	1.3360	1.3333	1.3333	1.336	4/3

Using data from [3, 15, 28]

All eyes are accommodated for distant vision (unaccommodated), except Schematic eye 2′, which accounts for accommodation in Schematic eye 2 for near vision (accommodated). Distances are in mm

[a]For the Reduced eye, the second focal point is 1.67 mm + 22.22 mm = 23.9 mm after the real anterior surface of the cornea

The cornea and aqueous humor are treated as one region in the *simplified schematic eyes* (Schematic eyes 2, 2′, and 3) so it has three refractive surfaces: at the cornea and at the two surfaces of the crystalline lens. Schematic eyes 2 and 2′ are the Gullstrand–Emsley models of relaxed and accommodated eyes and Schematic eye 3 is the revised relaxed eye model by Bennett and Rabbetts. The top part of Fig. 11.20 shows the location of the cardinal points for Schematic eye 2.

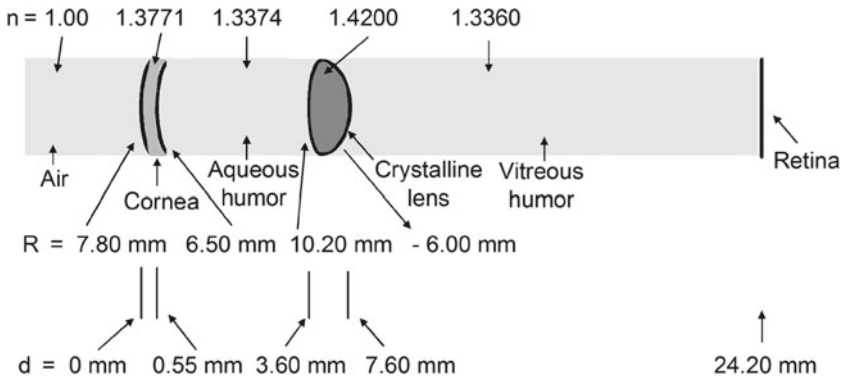


Fig. 11.19 Schematic eye 1

The simplest model is the *Reduced eye* (Fig. 11.21), which has only one refractive interface: at the "cornea"—which is actually 1.67 mm after the real cornea. This is seen along with the cardinal points for this Reduced eye in the bottom part of Fig. 11.20. For this eye all distances are relative to the single refractive interface (1.67 mm after the real anterior surface of the cornea), so the second focal point is 1.67 mm + 22.22 mm = 23.9 mm after this corneal surface.

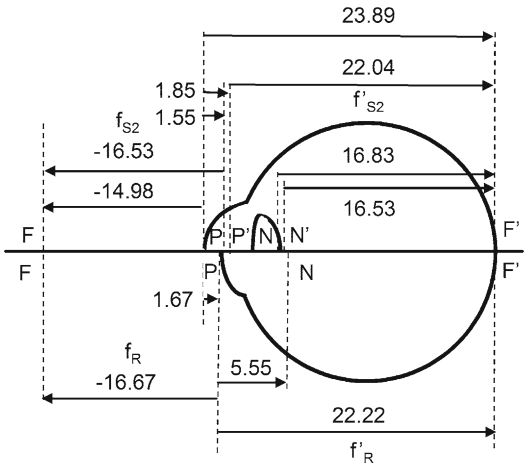


Fig. 11.20 Comparison of the cardinal points of the three-surface, relaxed Gullstrand–Emsley Schematic eye 2 in the *top* half and the Reduced eye in the *bottom* half. Distances are in mm (Based on [3])

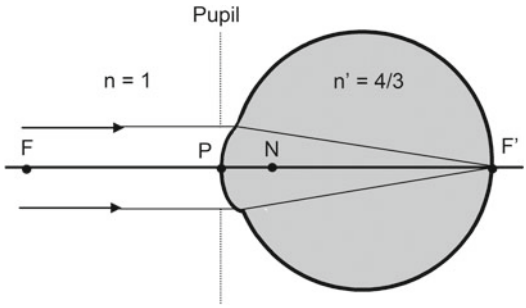


Fig. 11.21 Reduced eye (Based on [3])

Thin Lens Approximation of the Schematic Eye

Consider Schematic eye 1 with four interfaces (Table 11.1). The refractive power of each interface is

$$P_{i,i+1} = \frac{n_{i+1} - n_i}{R_{i,i+1}}. \tag{11.48}$$

We will initially ignore the distance the rays propagate between the refractive interfaces. For the air/anterior cornea interface:

$$P_{12} = \frac{1.3771 - 1.0}{0.0078\,\text{m}} = 48.35\,\text{D}. \tag{11.49}$$

For the posterior cornea/aqueous humor interface:

$$P_{23} = \frac{1.3374 - 1.3771}{0.0065\,\text{m}} = -6.11\,\text{D}. \tag{11.50}$$

For the aqueous humor/anterior crystalline lens interface:

$$P_{34} = \frac{1.4200 - 1.3374}{0.0102\,\text{m}} = 8.10\,\text{D}. \tag{11.51}$$

For the posterior crystalline lens/vitreous humor interface:

$$P_{45} = \frac{1.3360 - 1.4200}{-0.0060\,\text{m}} = 14.00\,\text{D}. \tag{11.52}$$

The refractive power of the cornea, ignoring its thickness, is (11.17)

$$P_{\text{cornea}} = P_{12} + P_{23} = 48.35\,\text{D} - 6.11\,\text{D} = 42.24\,\text{D}. \tag{11.53}$$

The refractive power of the crystalline lens, again ignoring its thickness, is

$$P_{\text{lens}} = P_{34} + P_{45} = 8.10\,\text{D} + 14.00\,\text{D} = 22.10\,\text{D}. \tag{11.54}$$

This shows that two-thirds of the refractive power is due to the cornea and one-third is due to the lens.

The total refractive power of the eye is

$$P_{\text{eye}} = P_{\text{cornea}} + P_{\text{lens}} = 42.24\,\text{D} + 22.10\,\text{D} = 64.34\,\text{D}. \tag{11.55}$$

The focal length is (11.21)

$$f = \frac{n_j}{\sum_{i=1 \text{ to } j-1} P_{i,i+1}} = \frac{1.336}{64.34\,\text{D}} = 0.0208\,\text{m} = 20.8\,\text{mm}. \tag{11.56}$$

The image from this compound lens falls on the retina, which is 24.20 mm from the anterior surface of the cornea and so this calculated focal length is not exactly correct. The compound lens is 7.6 mm long (anterior surface of the cornea to the posterior surface of the crystalline lens), so we would expect that the focal length is really measured for this type of compound lens from somewhere between the cornea and crystalline lens. We will estimate that it is from the middle (at the position 3.8 mm), so we would expect the image to fall 3.8 mm + 20.8 mm = 24.6 mm from the anterior surface of the cornea, compared to 24.20 mm. This agreement is surprisingly good. (Because the refractive power of the cornea and crystalline lens are not equal, the "starting point" is not exactly in the center, but this is a reasonable first guess.) We have ignored the propagation of light between the curved, refractive interfaces, which is not insignificant here, and will address it in the next section.

What happens when you swim in water? The refractive power of the first (air/anterior cornea) interface changes to

$$P_{12} = \frac{1.3771 - 1.331}{0.0078\,\text{m}} = 5.91\,\text{D}, \tag{11.57}$$

which is a loss of 42.44 D of refractive power. The refractive power of the cornea is 5.91 D + (−6.11 D) = −0.2 D, which means the cornea has essentially no refractive power under water. The total refractive power of the eye is only −0.2 D + 22.10 D = 21.90 D, and the eye sees very blurred images because the focused image would be beyond the retina. (Why can we see much better in water when wearing ordinary goggles?) The images in water are made even blurrier when the water is not perfectly still, because the movement of water causes local variations in the index of refraction.

More Exact Analysis of the Schematic Eye (Advanced Topic)

How important is the finite separation of the refractive elements in the eye, as represented by Schematic eye 1 (Fig. 11.19)?

Using (11.30), (11.49), and (11.50), the effective power of the cornea is

$$P_{\text{cornea}} = P_{12} + P_{23} - \frac{t_{\text{cornea}}}{n_{\text{cornea}}} P_{12} P_{23} \tag{11.58}$$

$$= 48.35\,\text{D} - 6.11\,\text{D} - \left(\frac{0.00055\,\text{m}}{1.3771}\right)(48.35\,\text{D})(-6.11\,\text{D}) = 42.36\,\text{D} \tag{11.59}$$

for a cornea thickness $t_{\text{cornea}} = 0.55\,\text{mm} = 0.00055\,\text{m}$ and refractive index $n_{\text{cornea}} = 1.3771$. (Remember, 1 D = 1/m.) This $+0.12\,\text{D}$ correction to $P_{12} + P_{23} = 42.24\,\text{D}$ from (11.53) is very small because the cornea is very thin. Using (11.31) and (11.32), the principal points of the cornea P_{cornea} and $\text{P}'_{\text{cornea}}$ are $-0.058\,\text{mm}$ and $-0.060\,\text{mm}$ to the right of the anterior surface of the cornea, meaning they are 0.058 mm and 0.060 mm to the left of this surface.

Using (11.30), (11.51), and (11.52), the effective power of the crystalline lens is

$$P_{\text{lens}} = P_{34} + P_{45} - \frac{t_{\text{lens}}}{n_{\text{lens}}} P_{34} P_{45} \tag{11.60}$$

$$= 8.10\,\text{D} + 14.00\,\text{D} - \left(\frac{0.0040\,\text{m}}{1.420}\right)(8.10\,\text{D})(14.00\,\text{D}) = 21.78\,\text{D} \tag{11.61}$$

for a crystalline lens thickness $t_{\text{lens}} = 4.0\,\text{mm} = 0.0040\,\text{m}$ and refractive index $n_{\text{lens}} = 1.420$. This includes a $-0.32\,\text{D}$ correction to $P_{34} + P_{45} = 22.10\,\text{D}$ from (11.54). Using (11.31) and (11.32), theprincipal points of thecrystalline lens P_{lens} and P'_{lens} are 2.42 mm and 2.60 mm to the right of the anterior surface of the lens (meaning they are inside the lens) and they are 6.02 mm and 6.20 mm to the right of the anterior surface of the cornea.

Combining these two lenses using (11.30), the effective power of this Schematic eye 1 is

$$P_{\text{eye}} = P_{\text{cornea}} + P_{\text{lens}} - \frac{t_{\text{aqueous humor}}}{n_{\text{aqueous humor}}} P_{\text{cornea}} P_{\text{lens}} \tag{11.62}$$

$$= 42.36\,\text{D} + 21.78\,\text{D} - \left(\frac{0.00608\,\text{m}}{1.3374}\right)(42.36\,\text{D})(21.78\,\text{D}) = 59.95\,\text{D}. \tag{11.63}$$

Referencing the six cardinal points to the anterior surface of the cornea (a positive number means the point is to the right of it and a negative number means it is to the left of it), the principal points P and P′ are at 1.59 mm and 1.91 mm. The first and second principal foci F and F′ are at $-15.09\,\text{mm}$ and 24.20 mm, which means they are respectively $f = 16.68\,\text{mm}$ to the left of P and $f' = 22.29\,\text{mm}$ to the right of P′. Using PP′ = NN′ and FP = N′F′, the nodal points N and N′ are at 7.20 mm and 7.52 mm, respectively. (Except for round-off error, these are the same as the values listed in Table 11.1.)

11.3.3 Accommodation

The controllability in the focal length of the eye crystalline lens, and the corresponding changes in overall eye focal length, is called *accommodation*. The eye crystalline lens is suspended by ligaments and the tension in these ligaments controls the curvature of the crystalline lens surfaces and consequently its focal length. These ligaments are attached to the ciliary muscles and the state of these muscles determines the tension in these ligaments (Fig. 11.22). When the ciliary muscles are relaxed (Fig. 11.22b), the tension in the suspensory ligaments is at a maximum, the crystalline lens flattens and the focal length is at a maximum. The lens equation shows that this flatter lens will produce clear images of distant objects. When the ciliary muscles are contracted, the ligaments are relaxed (Fig. 11.22a) and the crystalline lens assumes its more normal spherical shape. The larger curvature of the crystalline lens surfaces produces a shorter focal length, which produces clear images of nearby objects. Schematic eye model 2′ for near vision (accommodated) in Table 11.1 differs from Schematic eye models 2 for distant vision (unaccommodated) by a large change in the curvature of the anterior surface of the lens and a smaller change in the curvature of the posterior surface. In some people the change in the posterior surface may be very small and there may be a change in the effective refractive index of the crystalline lens in accommodation. (Analysis of accommodation is addressed further in Problems 4.5, 11.19, 11.21, and 11.49.)

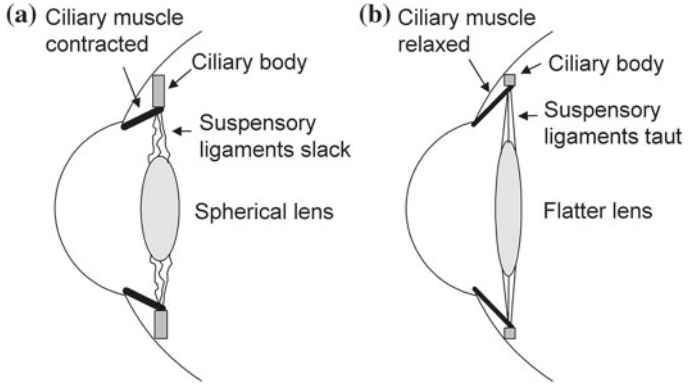


Fig. 11.22 Schematic of the physiological mechanism for accommodation by changing the crystalline lens focal length. **a** For imaging nearby objects, the ciliary muscles contract, which reduces the tension on the suspensory ligaments. This allows the lens to be rounder (its natural shape), which decreases the eye focal length and enables the eye to focus on nearer objects. **b** For imaging distant objects, the ciliary muscles relax, which increases the tension on the suspensory ligaments. This makes the lens flatter, which increases the eye focal length and enables the eye to focus on more distant objects. The ciliary muscle and suspensory ligaments encircle the eye, and are not as depicted here

Such tunability in the eye focal length should allow people to see objects clearly both far and near. The maximum distance from the eye where objects form sharp images is called the *far point* (FP)—and we would like our FP to be ∞. The nearest distance where objects are clear is the *near point* (NP)—and we would like our NP to be 25 cm (=10 in) for convenient reading and such.

How much accommodation do we need? First we need to find what the focal length of the eye f_{eye}—due to the combined focusing of the cornea and crystalline lens—needs to be to achieve far and near point vision (Fig. 11.22). In both cases the eye lensing system must image the object on the retina. In the Standard eye model, the effective eye lens acts as if it were in air a distance $L_{\text{eye}} = 17\,\text{mm} = 0.017\,\text{m}$ before the retina. At the far point, $\text{FP} = d_1 = \infty$ and $d_2 = L_{\text{eye}} = 0.017\,\text{m}$, so (11.3) shows that $f_{\text{eye}} = d_2 = 0.017\,\text{m}$ or $1/f_{\text{eye,FP}} = 1/0.017\,\text{m} = 58.8/\text{m} = 58.8\,\text{D}$. At the near point, $\text{NP} = d_1 = 25\,\text{cm} = 0.25\,\text{m}$ and $d_2 = L_{\text{eye}} = 0.017\,\text{m}$, so $f_{\text{eye}} = 15.9\,\text{mm} = 0.0159\,\text{m}$ or $1/f_{\text{eye,NP}} = 1/0.0159\,\text{m} = 62.8/\text{m} = 62.8\,\text{D}$. The necessary accommodation is

$$\frac{1}{f_{\text{eye,NP}}}\,(= 62.8\,\text{D}) - \frac{1}{f_{\text{eye,FP}}}\,(= 58.8\,\text{D}) = 4.0\,\text{D}. \tag{11.64}$$

Table 11.2 shows that people 45 years old and older tend to have less than this amount of accommodation, and therefore need to wear corrective lens to be able to see both near and far. If vision at the far point at infinity is clear (as it often is), then 55 year olds need to wear convex corrective lenses with (4.0 D−1.3 D =) 2.7 D correction to read a book held 25 cm from their eyes. This lack of appropriate accommodation is called *presbyopia* ("old eyes" or "old age vision"). Most of the loss of accommodation with age can be explained by the increase in the Young's moduli of the (asymmetric) crystalline lens, a flattening of the lens, and a decrease in the Young's modulus of the lens capsule with increasing age [11]. Typical amounts of accommodation cannot correct the loss of refractive power for seeing in water, even for very young people.

Table 11.2 Mean accommodation versus age

Age (years)	Mean accommodation (diopters, D)
8	13.8
25	9.9
35	7.3
40	5.8
45	3.6
50	1.9
55	1.3

Using data from [61]

11.3.4 Field of View and Binocular Vision

Our *field of view* of vision is quite amazingly large and our vision is quite good over this range, given the imaging distortions we will describe later. For each eye the field of vision is approximately an ellipse that is about 150° high and 210° wide. The binocular field of vision, seen by both eyes together, is approximately a circle that is 130° in diameter. It is difficult to achieve this large field of view in optical systems unless the imaging surface is curved, and not flat as in film, film plates, or charge-coupled device (CCD) arrays. This is related to the *curvature of field* aberration. A large field of view with minimal distortion is achieved in our eyes because of the spherical surface of the retina. Of course, we form sharp images only in the region we fixate on because that region images on the fovea. In humans, the ocular muscles can rotate the field of view by 90° in each eye.

In most people, the brain *fuses* the different retinal images from both eyes to give a single perceptual image. This *binocular* vision provides depth perception beyond that from the monocular vision clues from the sizes of objects, the occlusion of a farther object by a nearer one, perspective (such as parallel lines appearing to converge far away), and motion parallax (that nearby objects tend to move more than objects that are farther away as the viewer moves). Such fusion does not occur in people with *diplopia*, who instead perceive two separate images. In people with *strabismus* the motion of the two eyes is not coordinated well and they cannot fixate in a binocular manner on objects. (Some of their ocular muscles are surgically shortened to help trick the brain to maintain binocular vision.) From now on we will address only each individual eye.

11.3.5 Adjustments of Light Levels

So far we have focused on forming a sharp image on the retinal detector. For proper vision it is equally important that this image is neither too dim nor too bright—but exactly what does too dim or too bright mean? Conventional photographic emulsion film has a dynamic range over which it can detect signals. If the incident light is too weak, no image is observable, and if it is too strong, everything is bleached (all is white). It is similar in the charge-coupled detector (CCD) elements in digital cameras. The retinal rods and cones also have a dynamic range over which they can lead to high-quality perceived images. In a still camera, the light level incident on the detector needs to be adjusted by changing the aperture diameter, altering the length of exposure time, and possibly by using a flash. In a motion camera, it needs to be adjusted by changing the aperture diameter and possibly by using external lighting. Humans adjust the level of light needed for vision in several conscious ways (changing ambient light levels, by increasing them by turning on more room lights or by using flashlights and decreasing them outside by putting on sunglasses) and subconscious ways (adjusting apertures and the bleaching and recovering of detector elements).

Decreasing pupil size decreases light transmission to the retina. A change from the maximum to minimum pupil diameter, from roughly 8 to 2 mm, corresponds to a 16-fold change in area and a 16-fold decrease in the amount of light incident on the retina. In one typical example, the pupil diameter decreases from 7–4 to 3–2 mm as the luminance increases from 10^{-3}–10 to 10^{2}–10^{4} cd/m^2. (The units of luminance are described later in this chapter; see Fig. 11.52 and Table 11.4.) Even with the change in pupil size, the amount of light entering the eye and imaging on the retina is much larger at high light levels. In the cited range the luminance $\times$ the pupil area increases, from $(\pi/4)(7\,\text{mm})^2(10^{-3}\,\text{cd/m}^2) = 4 \times 10^{-8}\,\text{cd}$ to $(\pi/4)(2\,\text{mm})^2(10^{4}\,\text{cd/m}^2) = 3 \times 10^{-2}\,\text{cd}$; this is a range of a million. Either the dynamic range of light levels over which our photoreceptors are sensitive is extraordinarily large or the sensitivity of these retinal detector elements, the rods and cones, must also decrease with increasing light intensity.

Rods are primarily important in dark-adapted (scotopic) vision, and cones for light-adapted (photopic) vision. The most important factor in our ability to see under a very wide range of light intensities is the combination of the bleaching of rhodopsin and its subsequent recovery by the processes in Fig. 11.23, and changes inretinal networking; changes in pupil diameter have a much smaller effect. After bright light exposure, all-*trans* retinal is released from the photopigment and is converted to all-*trans* retinol, which is transported to an adjacent cell layer. The all-*trans* retinol is converted to 11-*cis* retinal, which returns to the photoreceptor and combines with opsin to reform rhodopsin. For bright normal lighting, much of the rhodopsin inrods is "bleached" and that in cones is relatively "unbleached." The recovery time for rodrhodopsin is about 20–30 min, and because vision is most sensitive when the supply of rhodopsin in rods is maximum, the adaptation of eyes to night vision takes up to 30 min (see Fig. 11.53 below).

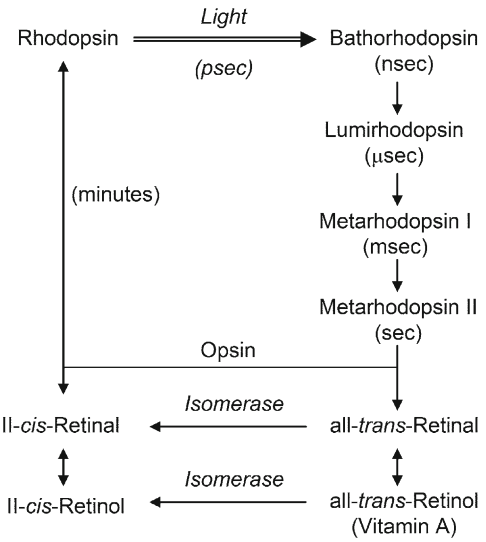


Fig. 11.23 After retinal exposure to light, rhodopsin in the rods reforms slowly by these chemical processes, with the time scales shown (Based on [17])

The variation of the fraction of available photopigment p with time is described by

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = -\frac{I(t)p(t)}{Q} + \frac{1-p(t)}{\tau}, \tag{11.65}$$

where $I(t)$ is the level of illumination on the retina, Q is the energy in a flash required to reduce the fraction of unbleached photopigment to 1/e, and τ is the regeneration time constant. There are separate relations for rods and cones, with different Q and τ; τ is approximately 2 min for cones and 6 min for rods. For steady-state illumination at level I_{ss} (with the left-hand side of (11.65) set equal to zero).

$$p(I_{\mathrm{ss}}) = \frac{I_0}{I_0 + I_{\mathrm{ss}}}, \tag{11.66}$$

where $I_0 = Q/\tau$. When the light is turned off at $t = 0$ and so $I(t) = 0$, p recovers from its initial value, say p_{i}, to its unbleached value $p_{\max}$ as

$$p(t) = p_{\mathrm{i}} + (p_{\max} - p_{\mathrm{i}})(1 - \exp(-t/\tau)). \tag{11.67}$$

In addition to this regeneration of rhodopsin during recovery that reverses bleaching, there are reversible changes in retinal networking that control the summation of retinal signals. This is discussed a bit further in Sect. 11.5.

We could express the current signal strength of the image going to the brain, S (at time t_{now}), as a function f of the light intensity I that is currently incident on the eye, the current pupil diameter d, and the intensity entering the eye in the previous 30 or so min: $S(t_{\mathrm{now}}) = f[I(t_{\mathrm{now}}), d(t_{\mathrm{now}}), I(t_{\mathrm{now}} - 30\ \mathrm{min} < t < t_{\mathrm{now}})]$. Clearly, this is a fairly complex function. It is used by the brain for conscious and subconscious feedback and control of vision, such as to control the pupil diameter $d(t_{\mathrm{now}})$ (as seen in Chap. 13). Detailed models of adaptation are presented in [26].

11.3.6 Limitations to Visual Acuity

There are several fundamental factors that limit *visual acuity* even in those with excellent vision. Under the best of circumstances the imaging process is limited by optical diffraction. There are additional limitations in the image formation process, called *aberrations* (just as in cameras, microscopes, and related imaging systems). Diffraction is the limiting factor for pupil diameters smaller than $\simeq$3 mm, whereas aberrations limit acuity for larger diameters. Moreover, the transfer of the image to the brain is limited in principle by the discrete and finite nature of the rod and cone photodetectors (*pixel size*) and by whether their outputs are interconnected to the same neurons.

Throughout this discussion we should keep in mind that the brain does amazing processing of this transferred image, and what we perceive is surprisingly much

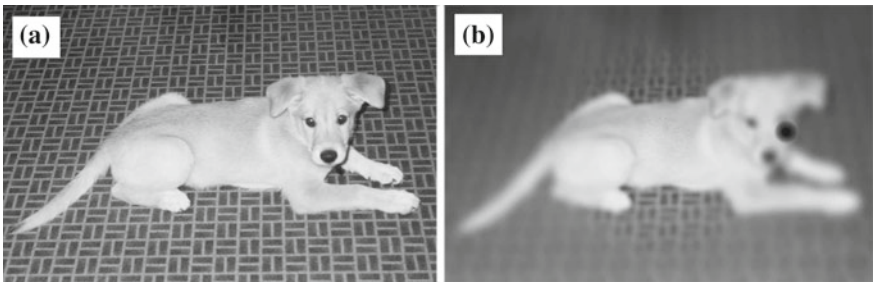


Fig. 11.24 **a** Photograph of Jenny as a puppy, which also represents the optical image on the retina (aside from inversion). **b** Simulation of this image from the *right eye* as sent to the brain, including the fuzziness away from the fovea and the lack of response at the blind spot. (This particular source is not very detailed in the *center*, which images at the fovea) (Inspired by the figure on page 48 in [41])

sharper than we would expect from optical imaging and the interconnection of signals from photodetector cells. The image in Fig. 11.24a is a photograph. It also represents the image of the same source on your retina, aside from image inversion. The image in Fig. 11.24b is a simulation of the retinal response of your right eye going to the brain, with this image on the retina. At and near the fovea (which images the center of the source) the response is sharp, but it becomes very fuzzy away from the fovea and there is no response at the blind spot. What we perceive is closer to the left image after inversion because our brains fill in details based on input from both eyes and information from rapid eye jiggling (microsaccades), with the jiggling itself removed during analysis [41].

Diffraction and Pixel Size

When water waves, say with a linear wave front and of wavelength λ, impinge on a slit with width D, the transmitted wave spreads out in angle; this is diffraction (Fig. 11.25). The smaller the slit, the wider the diffraction angle. At a screen placed a distance z downstream there is a sinusoidal-like pattern of waves, whose magnitude gets progressively smaller away from the center of the pattern. The width of the central region is $d = z(2\lambda/D)$, where $2\lambda/D$ is called the diffraction angle. We can run the experiment in reverse and some how take these water waves with a linear wave front and send them through a "water lens" that focuses them to an angle $2\lambda/D$. After traveling a distance z the wave will converge to a minimum width D given by this same equation, with $D = z(2\lambda/d) = 2(z/d)\lambda$. This distance z is then the focal length f of the lens, so $D = 2(f/d)\lambda$.

This wave picture also describes the propagation of light. A lens of diameter d and focal length f can focus a light beam of wavelength λ to a diameter no smaller than a diameter D, given by

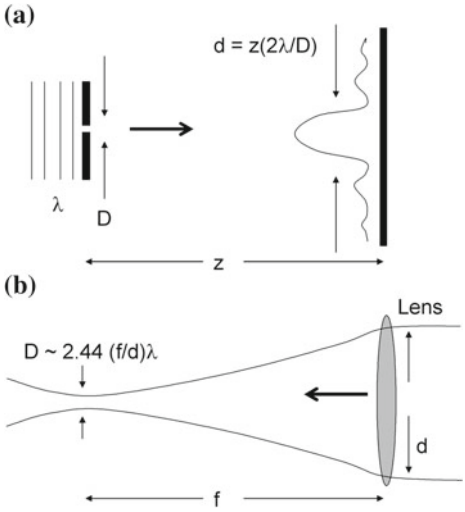


Fig. 11.25 Diffraction of water waves in **a** applied to light in **b**

$$D \simeq 2.44 \, \frac{f}{d} \, \lambda. \tag{11.68}$$

This is known as the diffraction-limited spot size, because the diffraction of light waves is the fundamental process that limits how small this focus can be even with a "perfect" lens. (d actually refers to the smaller of the lens diameter and the diameter of the aperture (pupil) that may be before or after it. The numerical factor in this equation is instead 1.22 when d is replaced by the appropriate radius.) The quantity f/d is also known as the *f-number* or $f\#$ of the lens or imaging system. Equation (11.68) can also be expressed as $D \simeq f(2.44\lambda/d)$, where $2.44\lambda/d$ is the (full) diffraction angle. This is closely related to the Rayleigh criterion for distinguishing between the diffraction-limited images of nearby points or angular regions; this is discussed further in Problem 11.33. Our earlier tracing of rays was within the *geometric optics* limit, which becomes more and more valid as λ decreases and approaches 0, which causes D to decrease toward 0.

For an eye crystalline lens of diameter of ≈8 mm and $f \approx 17$ mm (Standard eye model), this limit is ≈2.5 μm for 500 nm light. Because the crystalline lens is not perfect, this image diameter is measured to be larger, ∼11 μm. However, because this image is not uniform in brightness—it is brightest in the center and shades off nearer the edges—the resolution is a bit better than 11 μm. In the fovea the average diameter of cones is ≈1.5 μm, so images separated by ∼2 μm on the fovea can sometimes be barely resolved. We saw earlier that this corresponds to the normalvisual acuity in discriminating between point light sources separated by 25 s of arc (corresponding to barely distinguishing two spots that are 1.5–2.0 mm apart and 10 m away). (See the discussion of measuring visual acuity below.)

Because the fovea is ∼300 μm in diameter, this high acuity occurs only for 1° of the visual field, corresponding to a width of 20 cm a distance 10 m away. Just

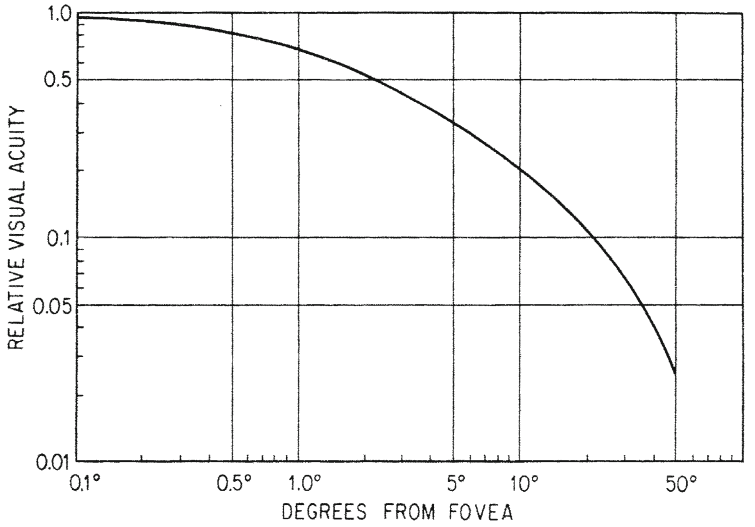


Fig. 11.26 The variation of visual acuity with retinal position relative to the fovea. (From [54]. Reprinted with permission of McGraw-Hill)

outside the fovea, the visual acuity of the retina is worse by a factor of 5–10, and becomes progressively even worse toward the periphery (Fig. 11.26). This occurs because many rods and cones are connected to the same optic nerve fiber outside the fovea [17].

Aberrations in Image Formation

The imaging elements in the eye are quite good, but they have imperfections. Two of the more important sources are spherical and chromatic aberration.

In the paraxial limit all rays from the same point image at the same place. Rays making larger angles to the optic axis, image at different places. Such *monochromatic aberrations* depend on the details of the imaging surface and are present even with one color of light. *Spherical aberration* (SA) is due to the inherently imperfect imaging by a spherical surface. Only paraxialrays near the optic axis image according to (11.9) at the nominalfocal length f. Horizontal light rays farther from the optic axis (a larger distance y—or half aperture height—away, corresponding to zonal and marginal rays) are refracted by a spherical convex lens more than is predicted by (11.8) and therefore cross the optic axis before the rays nearer the axis cross it (Fig. 11.27). With this definition, SA is positive for spherical convex lenses and negative for spherical concave lenses. The distance along the optic axis between the places where these zonal and marginal rays and the paraxial rays hit the optic axis is the *longitudinal* (or axial) spherical aberration L·SA. The height above or below the optic axis a ray hits a screen placed at this focal length is the *transverse* (or lateral) spherical aberration

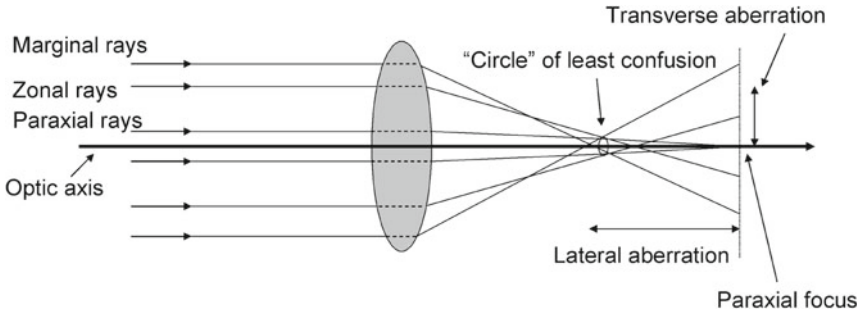


Fig. 11.27 Ray tracing and imaging with spherical aberration present

T·SA. Longitudinal and transverse spherical aberration become larger rapidly with semiaperture size y, as $\propto y^2$ and $\propto y^3$, respectively. The image is sharpest somewhere between f and $f - \text{L·SA}$, where the image is called the *circle of least confusion*.

Spherical aberration can be limited by using compound lenses (multiple lenses in which the net effect of the aberration is negated), lenses with aspherical surfaces, or simple lenses preceded by apertures that transmit light only near the optic axis. (However, smaller apertures also limit the amount of transmitted light.) Notably, spherical and other aberrations can be greatly affected by the lens shape. From (11.18) it is clear that you can make a lens with a certain focal length by using infinite sets of different pairs of R_{12} and R_{23}. Figure 11.28 shows that for parallel rays incident on a positive lens, spherical aberration is minimized for a convex anterior surface and nearly planar posterior surface (so the rays make nearly equal angles to both surfaces). The eye crystalline lens is biconvex ($R_{12} > 0$ and $R_{23} < 0$ as in Fig. 11.11), but the posterior surface is much flatter than the anterior surface, and this decreases the usual positive spherical aberration. Man-made lenses usually have a uniform index of refraction. Nature has decreased spherical aberration in the human crystalline lens even further by decreasing the index of refraction from its center in all three directions and decreasing the magnitude of the radius of curvature towards the margin of the lens. Both variations reduce the surface refracting power at the lens margins to decrease SA. In fact, they overcorrect for the SA to compensate for the undercorrected SA at the anterior surface of the cornea. Spherical aberration in the eye decreases with accommodation for more distant vision.

There are several other monochromatic aberrations. Off-axis object points image to a comet-shaped image because the peripheral rays focus closer to the optic axis than the central rays. This is called *coma*. Coma is also present for object points on the optic axis when the optical elements are not centered and are, in fact, tilted relative to each other, and this occurs in the eye. Points in the object plane that are far from the axis are farther from the lens than those near the optic axis, so they image at shorter distances after the lens. The surface of the image is not planar but curved (Petzval surface), even for rays passing through the center of the lens for

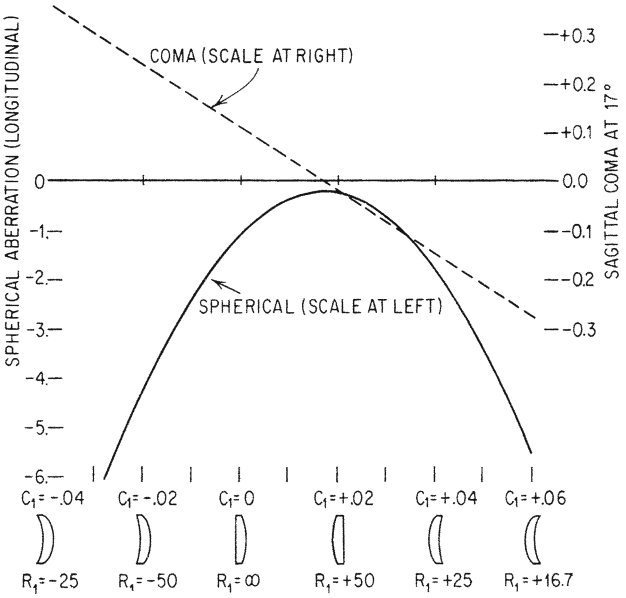


Fig. 11.28 Spherical aberration and coma as a function of lens shape for a 100-mm focal length lens (with a stop at the $f/10$ lens, covering a $\pm 17°$ field). The radius of curvature of the *left* interface in mm is labeled as R_1—which is given in the text as R_{12}, and its reciprocal is shown as $C_1 = 1/R_1$ (From [54]. Reprinted with permission of McGraw-Hill)

this *curvature of field* aberration. A square grid images to the distorted grid shapes with *barrel distortion* or *pincushion distortion*, generally for negative and positive spherical lenses, respectively, because the lateral magnification of the lens varies from its center to periphery.

Moreover, simple, single component lenses have different focal lengths at different wavelengths, because the index of refraction of the lens material differs at different wavelengths. This is known as *dispersion*. Therefore the image can be sharp at one wavelength but a bit blurred at another because of this *chromatic aberration* (CA) (Fig. 11.29). This is common in glass lenses because blue light has a larger refractive

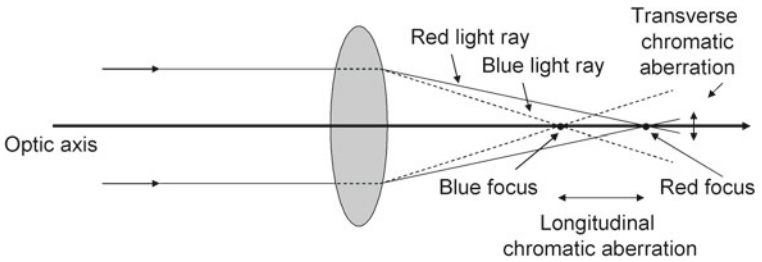


Fig. 11.29 Ray tracing and imaging with chromatic aberration present

Table 11.3 Refractive indices (n) in the ultraviolet (380 nm), visible, and near infrared (780 nm), and, in the last row, the dispersion for ocular media

Wavelength (nm) (spectral line)	Humors	Crystalline lens
380	1.3450	1.4334
480.0 (F′)	1.3378	1.4221
587.6 (d)	1.3333	1.4160
643.8 (C′)	1.3315	1.4138
780	1.3285	1.4101
Dispersion constant	52.9	50.1

Using data from [3]

index than red light in glass, and consequently glass lenses have a shorter focal length in the blue (Problem 11.38). The distance between where rays at two wavelengths (such as in the red and blue) hit the optic axis is the *longitudinal* (or axial) chromatic aberration L·CA [2]. The vertical distance between the images of the two colors on a screen placed at the focal length is the *transverse* (or lateral) chromatic aberration T·CA or lateral color. In camera and microscope lenses this is corrected by using compound lenses composed of lenses made from different types of glass—with different dispersion. The human aqueous and vitreous humors and the crystalline lens have small, yet significant, dispersion, as seen in Table 11.3. In the crystalline lens this chromatic aberration contributes ∼2 D change in optical power across the visible, as described by

$$D(\lambda) = p - \frac{q}{\lambda - c} \tag{11.69}$$

for the defocus in D, with wavelength λ (in μm), $p = 1.7312$, $q = 0.63346$, and $c = 0.21410$, relative to that at 578 nm [58]. The overall change in the refractive power of the schematic eye is plotted in Fig. 11.30. The dispersion constant, defined as $[n(\lambda = 587.6\,\text{nm}) - 1]/[n(\lambda = 480.0\,\text{nm}) - n(\lambda = 643.8\,\text{nm})]$, and related matters are discussed in Problems 11.38–11.41.

Quantitative Evaluation of Image Acuity

Because of optical diffraction and these other imaging imperfections, a point really images to a blur with diameter w_B. Optical diffraction contributes $\sim 2.44(f/d)\lambda$. Spherical aberration contributes $\sim 2T \cdot SA$ to w_B (suitably averaged from the axis to y above it) and chromatic aberration contributes $\sim 2T \cdot CA$ to it (suitably averaged over the wavelength spread). A more quantitative evaluation of image distortion is often needed than this poorly defined blur diameter. This is provided by measuring the image from a very thin line source to obtain the *line spread function* (Fig. 11.31) or a point source to obtain the *point spread function*. These functions provide the relative image intensity versus distance or visual angle in one or two dimensions,

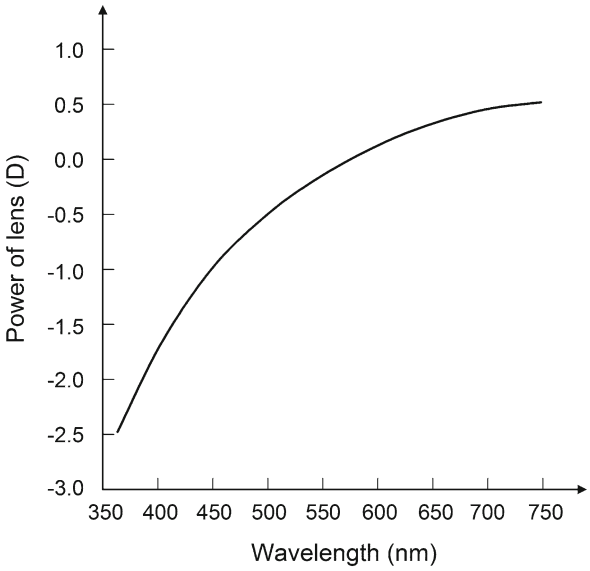


Fig. 11.30 Chromatic aberration of the human eye lens as quantified by the optical power that must be added to the human eye lens to bring the eye in the same focus as 587 nm, depicted as the *curve* given by (11.69) (Based on [63])

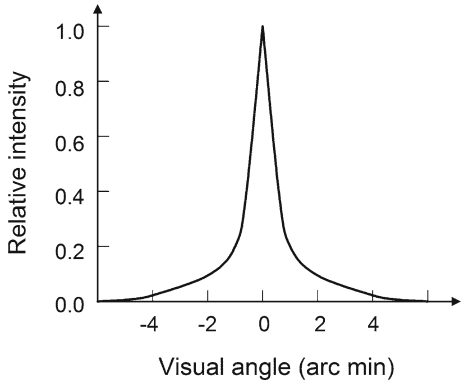


Fig. 11.31 The human line spread function for an eye with 3.0 mm pupil diameter, using a model in [64] (see Problem 11.36) that is based on experiment. (Based on [63, 64])

respectively, and have the same full width $\sim w_B$. Alternatively, we can image a source with an intensity that periodically varies laterally (in the y direction) with spatial frequency k, such as by $(1 + \sin ky)/2$ or the square wave variation shown in Fig. 11.32b, and see how much of the modulation remains in the image. For $k \ll 1/w_B$ there is much modulation (11.32d), while for $k \gg 1/w_B$ there is very little modulation (Fig. 11.32e). Figures 11.31 and 11.32 track these changes as a function of position. They could also be tracked in terms of the spatial frequency k. This modulation variation versus k is called the *optical transfer function* or, for the symmetric systems we will assume here, the *modulation transfer function*; they have widths $\sim 1/w_B$. Figure 11.33 shows this function for the eye with the line spread function of Fig. 11.31. This inverse relationship between the spread in real space

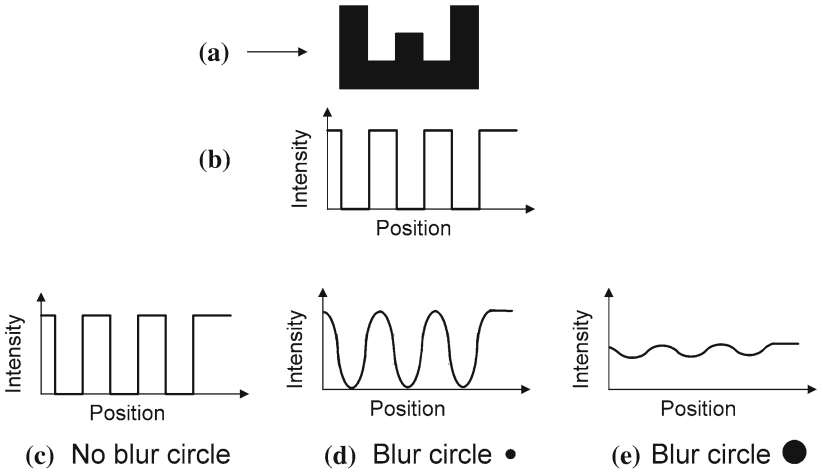


Fig. 11.32 **a** The horizontal letter E on it side, and **b** the light level in the direction of the *arrow* in **a** showing a periodic square wave object. This leads to potential images on the retina for **c** ideal imaging, **d** imaging with a small blur spot, and **e** imaging with a large blur spot

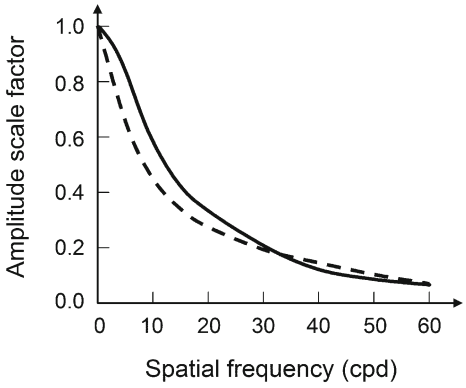


Fig. 11.33 Modulation transfer function of the human eye, as measured in [67] (*dashed curve* drawn through data points (which are not shown), as drawn in [63]) and compared to the predictions using the human line spread model shown in Fig. 11.31 (*solid curve*). The abscissa axis is in units of cycles per degree (cpd) (Based on [63])

(spread functions) and in spatial frequency space (transfer functions) is similar to that in the Chap. 10 analysis of the voice intensity in time versus frequency. In both cases the inverse functions (time vs. frequency there and space vs. spatial frequency here) are related by Fourier analysis as in (10.42) and (10.43) (where here t would be replaced by x, $2\pi f$ by k, and the summation by an integral).

Effects of Changing Pupil Size, Including Depth of Field

The human pupil can change from $\simeq$8 to $\simeq$2 mm in diameter. Decreasing the pupil size affects vision in several ways.

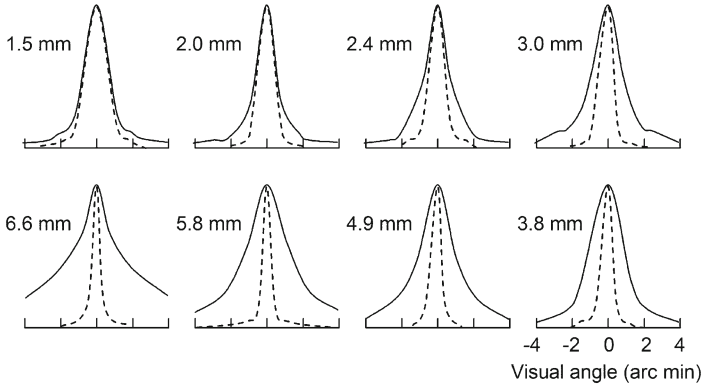


Fig. 11.34 Line spread function for the human eye for different pupil diameters. The *solid curves* are the total functions. The *dashed curves* are the line spread functions for those apertures assuming diffraction-limited conditions. (Based on [7, 63])

1. Decreasing pupil size decreases light transmission to the retina, as described above.
2. The size of the diffraction-limited spot varies inversely with aperture diameter (11.68). With the pupil closed all the way to a 2 mm diameter, this diffraction-limited spot diameter is $\approx 10\,\mu$m. Figure 11.34 shows that the contribution to the line spread function by diffraction increases with decreasing pupil size as expected and that it becomes the dominant contribution to the blur at small pupil sizes.
3. Decreasing pupil size lessens the effect of lens aberrations. Transverse spherical aberration varies as the cube of the distance of rays from the optic axis. Figure 11.34 shows that contributions to the line spread function other than diffraction, i.e., due to these and related aberrations, increase remarkably with pupil diameter and become the dominant contribution to the blur at large pupil sizes.
4. The range in object (or source) distances over which good images are formed is called the *depth of field*. This depth increases by decreasing the size of the aperture (pupil) before the crystalline lens. "Conjugate" to this is the range of image distances over which the images over the depth of field form, and this is called the *depth of focus*. Figure 11.35 shows that the blur on the retina due to the different image locations with different source distances decreases with pupil diameter—and this is one manifestation of the depth of field. Closely related to this is the change in image distance with source distance, as plotted in Fig. 11.36; the depth of focus is related to the slope of the curves, so it is large for distant sources and small for nearby objects (and this is not ideal). This can also be easily understood using simple image formation using (11.3).

The pupil loses its ability to dilate with age. The maximum pupil diameter decreases from, $\sim$10 mm at 1 year, to 6 mm at 20 year, to 3 mm at 50 year, and to 1 mm at 80 year.

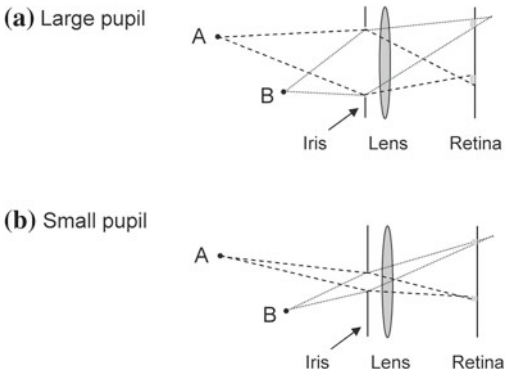


Fig. 11.35 Schematic showing the image blur for two object distances, that alternately image (*A*) before or (*B*) after the retina, for a relatively **a** large pupil and **b** small pupil. This provides an assessment of the depth of field versus pupil diameter because the blur increases with pupil diameter

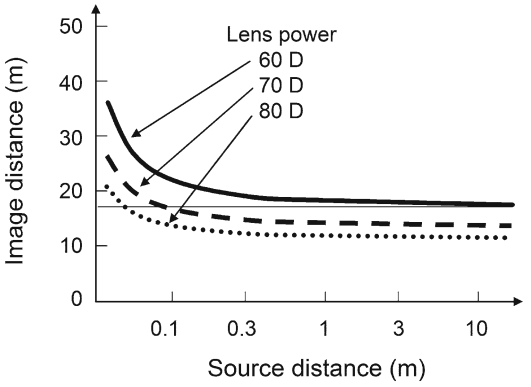


Fig. 11.36 Image distance versus source distance for different lenses, which provides information on the depths of field and focus of the human eye. An extra *horizontal line* is shown for a distance of $\simeq$17 mm, which is the focal length of the Standard eye in air (Based on [63])

11.3.7 Imperfect Human Vision

Visual acuity (VA) in human vision is frequently tested by reading black letters (that have 4% reflectance) on a white background (84% reflectance) from a Snellen chart placed 20 ft (or 6 m) away under good light illumination (480 lux is a standard, and some say 800–1,000 lux) (Fig. 11.37). (These units will be explained later in this chapter). For each eye, the vision is said to be 20/z if the smallest letters you can read at 20 ft are what people with "perfect" 20/20 vision (or 6/6 vision in the metric system) in that eye can read z feet away. This is a test designed to assess near-sightedness (or more precisely, lack of far-sightedness); the optical object infinity is considered to be 20 ft. Visual acuity is defined as the reciprocal of the angular size (in minutes of arc at 20 ft) of one of the elements of the letters on the chart. The letter E has three horizontal solid bars and two horizontal open bars (Figs. 11.32a and 11.37). In the 20/20 line, each of these is an element that subtends 1 min of arc and the whole letter subtends 5 min of arc at the eye at a distance of 20 ft. Each bar in the E is about 1.7 mm wide and the E is 8.8 mm high in the 20/20 line. Someone with 20/40 vision would be able to resolve elements of no less than 2 min of arc and letters of 10 min

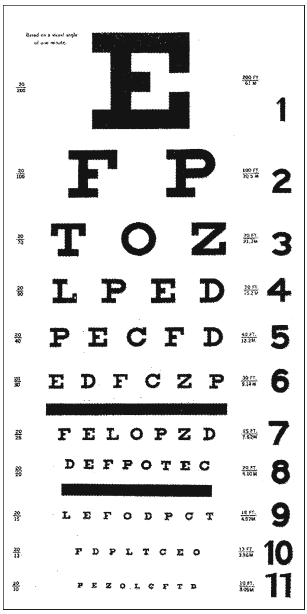


Fig. 11.37 The Snellen eye chart used to test visual acuity. The label of the line with the smallest letters you can read at 20 ft is your vision for that eye, with lines 1–11, respectively, corresponding to 20/200, 20/100, 20/70, 20/50, 20/40, 20/30. 20/25, 20/20, 20/15, 20/13, and 20/10 (From [43])

of arc from the eye. Clearly, this is closely related to the ability to view square wave patterns, as in Fig. 11.32. Normal VA is 1.0 (for 20/20 vision) under normal indoor lighting conditions and decreases with poorer lighting (Fig. 11.38). Someone with 20/40 vision has a VA of 0.5. Under ideal conditions, some people have a VA of 2 or 3. Figure 11.26 illustrates that visual acuity is worse in the periphery than in the fovea. (Can you use the results from the test in this figure to estimate the dimensions of the fovea?) The analogous test for far-sightedness (i.e., lack of near-sightedness) is the Jaeger test, in which one reads lines of various font sizes at normal reading distance.

Emmetropes have perfect imaging (VA $\simeq 1$), meaning their eyes need a correction less than a stated small amount, such as 0.5 or 1.0 D. They have a FP at infinity and a NP at 25 cm, with at least 4 D of accommodation (Fig. 11.39a). All other people are *ametropes*, and need corrective lenses to see clearly.

People with *myopia* are "near-sighted," meaning that their vision 25 cm away (their NP) is fine, but it needs correction for objects near infinity (VA < 1); the smallest Snellen eye chart line they can read is above the 20/20 line. They have good vision for objects from 25 cm to their far point, which is $<\infty$. Distant objects ($d_1 = \infty$) form images in the eyeball "before" the retina (Fig. 11.39) and negative lenses are needed to project the image on the retina. If the person has 4 D or more of accommodation, the image at 25 cm will also be fine with these lenses. Myopia occurs because the eyeball is too long (axial myopia, which is the most common reason) or the cornea is too curved and there is too much refractive power (refractive myopia).

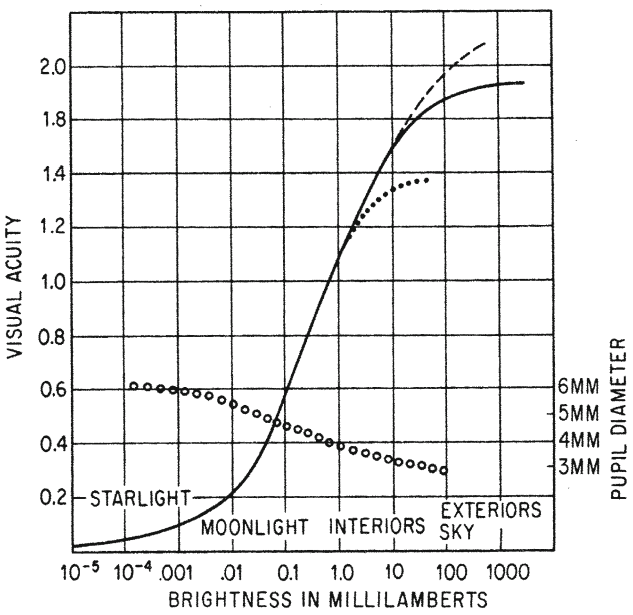


Fig. 11.38 Visual acuity (in 1/min) and pupil diameter (*circles*) as a function of object brightness. The *dashed* and *dotted* acuity lines are, respectively, for increased and decreased surround brightness of 1 milliLambert. (See Table 11.4: 1 milliLambert = 3.183 nit = 3.183 cd/m^2). (From [54]. Reprinted with permission of McGraw-Hill)

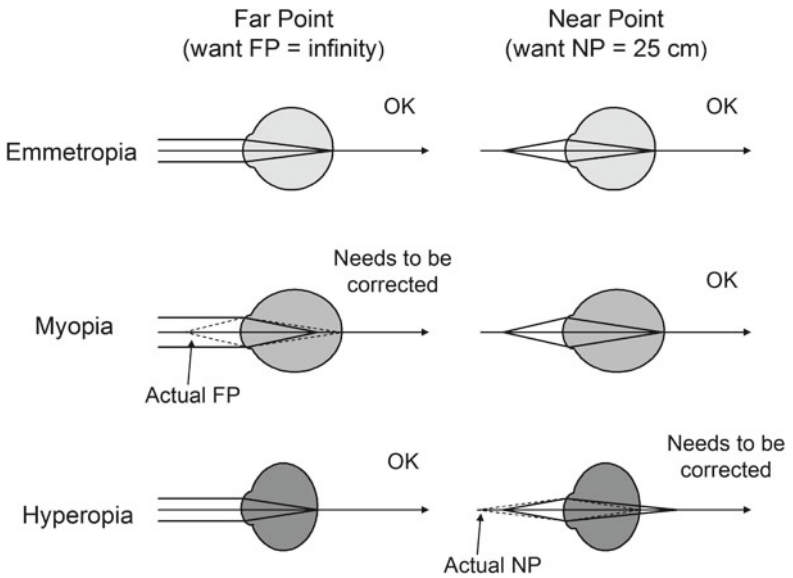


Fig. 11.39 Imaging in emmetropic, myopic, and hyperopic eyes. For the latter two both possible reasons for the conditions are illustrated: a long eyeball and short focal length for myopia and a short eyeball and long focal length for hyperopia

"Frequent" and "excessive" eye accommodation when we focus on nearby objects indoors or when we read screens (computers and smart phones screens and televisions), books and documents seems to contribute to myopia by somehow making the eyeballs longer. This environmental factor is clear because of the higher rates of myopia among those engaging in such activities, such as university students, compared to those who do not, such as farmers and seaman, both across a population in a generation and across generations [31, 37, 38, 49, 60, 70]. It can be congenital-onset (starting in children aged 0–6 years, with prevalence ∼2%), youth-onset (6–20 years, ∼18%) and adult-onset (20 years and older, ∼10%), and can progress during adulthood. The exact mechanism for the eyeball elongation is not understood well [31, 68]. One potential contributing factor for myopia is the mechanical stress (Chap. 4) on the posterior sclera due to the muscles that control the eye motion—particularly the oblique muscles—and high vitreous pressure [14]. With possibly lessened tension in the sclera (perhaps due to weakening by the muscles) or higher pressure, the local radius of curvature of the posterior eyeball decreases in equilibrium according to the Law of Laplace (7.9), and this bulging leads to myopia.

People with *hyperopia* are "far-sighted," meaning that their vision at infinity (their FP) is fine, but it needs correction with objects that are near. They have good vision for objects from their near point, which is >25 cm, to their far point at ∞. Near objects ($d_1 = 25$ cm) form images "beyond" the retina (Fig. 11.39) and positive lenses are needed to project the image on the retina. If the person has 4 D or more of accommodation, the image at ∞ will also be fine with these lenses. Hyperopia occurs because the eyeball is too short (axial hyperopia, which is the most common reason) or the cornea is too flat and there is too little refractive power (refractive hyperopia).

With the optic axis (horizontal line in each part of Fig. 11.39) called the z-axis, the lateral extents of the object, lens, and image are described by the x and y axes. In everything we have said so far the x and y axes are perfectly interchangeable, and they still would be interchangeable after being rotated by any arbitrary angle about the z-axis. This is no longer true for people with *astigmatism*. Astigmatism is caused by ocular optics that lack rotational symmetry—often because their corneas have unequal curvatures in the x and y axes (a *toric* cornea; a toric lens is also possible) or because an optical element is tilted or displaced from the optic (symmetry) axis. There is a plane (or *meridian*) containing the optic axis in which there is a maximum optical power and a plane normal to that with minimum optical power, for *regular astigmatism*. There is a tendency, but not certainty, for these meridians to be rotated near the transverse and sagittal planes. The power of the vertical (sagittal) meridian is greater for *with-the-rule astigmatism* and that along the transverse meridian is greater for *against-the-rule astigmatism*. In *oblique astigmatism* the highest power meridian is neither vertical nor horizontal. Rarer is *irregular astigmatism*, for which the principal meridians are not normal to each other; this condition is caused by trauma, surgery, or disease. Only regular astigmatism can be corrected by spherocylindrical lenses (see below).

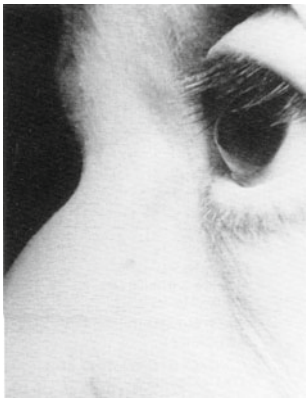


Fig. 11.40 Eye of one with severe keratoconus (pre-1973), showing a very conical cornea that protrudes out much (in a pointed, nonspherical manner) and that exhibits some drooping. The conical anterior surface of the cornea forms a very poor image. Nowadays, patients usually have a corneal transplant before keratoconus reaches this stage (From [20]. Used with the permission of Jack Hartstein, M. D.; also see [57])

Keratoconus is a condition in which the cornea thins near the center to form a cone and can be locally wavy because of corneal scarring (Fig. 11.40).

Visual efficiency (VE) is sometimes calculated for legal and industrial purposes by using the results of visual acuity, visual field, and ocular motility (diplopia and binocular fields) measurements for corrected eyes. Each of these three measurements is described by a factor ranging from 0 to 1, with 1 representing no loss of vision. They are multiplied together to give VE for an eye. VE for two eyes is a weighted sum of that for each eye, weighting the better eye (higher VE) by 3 and the poorer eye by 1. The visual loss is 1 − VE. The visual acuity factor is an average from the Snellen (far) and Jaeger (near) vision tests. For example, the factors from the Snellen test are 1.0 for eyesight corrected to 20/20 vision, 0.75 (for 25% loss) for 20/50, 0.50 for 20/100, and 0.10 for 20/400. (This scale is different from that used for the VA measurement of the uncorrected eye.) The visual field factor is the fraction of remaining visual field. The ocular motility factor is the fraction of remaining coordinated ocular movement (which is the ability to see one object as one and not two).

Absence of the photopigment in the three cone cells causes various types of color blindness, which affects about 10% of men and 1% of women. *Dichromats* are missing one cone photopigment. *Protanopes*, *deuteranopes*, and (the rarer) *tritanopes* do not have the long-, medium-, and short-wavelength photopigments, respectively. *Monochromats* lack two or all three of these cone photopigments and have no color vision. Color blindness is tested using pseudoisochromatic plates (for example, see [51]).

11.3.8 Correction of Vision by Eyeglasses, Contact Lenses, and Other Means

We will now analyze how eyeglasses, soft contact lenses, and hard/gas-permeable contact lenses can be used to correct vision [20, 32, 33, 35, 40, 57]. The aim is to achieve perfect vision, using a compound lens consisting of the eye (Standard eye model) and the corrective lens, with the far point at FP = ∞ ($f \sim 17.0$ mm) and the near point at NP = 25 cm ($f \sim 15.9$ mm).

In eyeglasses (or spectacles), the distance between the back (ocular) surface of the lens and the cornea is 12–15 mm; if they were closer the lenses would be brushed by eyelashes during blinking. The topography of the lens surfaces is chosen so the eye encounters the same prescription for any direction the eye gazes through the spectacles. The front surface is usually spherical and the back surface toric. Glass lenses for eyeglasses are usually made of ophthalmic crown glass, which has a refractive index 1.523. Higher index material is sometimes used to reduce the thickness and weight of high power eyeglasses, such as with high index glass with indices >1.70. Because impact resistant eyeglasses require thicker glass, lighter plastic lenses are commonly used for safety glasses and more generally for comfort. CR39 plastic, with a refractive index of 1.498, is often used; polycarbonate, with a high index 1.586, is highly impact resistant, although it is soft and has large chromatic aberration.

Soft contact lenses adjust to the shape of the cornea, while hard/gas permeable contacts do not. These soft lenses are composed of hydrogel, a gel-like material that is soft and flexible when it is hydrated, and this material often includes 2-hydroxyethyl methacrylate (HEMA). The inner surface of a soft contact lens molds to the outer surface of the cornea, so the corneal topography and the controllable lens thickness (from the lens center) determine the radius of curvature of the outer lens surface, and consequently the refractive power of the imaging correction. The eye tolerates a soft lens, $\simeq$13.0–14.5 mm in diameter, which is larger than the cornea and which extends to the sclera; this larger size optimizes lens centering and stability. The center thickness of soft contact lenses start from $\simeq$0.03 mm. Most of the oxygen needed by the cornea is supplied by oxygen diffusion through these very thin lenses. Because of this large size, there is very little motion of oxygen-rich tears to the cornea during blinking, and this motion supplies only $\sim$4–6% of the total oxygen to the cornea.

Rigid or hard lenses are much less pleasant to wear, but are necessary for irregular corneas—as for those with keratoconus (Fig. 11.40) or high levels of astigmatism. Rigid or hard lenses have traditionally been composed of poly(methyl methacrylate) (PMMA). Unfortunately, the oxygen needed by the surface of the cornea cannot penetrate the lens and can be supplied only by tears that slip to the cornea as the lens moves on the cornea during blinking. Because this level of oxygen is lower than needed, the corneas become irritated and such rigid lenses cannot be worn for long continuous stretches of time.

Gas permeable contacts are rigid lenses composed of materials such as silicone acrylate—a combination of silicone and PMMA—or fluorocarbonate silcone

acrylate, that let oxygen diffuse to the cornea. This increases the amount of oxygen reaching the cornea to about 10× that by tears alone. Because such lenses can be worn for most of the waking hours, ∼10–12 h, without excessive discomfort, gas permeable contact lenses have replaced PMMA hard contacts in most cases; they should not be worn for longer times. These gas permeable lenses are 8–10 mm in diameter and only 0.15 mm thick (with thickness starting at 0.10 mm), and (should) remain over the cornea, over most of the iris. The radius of curvature of the posterior surfaces of a rigid contact lens is chosen to best fit the anterior surface of the cornea. The anterior surface is chosen to produce the refractive power needed to obtain a perfect image on the retina. (If the topography of the posterior surface of the hard contact lens is different than that of the cornea anterior surface, in some cases it will shape this surface and change the refractive power in this way too. However, when there is a poor fit—especially when fitting to the nonspherical corneas in keratoconus as seen in Fig. 11.40, dust particles can be swept by tears to the region between the contact lens and cornea and this is very painful.) Figure 11.41 shows a three-point fit, which is ideal for keratoconus and similar eye disorders.

In *keratometry* (which means "cornea measurement") the curvature of the anterior surface of the cornea is determined by looking at the reflection giving Purkinje image I (Fig. 11.42) (Problem 11.73). This curvature is used to determine the posterior surface (base curve) of the contact lens, and consequently to "fit" it. It is often converted to a *K reading or number* (in D); this is the refractive power of a curved interface with this measured radius, between air and a material with the refractive index of tears (and not the refractive index of the cornea) (11.10).

Why don't contact lenses fall off your eye? When you look down with open eyes, your contact lenses do not fall off because the force due to the surface tension between the contact lens and tears exceeds the downward force of gravity on the lens (Problem 11.76). *Why do contact lenses remain centered on the pupil when your head is upright and do not slide off to the sclera (the white portion)?* This is particularly important for small lenses that cover most of the iris but do not extend to the sclera. Figure 11.43 shows some of the relevant forces. Gravity pushes the lenses down. Blinking of the eye lid pushes the lens downward and then upward in the blinking cycle. Viscosity of the tear volume between the cornea and contact lens resists downward motion. Slipping of the lens to a region of different curvature (i.e., the sclera, for the smaller gas-permeable lenses) is resisted by friction and changes in surface tension. Surface tension depends on tear viscosity, the topography of the cornea, and how well it is matched by the contact lens fit.

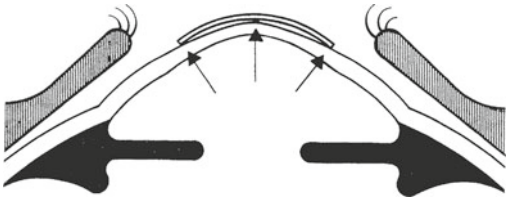


Fig. 11.41 Three point fit of contact lens—apical (apex) touch to the cone plus peripheral touch—for an eye with keratoconus. (Reprinted from [57]. With permission of Elsevier)

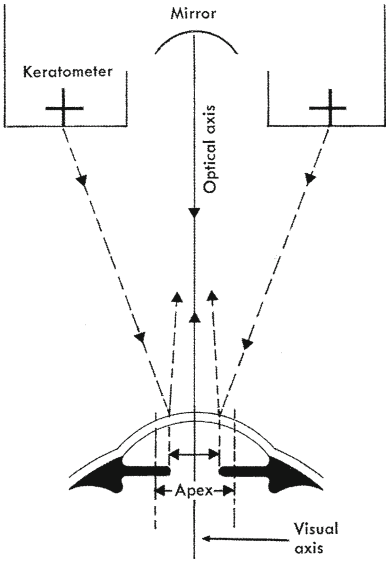


Fig. 11.42 Optical paths in a keratometer (Reprinted from [57]. Used with permission of Elsevier)

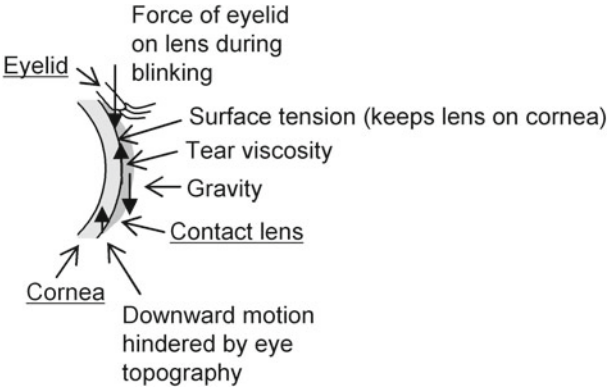


Fig. 11.43 Forces affecting contact lens position (Based on [40])

Permanent changes in the optical power of the eye can be made by changing the curvature of the anterior surface of the cornea by (1) making four or eights radial slit incisions in it with a scalpel to flatten it (which is called *radial keratotomy* or *RK*), which will decrease its optical power and can help correct myopia only or (2) sculpting its surface with an excimer laser by either (a) *photorefractive keratectomy* (*PRK*) or (b) *laser assisted in situ keratomileusis* (*LASIK*), which can either decrease (for myopia) or increase (for hyperopia) its curvature. In both PRK and LASIK, the middle layer of the cornea, the stroma, is exposed before it is sculpted by the laser

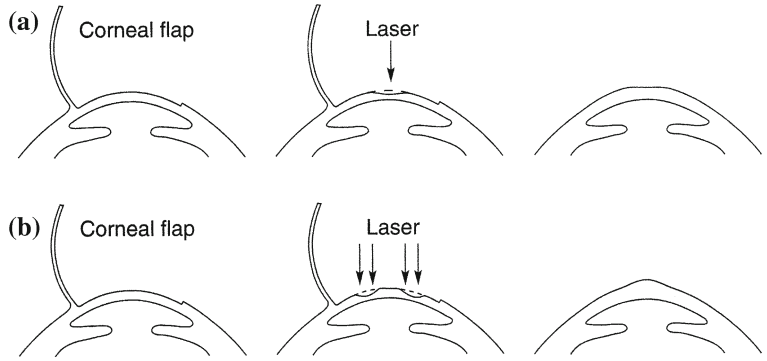


Fig. 11.44 In LASIK surgery the central cornea is made **a** more flat to correct for myopia and **b** more curved to correct for hyperopia. (From [52]. Reprinted with permission of McGraw-Hill)

via vaporization. In PRK, the epithelium (the top layer of the cornea) is scraped away by laser vaporization before the laser sculpts the stroma, while in LASIK a flap of the top 20% of the cornea is cut using a knife called a microkeratome and this flap is then folded to one side. The exposed cornea is sculpted with the laser to change the radius of curvature of the anterior surface of the cornea and the flap is flapped back to its original position after surgery (Fig. 11.44). Such cornea surgery cannot correct for the decreasing accommodation with age, so some sort of corrective lenses are eventually necessary.

Temporary changes in the optical power of the eye are routinely made using corrective lenses: eyeglasses and contact lens. We first describe a scientific way to correct myopia and hyperopia by correcting the far and near points respectively, and then the practical way it is really done.

Myopia

A person with myopia has an adequate near point (at 25 cm), but a far point that is too near. Say the FP = 53 cm from the eye, when the eye has accommodated as well as it can for infinite distances. The eye can image an object that is 53 cm away, but needs a corrective lens so with this accommodation it can see a farther distance, ∞ away (Fig. 11.45). The corrective lens is D to the left of the eye, with $D \simeq 1.2\,\text{cm}$ for eyeglass lenses and $D \simeq 0\,\text{cm}$ for contact lenses. We want this lens to take an object ∞ to the left of it and image it a distance $D - 53\,\text{cm}$ to the right of it (which is $53\,\text{cm} - D$ (>0) to the left of it). Using this backward propagation approach and (11.25), with f_1 replaced by $f_{\text{corrective}}$, this condition is

$$\frac{1}{\infty} + \frac{1}{D - \text{FP}} = \frac{1}{f_{\text{corrective}}} = P_{\text{corrective}}. \tag{11.70}$$

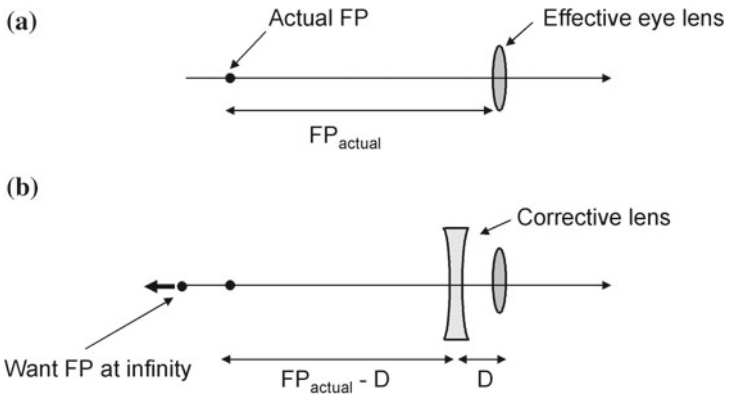


Fig. 11.45 **a** The problem in myopia and **b** the goal of choosing corrective lenses to correct it. The effective eye lens is that from the Standard eye model

So we find

$$f_{\text{corrective}} = D - \text{FP} \tag{11.71}$$

and $f_{\text{corrective}} = D - 53\,\text{cm}$ here. For eyeglasses, we see that $f_{\text{corrective}} = -51.8\,\text{cm} = -0.518\,\text{m}$ and the corrective power is $P_{\text{corrective}} = 1/f_{\text{corrective}} = -1.93\,\text{D}$. For contact lenses, we find that $f_{\text{corrective}} = -53\,\text{cm} = -0.53\,\text{m}$ and $P_{\text{corrective}} = 1/f_{\text{corrective}} = -1.89\,\text{D}$. These are negative lenses, with different prescriptions for the eyeglass and contact lenses.

Hyperopia

A person with hyperopia has an adequate far point (at ∞), but a near point that is too far away. Say NP = 190 cm from the eye, when the eye has accommodated as well as it can for near distances. The eye can image at $x = 190\,\text{cm}$ away, but needs a corrective lens so with this accommodation it can see a nearer distance, 25 cm away (Fig. 11.46). We want the corrective lens to take an object $25\,\text{cm} - D$ to the left of it and image it a distance $D - 190\,\text{cm}$ to the right of it (which is $190\,\text{cm} - D$ (>0) to the left of it). Using (11.3), this condition is

$$\frac{1}{25\,\text{cm} - D} + \frac{1}{D - \text{NP}} = \frac{1}{f_{\text{corrective}}}. \tag{11.72}$$

For eyeglasses, we see that $f_{\text{corrective}} = +27.2\,\text{cm} = 0.272\,\text{m}$ and $1/f_{\text{corrective}} = +3.67\,\text{D}$. For contact lenses, we find that $f_{\text{corrective}} = +28.8\,\text{cm} = 0.288\,\text{m}$ and $1/f_{\text{corrective}} = +3.47\,\text{D}$. These are positive lenses, with quite different prescriptions for the eyeglass and contact lenses.

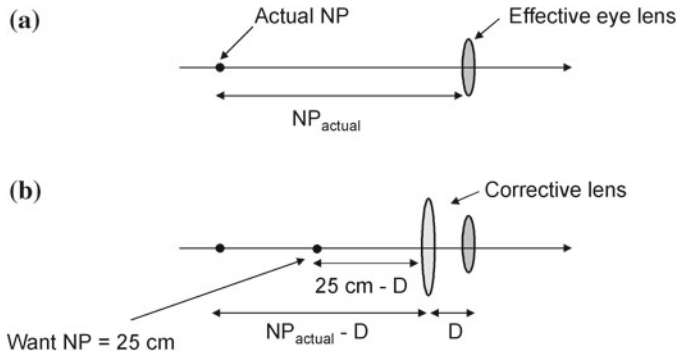


Fig. 11.46 **a** The problem in hyperopia and **b** the goal of choosing corrective lenses to correct it. The effective eye lens is that from the Standard eye model

Corrective Lens Prescriptions and Differences in Prescriptions for Contact Lenses and Eyeglasses

In practice, corrective lenses are prescribed by optimizing vision with different lenses placed in front of the eye, near where eyeglass lenses normally sit, and so the correction obtained is that for eyeglasses. As noted, correcting the same eye by contact lenses and eyeglasses requires different prescriptions because the eyeglasses are usually worn about 1.2–1.5 cm anterior to the contact lenses. Using the effectivity formula, (11.20), with $D = 1.2\,\text{cm}$ and $n = 1$, we see that

$$P_{\text{eyeglasses}} = \frac{P_{\text{contact lens}}}{1 + (1.2\,\text{cm})\,P_{\text{contact lens}}}. \tag{11.73}$$

This equation can be inverted (as you should show) to give

$$P_{\text{contact lens}} = \frac{P_{\text{eyeglasses}}}{1 - (1.2\,\text{cm})\,P_{\text{eyeglasses}}}. \tag{11.74}$$

where the eyeglasses are still 1.2 cm in front of the corneas.

An eyeglass prescription of $-5.00\,\text{D}$ for a myopic eye becomes the weaker (in magnitude) prescription of $-5.00\,\text{D}/[1 - (0.012\,\text{m})(-5.00\,\text{D})] = -4.72\,\text{D}$ for contact lenses. In contrast, a contact lens prescription of $-5.00\,\text{D}$ for a myopic eye becomes the stronger prescription of $-5.00\,\text{D}/[1 + (0.012\,\text{m})(-5.00\,\text{D})] = -5.32\,\text{D}$ for eyeglasses.

An eyeglass prescription of $+5.00\,\text{D}$ for a hyperopic eye becomes the stronger prescription of $+5.00\,\text{D}/[1 - (0.012\,\text{m})(+5.00\,\text{D})] = +5.32\,\text{D}$ for contact lenses. In contrast, a contact lens prescription of $+5.00\,\text{D}$ for a hyperopic eye becomes the weaker prescription of $+5.00\,\text{D}/[1 + (0.012\,\text{m})(+5.00\,\text{D})] = +4.72\,\text{D}$ for eyeglasses.

Note that in each case the contact lens prescription is higher in an algebraic sense than that for the corresponding eyeglasses. We also saw this when we calculated the prescription for correcting myopia by using (11.71) but not when we corrected for hyperopia by using (11.72). There is really no inconsistency. The result for correcting the near point for myopia is actually the effectivity formula as derived for parallel beams entering a lens. (Show this.) The result for correcting the far point for hyperopia is very different, and does not correspond to parallel beams imaging to the same point for eyeglasses and contact lenses. Which is correct? As a practical matter, corrections are measured for eyeglasses and are modified for contact lenses using the effectivity relation, so both prescriptions allow you to image objects that are far away.

Examples of eyeglass and contact lens prescriptions are given below.

Analysis Using Vergences and Imaging Effectivity

Let us examine corrections for the Reduced eye in Table 11.1 by using the notation of vergences [52]. (We will ignore the 1.67 mm distance between the cornea and the refractive surface in this model.) From (11.10), the refractive power is $P_{12} = (1.333 - 1.00)/0.00555\,\text{m} = 60.00\,\text{D}$. For an object at infinity ($d_1 = \infty$), the object vergence $U = -n_1/d_1 = 0$, so the image vergence, from (11.14), is $V = U + P_{12} = 0\,\text{D} + 60.00\,\text{D} = 60.00\,\text{D}$. Because $V = n_2/d_2$ and $n_2 = 1.333$, the image will be at $d_2 = n_2/V = 1.333/60.00\,\text{D} = 22.22\,\text{mm}$ (using 1/1 D = 1 m). For an emmetrope, the retina is at 22.22 mm.

For an eye with myopia with the same refractive power but with the retina 23.22 mm away, an object at infinity will still image 22.22 mm after the interface, or 1.00 mm anterior to the retina. Now we want $U = n_2/d_2 = 1.333/23.22\,\text{mm} = 57.41\,\text{D}$, so the image will fall on the retina and vision will be corrected if you place a contact lens with refractive power $57.41\,\text{D} - 60.00\,\text{D} = -2.59\,\text{D}$ at the interface. Using the effectivity relation (11.20) with $n = 1$, the correction with eyeglasses 1.2 cm before the contact lenses is $-2.59\,\text{D}/(1 + (0.012\,\text{m})(-2.59\,\text{D})) = -2.67\,\text{D}$.

For an eye with hyperopia with the same refractive power but with the retina 21.22 mm away, an object at infinity will again image 22.22 mm after the interface, which is now 1.00 mm posterior to the retina. Now $U = n_2/d_2 = 1.333/21.22\,\text{mm} = 62.82\,\text{D}$, so the image will fall on the retina and vision will be corrected if you place a contact lens with refractive power $62.82\,\text{D} - 60.00\,\text{D} = 2.82\,\text{D}$ at the interface. Again using the effectivity relation, the correction with eyeglasses is $2.82\,\text{D}/(1 + (0.012\,\text{m})(2.82\,\text{D})) = 2.73\,\text{D}$.

Astigmatism, Keratoconus, Presbyopia, Prescriptions, and Diplopia

For people with astigmatism, corrective lenses composed of spherical lenses can provide sharp imaging in one direction in the xy plane but not in any other direction. Vision can be corrected with lenses having unequal radii of curvature in the x and y axes, as defined by a suitable rotation about the z-axis (optic axis). Cylindrical lenses are curved in one direction and flat in the other (Fig. 11.47). The corrective lenses for those with astigmatism have a certain base level of curvature (like that of a spherical lens) plus added curvature in one direction (like that of an added cylindrical lens,

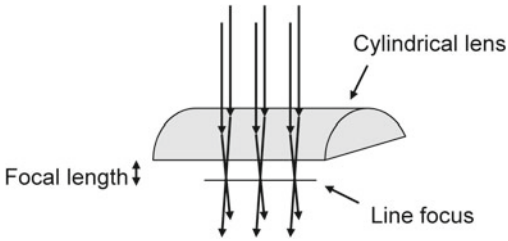


Fig. 11.47 A cylindrical lens

rotated the correct way), and as such are called spherocylindrical lenses. Because of the need for this specific correction axis, correction for astigmatism has been possible with eyeglasses or hard/gas permeable contact lenses, but not with soft contact lenses. However, astigmatism can be corrected with new toric soft contact lenses that are oriented on the eye in a stabilized manner. One way to achieve this is by the action of the lids during blinking on the superior-to-inferior position thickness gradient.

A possible eyeglasses prescription for a nearsighted person with astigmatism is

$$\begin{aligned} \text{O.D.} \quad & -3.00 + 2.25 \times 175 + 1.25 \\ \text{O.S.} \quad & -2.50 + 1.75 \times 160 + 1.25, \end{aligned}$$

where O.D. (oculus dexter) stands for the patient's right eye and O.S. (oculus sinister) for the left eye. The first number for each is the refractive power of the spherical lens corrections, $-3.00\,\text{D}$ for the right eye and $-2.50\,\text{D}$ for the left. The second term for each is the refractive power of the cylindrical lens correction (and so the lens is a spherocylindrical lens when this term is not zero) and orientation for astigmatism for each eye. For the right eye, 2.25 D is added with the cylindrical axis at an angle 175° from the horizontal in the manner shown in Fig. 11.48. As seen in this figure, along one axis (the cylindrical axis—which has no added cylindrical refractive power), the refractive power is that of the spherical lens, $-3.00\,\text{D}$. Along the perpendicular axis, the refractive power is $-3.00\,\text{D} + 2.25\,\text{D} = -0.75\,\text{D}$. For the left eye, 1.75 D is added with the cylindrical axis at an angle 160° from the horizontal. As presented, this is a prescription for bifocals. The added 1.25 D for each lens is in the lower, bifocal

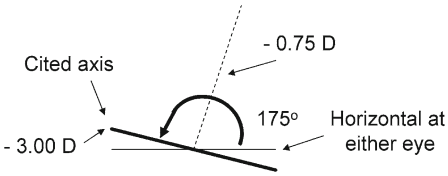


Fig. 11.48 Angles used in prescription for corrective lenses for astigmatism

region, to compensate for insufficient accommodation for reading, and is needed because of presbyopia. When corrections for astigmatism and/or presbyopia are not needed, they are not designated on the prescription.

One major stimulus to the development of (initially hard) contact lenses was the need to correct the blurs in those with keratoconus (Fig. 11.40). For people with keratoconus, eyeglasses and soft contact lenses can correct for the average optical deviation of the cornea, but cannot produce sharp images on the retina. Blurred or double images would still form on each retina with either correction, so sharp vision is possible only with hard/gas permeable contact lenses. Again, the inner radius of curvature is chosen to optimize the fit to the cornea, which is often not perfect (Fig. 11.41). Tears fill the gaps between the wavy cornea and the rigid inner lens surface (Fig. 11.49). Figure 11.49 shows how a gas permeable (semirigid) contact lens made of fluorosilane acrylate or silicon acrylate with spherical anterior and posterior surfaces can correct for the unequal radii of curvature in the cornea in astigmatism and myopia or hyperopia, and the waviness in the cornea in keratoconus. The tears (refractive index $n \approx 1.33$) very nearly index match the gap between the contact lens ($n \approx 1.4$) and the cornea ($n \approx 1.37$). Equation (11.16) shows that the refractive powers of the posterior surface of the contact lens/tears and tears/anterior surface of the cornea are very small because of the nearly matching indices of refraction, so the effects of the imperfections in the cornea are minimized.

Prescriptions for contact lenses provide optical information similar to those for eyeglasses—after correction for their different placement relative to the cornea, plus information needed to fit the lens well. One example is

$$\begin{aligned} &\text{O.D.} \quad 8.6/14.0/-1.00 \\ &\text{O.S.} \quad 8.6/14.0/-1.50 \end{aligned}$$

which indicates base curves of 8.6 mm and lens diameters of 14.0 mm for both eyes, and optical powers of −1.00 D for the right eye and −1.50 D for the left eye. Given

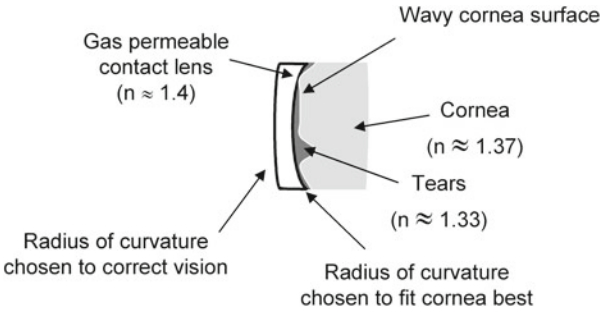


Fig. 11.49 Gas permeable (semirigid) contact lenses for correcting conditions with nonspherical corneas, such as keratoconus, with tears performing refractive index matching

the large diameter of this lens, it is clearly a soft contact lens. Astigmatism could be corrected with a rigid contact lens or a toric soft contact lens.

We have already described how people with *presbyopia* do not have sufficient accommodation (see Table 11.2). For those with good far vision, corrective lenses correct the near point for reading ("reading glasses"). Bifocals have been commonly used by those who also need correction at infinity, where the main lenses are useful for far vision and the small "bifocal" parts are for reading (as in the above eyeglasses prescription). In a trifocal, there is also correction for intermediate distances. Eyeglasses are also available and becoming more widely used with a more continuous change in optical power from the upper region (for long-range vision) to the lower region (for reading), using *progressive lenses*, which do not have the bifocal discontinuity. There are several ways a contact lens wearer can adapt to insufficient accommodation. One is to use reading glasses over the contact lens. Rigid contact lenses shaped to be bifocals, called *alternating vision lenses*, can be put into place by the lower eyelid when the wearer looks down; however, these are difficult to fit well. Corrections for near and far vision are built everywhere with *simultaneous vision* rigid or soft lenses, so the wearer sees sharp and blurred images of near and far images. In *monovision*, one contact lens is corrected for distance and the other is corrected for reading, so that one eye always sees a sharp image and the other a blurred image. Many people adapt to this well.

Diplopia (double vision) can often be corrected by using eyeglasses with prismatic corrections, which merge the two images; this can also be useful in treating some forms of strabismus [16, 62]. Prisms deflect the angle of incident light due to Snell's Law at the outer surface, so this merging can occur by using different prismatic corrections in the two eyeglass lenses (see Problems 11.77–11.79). The "lenses" of such eyeglasses can be more traditional ground-in prisms, which are single prism elements over the entire spectacle lenses, as outlined by the dashed lines in Fig. 11.50. They can be very thick and heavy for stronger prismatic corrections. Moreover, they noticeably disperse the spectrum of light because the refractive index of the prism materials varies with wavelength, which is desirable in some applications but not in this one. The same angle corrections can be made with Fresnel prisms, which are thinner corrugated prismatic surfaces, as depicted by the unbroken lines in Fig. 11.50. These types of lenses are much lighter (why?), especially when plastic is used instead

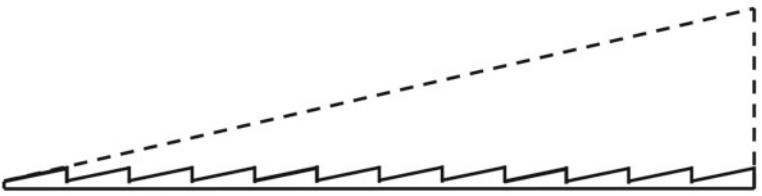


Fig. 11.50 Cross-section of the prismatic correction with a ground-in prism (*dashed line*) or a Fresnel prism (*solid line*) (Based on [62])

of glass (why?), and there is much is much less noticeable spectral dispersion because the path length in the prismatic regions is relatively short. Plastic membrane Fresnel lenses can be mounted on normal eyeglass lenses; however, they cause some loss of visual acuity.

Magnification by Corrective Lenses

A magnifying glass is a lens with positive focal length that you try to place in an optimal position between the source and your eye. Similarly, the positive focal length eyeglass lenses used to correct hyperopia produce a larger, magnified image on the retina and the negative focal length lenses used to correct form myopia produce a smaller, minified image on the retina. Contact lenses neither magnify nor minify the image because they are placed on your cornea.

11.4 Types of Vision Impairment

One type of vision impairment is due to imperfect imaging on the retina. We have already discussed its origin and how to correct it. Imperfect transmission of the optical components of the eye, retinal damage, and optic nerve damage are three other sources of vision problems.

The optical properties of each component of the healthy eye are fairly uniform, so a high-quality image is transmitted with relatively little scattering. Scarring of the cornea due to corneal injury or disease and clouding of the crystalline lens due to cataracts lessen image quality and the amount of light imaged on the retina. Scar tissue on the cornea makes any transmitted light appear to have a halo. The only solution to this problem is to remove the damaged area, usually a disk with 6–8 mm diameter, and to transplant a section from another cornea. The treatment for cataracts is removal of the cloudy crystalline lens, with replacement by a plastic lens.

There are four major modes of retinal damage. (1) Almost a half of diabetics suffer from diabetic retinopathy, in which retinal capillaries swell and leak a bit. This sometimes progresses to *neovascularization*, in which abnormal blood vessels form that do not supply the retina with enough oxygen and are fragile and bleed into the vitreous humor, thereby hindering the transmission of light to the retina. Neovascularization can also scar and detach the retina. Argon ion lasers are sometimes used to photocoagulate these thin fragile vessels periodically. (2) The macula is the central region of the retina, $\sim$5 mm in diameter including the fovea, where the central region of the object you are viewing is imaged. *Macular degeneration* is most prevalent in older people, with the thinning of the cone cells and sometimes the formation of small new blood vessels; this is similar to neovascularization and can be treated as such. (3) In a *detached retina*, the retina physically detaches from the underlying pigment epithelium, and sometimes tears. Because it has moved from the optimized imaging surface, the image formed on the detached part of the retina is poor. Furthermore, the regeneration of the visual pigments in the detached part of the retina cannot occur

because it is not in contact with the pigment epithelium—and this can lead to blindness. The detached retina can be reattached by applying a cooling or heating probe to the outside of the eyeball in the region of the detachment. (4) *Retinitis pigmentosa* is a hereditary disease that first attacks rods (hurting night vision) and later cones, thereby leading to blindness.

Optic nerve damage can result from the increase in the aqueous humor pressure in glaucoma. The overall increase in intraocular pressure in a person with glaucoma compresses the blood vessels that nourish the retina and also increases the pressure on the head of the optic nerve. Both effects cause the optic nerve fibers to degenerate and can result in blindness. The risk of glaucoma increases exponentially with intraocular pressure (IOP), with it being 40× greater for a person with an IOP of 40 mmHg than one with a relatively normal value of 15 mmHg.

High intensities of light entering the eye can affect various ocular elements causing temporary or permanent impairment of vision [53]. *Photokeratitis* is a painful but usually transient 1–2 day irritation of the cornea, due to photochemical changes caused by ultraviolet light ($\lambda = 180 - 400$ nm). For obvious reasons, it is also known as *welder flash* or *snow blindness*. (Remembering the discussion about the seemingly formidable nature of medical terminology in Chap. 1, photokeratitis merely means light (photo) + cornea (kerat) + inflammation (itis), or cornea inflammation caused by light. Similarly, keratoconus , noted earlier, merely means conical cornea.) Cataracts can also be caused by photochemical injury of the crystalline lens by chronic exposure to ultraviolet light, from 295 to 325 nm and perhaps to 400 nm. The retina can be damaged by photochemical injuries by blue light, 400–500 nm, *photoretinitis*, as can occur by looking briefly at the sun (*solar retinitis*). The retina can also be damaged by thermal injuries from high intensity—and often pulsed—lasers, from 400 to 1,400 nm. The high coherence of lasers leads to very small foci on the retina, near the diffraction limit given by (11.68). This high intensity leads to damage by highly localized absorption and subsequent localized heating. Consequences are particularly severe when the person looks at the laser—which can occur automatically when the eye tracks to the bright source—because then the laser focuses on the fovea, which can become permanently damaged. If this occurs, the person has no sharp vision in the affected eye, but only fuzzy peripheral vision. Near infrared light, ∼800–3,000 nm, can thermally damage the crystalline lens. Also, the corneas of glassblowers and steelworkers were once prone to thermal injuries (burns) due to long-term exposure of high intensities of longer wavelength electromagnetic radiation, ∼1,300 nm–1 mm.

11.5 Connections to Visual Perception

Consider the image formed on the retina as the input data for vision. The first factor that affects how this image is processed is the spatial variation of rod and cone detectors across the retina (Fig. 11.3) and whether these cells are connected individually or in an interconnected manner to ganglion cells. Much information processing

occurs in the brain [22–24, 63, 65]. For example, an upright object forms an inverted image on the retina, but is perceived to be upright by the brain. If you wear prismatic eyeglasses that invert the object, you will first perceive an inverted image, but then after a week or so you will again perceive the object as being upright (until you take the glasses off).

The perception of brightness is another example of how the image on the retina is processed: brightness increases only sublinearly with the intensity of the light source. This is an important example of the more general Steven's Law, (1.6) $P = K(S - S_0)^n$ that describes how the perceived strength P varies with the intensity of a stimulus S for a given sensation above a threshold S_0. Experiments show that $n = 0.33 - 0.5$ for brightness (Table 1.15).

Another factor in brightness perception is color. Human sensitivity to different wavelengths is described by the standard luminosity curve $L(\lambda)$ (Fig. 11.51), with $L(\lambda = 555\,\text{nm}) = 1$ describing the highest sensitivity and $L < 1$ for all other wavelengths. Light with a power of 1 W (W being a *radiometric* unit, see next paragraph) at the peak psychological sensation, 555 nm, corresponds to 683 lumens (lumens being a *photometric* unit). A power of 1 W produces a response of $683L(\lambda)$ lumens, which is <683 W away from the peak response wavelength 555 nm. This response curve is for normal levels of illumination (*photopic* or diurnal vision) and is representative for vision dominated by the cones. Figure 11.51 shows the absolute (non-normalized) photopic luminosity curve. Under dark (*scotopic* or twilight) conditions, there is a blue shift in this curve to about 507 nm, at which 1 W gives a sensation of 1,700 lumens (Fig. 11.51). (This relative peak response is much larger than that at the peak of the

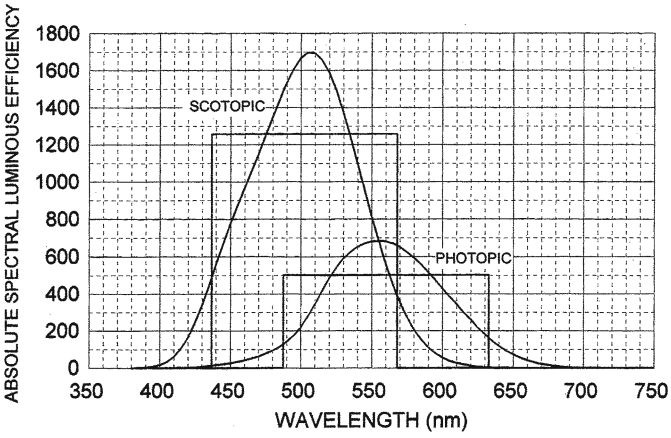


Fig. 11.51 Absolute spectral luminous efficiency of the eye for photopic vision (for normal levels of illumination, also known as the "luminosity curve") and scotopic vision (for *dark* conditions). The higher the number the more the eye is sensitive to a given light power or intensity at this wavelength. In some presentations each curve is given with its peak normalized to 1. The rectangles associated with each curve are approximations to the real curves (From [46]. Reprinted with permission of McGraw-Hill)

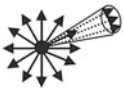
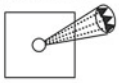
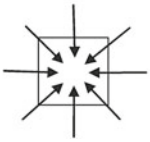
	Definition	Common photometry units	Radiometric equivalent (units)
Luminous power 	Total light power produced by a source	lumens	Radiant power (joules/second = watts)
Luminous intensity 	Light power produced in a solid angle by a point source	lumens/steradian candelas (1 lumen/steradian = 1 candela)	Radiant intensity (watts/steradian)
Luminance 	Luminous intensity per unit projected area of an extended source	candelas/square meter foot-lamberts	Radiance (watts/steradian-square meter)
Illuminance 	Luminous power falling on a surface	lumens/square meter lumens/square foot	Irradiance (watts/square meter)

Fig. 11.52 Terms and units in photometry (in common units column) and radiometry. (Based on [51])

photopic curve, but, of course, light levels are much lower for scotopic conditions.) This blue shift, the *Purkinje shift*, reflects the spectral sensitivity of rods.

There are two different sets of units used in studies of vision to denote the strength of the light beams (Fig. 11.52) [6, 44, 46, 47, 51]. Optical studies involving physical systems use MKS-SI units, which are also called *radiometric* units. In contrast, *photometric* units are scaled to account for the level of human response under normal, photopic conditions. Each set describes the same physical parameters, such as power from a source (either total or that per unit solid angle) or power incident onto a surface (either that incident per unit solid angle per unit area or per unit area). (All of the different terms can make understanding this a bit confusing.)

The units for the following radiometric parameters are:
Energy or radiant flux, or radiant power: for emitted energy/time, in W
Energy or radiant intensity: for emitted power/unit solid angle, in W/sr
Energy or radiant emittance: for emitted power/unit area, in W/m^2
Radiance: for incident power/unit solid angle-unit area, in W/m^2-sr
Irradiance: for incident power/unit area, in W/m^2

where sr stands for steradian. (All of angular space is covered by 4π steradians.) When working with lasers, variations with solid angle are usually not important and the relevant parameters are simply *power* for energy/time (W) and *intensity* for power/unit area (W/m^2).

Photometric units are normalized by the standard (photopic) luminosity curves, with L being 1 at 555 nm and smaller at other wavelengths. Photometric terms often involve the adjective *luminance*, which corresponds to the term *radiance* sometimes used in radiometric units. They are based on the *luminous intensity* unit, the candela (cd), which is light emitted at 555 nm with an energy intensity of 1/(683 W/sr). The *luminous flux* is emitted light in units of the lumen (lm), which is 1 cd-sr (and so it corresponds to 1/683 W for 555-nm light). Incident light has units of nit = cd/m^2 = lm/m^2-sr for luminance and lux (lx) = lm/m^2 for illuminance.

The units for the following photometric parameters are:
Luminous flux or power: for emitted energy/time, in lumen (lm)
Luminous intensity: for emitted power/unit solid angle, in candela (cd) = lm/sr
Luminance: for incident power/unit solid angle–unit area, in nit = cd/m^2 = lm/m^2-sr
Illuminance, illumination (brightness) or luminous flux density: for incident power/unit area, in lux (lx) = lm/m^2

For a more complete set of units see Table 11.4.

The luminous flux density of the sun at noon at the equator is about 10^5 lux, while that from a full moon is about 0.2 lux. This luminous flux density in daylight is 10^4 lux, on an overcast day 10^3 lux, on a very dark day 10^2 lux, during starlight 10^{-3} lux, and on a moonless overcast night 10^{-4} lux. The crossover from photopic to scotopic vision occurs near 1 lux. Artificial illumination is usually about

Table 11.4 Equivalent photometric units for luminance and illuminance

Luminance
1 nit = 1 cd/m^2
1 stilb (sb) = 1 cd/cm^2 = 10^4 nit
1 apostilb (asb) = $1/\pi$ cd/m^2 = $1/\pi$ nit = 0.3183 nit
1 lambert (L) = $1/\pi$ cd/cm^2 = $10^4/\pi$ nit = 0.3183 nit = 1,000 mL (milliLambert)
1 footlambert (fL) = $1/\pi$ cd/ft^2 = $10.764/\pi$ nit = 3.426 nit
1 candela/ft^2 (cd/ft^2) = 10.764 nit
1 equivalent phot = 1 lambert (L)
1 equivalent lux = 1 blondel = 1 apostilb (asb)
1 equivalent footcandle = 1 footlambert
Illuminance
1 lux (lx) = 1 lm/m^2
1 phot (ph) = 1 lm/cm^2 = 10^4 lx
1 milliphot (mph) = 10^{-3} lm/cm^2 = 10 lx
1 footcandle (fcd, fc, or ft-cd) = 1 lm/ft^2 = 10.764 lx

100–2,000 lux. Light bulbs are rated by the input electrical power in W and by their light output in lumens. Now old-style 100 W incandescent light bulbs produce about 1,600 lumens. This is about 2.4 W of visible light power, which shows that such light bulbs are only about 1–3% efficient. (The performance of the newer and more efficient compact fluorescent and LED light bulbs is often characterized by their electrical wattage, the number of lumens they produce, and the wattage of the incandescent bulbs that produce the same number of lumens.) The illuminance a meter away from this light bulb is 1,600 lumens divided by the total surface area of a sphere with 1 m radius, so it is $1{,}000\,\text{lumens}/(4\pi \times 1\,\text{m}^2) = 127\,\text{lux}$. For simple orientation and visual tasks 30–100 lux are needed, for common visual tasks 300–1,000 lux are needed, and for special visual tasks 3,000–10,000 lux are needed [47]. Table 11.5 shows recommended lighting levels for specific tasks.

A well-lit room has 200–500 lux (or 20–50 footcandles), and we usually need about 50 lux for reading. A dark-adapted human eye can see about 10^{-9} lux. If the pupil has a radius r of 2 mm, then the luminous flux through the pupil is $\pi(2 \times 10^{-3}\,\text{m})^2(10^{-9}\,\text{lm/m}^2) = 10^{-14}\,\text{lm}$. This corresponds to about 1.5×10^{-17} W of light. Using (11.1), a visible photon with 600 nm wavelength has an energy of $\approx 3 \times 10^{-19}$ J. This means that a dark-adapted eye can see ~30 photons/s.

The eye has excellent vision over a billion-fold change in intensity. Adaptation to the dark can take up to 30 min (Fig. 11.53). In the first 5–10 min sensitivity improves by about a factor of 50 during adaptation in the cone-dominated photopic regime. In the next 20 or so minutes, sensitivity improves by about a factor of 1,000 during adaptation in the rod-dominated scotopic regime. These are determined by the regeneration of rhodopsin and changes in retinal networking that control the summation of retinal signals. The eye adjusts to higher fluxes of light within 3–5 min, with a decrease in the gain achieved in retinal networking at lower light levels.

The color content of light entering our eyes depends on the illumination and reflectance of objects (Figs. 11.54 and 11.55) and the absorption by each cone. Figure 11.56 shows the degree of stimulation of each cone for different monochro-

Table 11.5 Recommended lighting levels

Task or location	level (lux)
Dance hall, residence entrance	50
Auditorium, church/synagogue	150
Cafeteria, chemical laboratory, eye examining room	500
Court room, reading, classroom	700
Close work laboratory	1,000
Office, kitchen	1,500
Autopsy table	10,000
Operating table	25,000

Using data from [40, 47]
The illuminance levels are given in lux; values in footcandles are ≈0.1× these values, so 1,000 lux ~100 footcandles

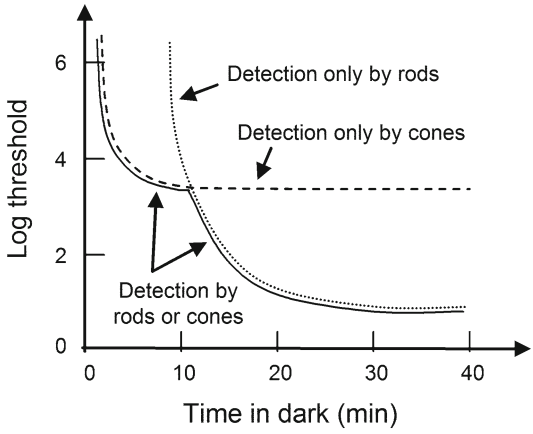


Fig. 11.53 Representative *dark* adaption curve, showing contribution from rods only (*dotted line*), cones only (*dashed line*), and from rods and cones (*solid line*). Typically, the subject is pre-exposed for several minutes to wide-angle white light at 1,500 mL. Typical vertical scales for threshold flux are in the pL–μL range, for various ranges of narrow-angle illumination (Based on [6, 51])

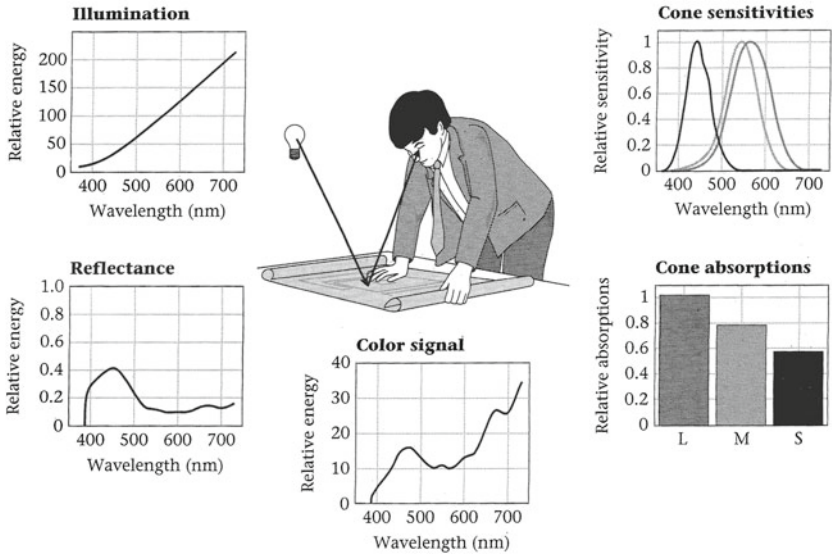


Fig. 11.54 The relative excitation of the cones—"Cone absorptions"—depends on the spectral distribution of light entering the eye—"Color signal"—and "Cone sensitivities," while the "Color signal" is the product of spectral power distribution of the light source—"Illumination"—and the "Reflectance" of the surface (From [63]. Used with permission)

matic lights [55]. The visual perception of color is a fairly complex function of its spectral components. Different mixtures of three different colors of lights can produce the same visual sensation of color and brightness (*metameric matches*) if they produce the same excitations of the three cones.

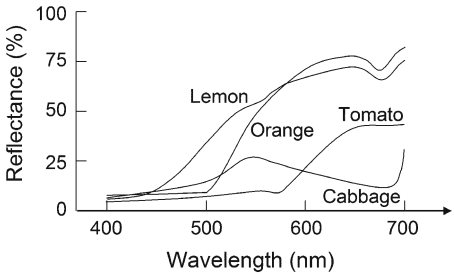


Fig. 11.55 Reflectance *curves* of common foods (Based on [13])

The perception of color by using different types of light was once described by the *R,G,B system*, which uses varying mixtures of real primaries: red (645 nm), green (526 nm), and blue (444 nm) to match colors. Because the amounts of each (called the *tristimulus values*) needed for matching (*color matching functions*) are negative in some cases, this system has been supplanted by the *CIE Color Specification System*, with "imaginary" primaries X, Y, Z (Fig. 11.57) [12, 25, 45, 69]. (CIE stands for Commission Internationale de L'Éclairage.) The amount of each primary needed to make a monochromatic color (spectral or monochromatic hue) is given by the x (for the amount of X) and y (for Y) on the arc perimeter of the *chromaticity diagram* in Fig. 11.58, with z (for Z) given by $1 - x - y$. (Also, see Fig. 11.59.) Non-monochromatic colors are shown inside the diagram. (The X, Y, Z "primaries" are outside of this curve because they are not real. While this system is mathematically consistent and correct, it is physically unappealing.) Figure 11.60 shows an example of using the chromaticity curve. Mixing monochromatic colors, 490 and 557 nm here, forms a mixture that has a color with a *dominant perceived wavelength* (or hue), which is 550 nm here, as described in the figure.

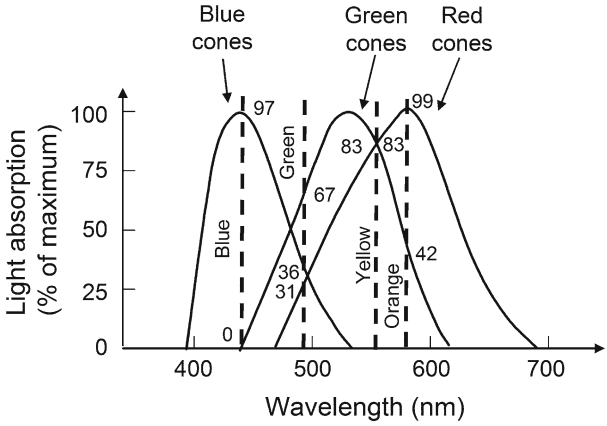


Fig. 11.56 Degree of stimulation of each cone for monochromatic *blue*, *green*, *yellow*, and *orange light* (Based on [17])

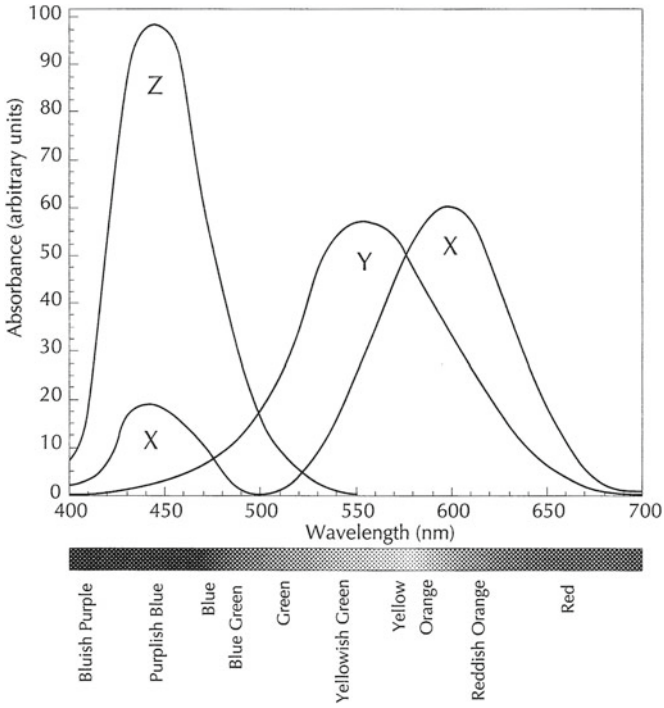


Fig. 11.57 Color matching functions for the imaginary primaries X, Y, and Z. They were arbitrarily chosen for the 1931 CIE scale, and are similar to the spectral responses of the cones—but with obvious differences, such as the short-wavelength second lobe of the X curve (From [12])

Another factor in visual perception is response time. The *critical flicker fusion frequency* (CFFF or CFF) is the fastest rate of light pulses a person perceives with no fluctuation in light intensity (and so there is no flickering). It never exceeds 60 Hz even at very high light intensities, and is usually much lower <20–40 Hz. The CFF in the fovea is about 50 Hz, as limited by the response recovery of cones. It is about 20 Hz about 20° outside the fovea. At low light levels it decreases to about 10 Hz. Every other horizontal line in a television is refreshed every 1/60 s (one cycle in 60 Hz), so a new picture appears every 1/30 s. If our response were faster under normal conditions of watching television, the refresh rate would have to be faster to avoid perceived flickering. People with particularly fast visual responses sometimes see—and are bothered by—the flickering from fluorescent lights.

11.6 Vision in Other Animals

One can appreciate better the optical design features of the human eye by comparing it to that of other animals. Primitive eyes, as in planaria, consist of a series of photoreceptor cells in pits of pigmented cells known as eye spots. In more complex

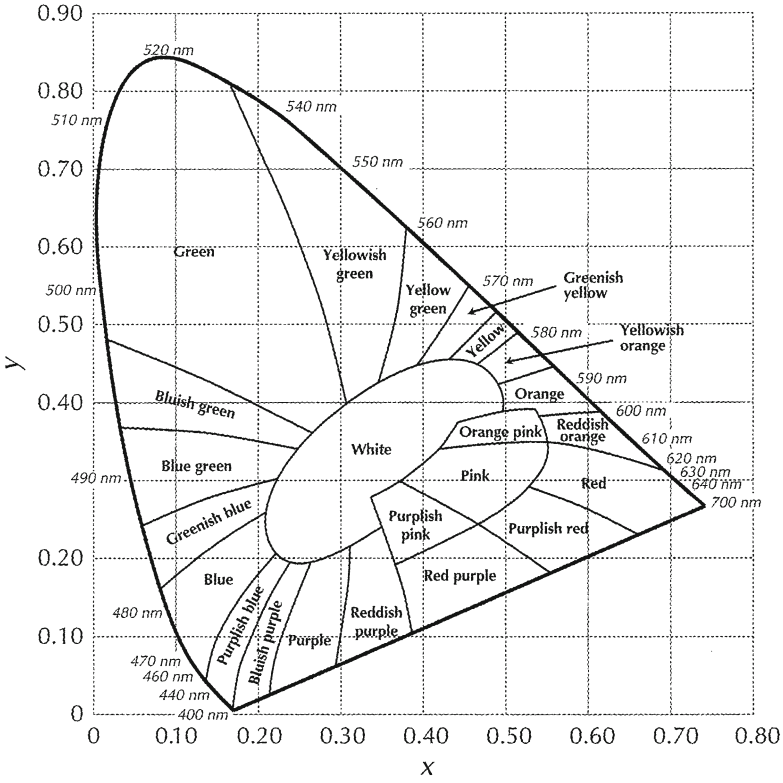


Fig. 11.58 CIE chromaticity curve (outer perimeter), with *color* labels and regions added inside. This is the version from 1931, which is still widely in use (From [12])

eyes, light is collected and imaged onto an array of photoreceptor cells. In camera or simple eyes, there is one imaging element, either a pinhole (as in a pinhole camera discussed in Problem 11.84)—such as found in the nautilus—or a combination of the cornea and a crystalline lens—as is found in all vertebrates. In compound eyes, there are a series of lenses that image those local regions onto photosensitive detection cells. Vertebrates have only camera eyes, while invertebrates can have camera and/or compound eyes. Different features are optimized in different animals.

1. *Night versus day:* The eyes of diurnal (active during the day) animals such as lizards and chipmunks are optimized for visual acuity. They have cone-dominated retinas (because cones are smaller than rods) with very little interconnection of neurons leaving the eye (i.e., few convergences), and large eyes. The eyes of nocturnal (active during the night) animals have large corneas and crystalline lenses (to gather much light) and only rods (because they are more sensitive thancones). In some animals, such as cats, there is a mirror-like layer behind the

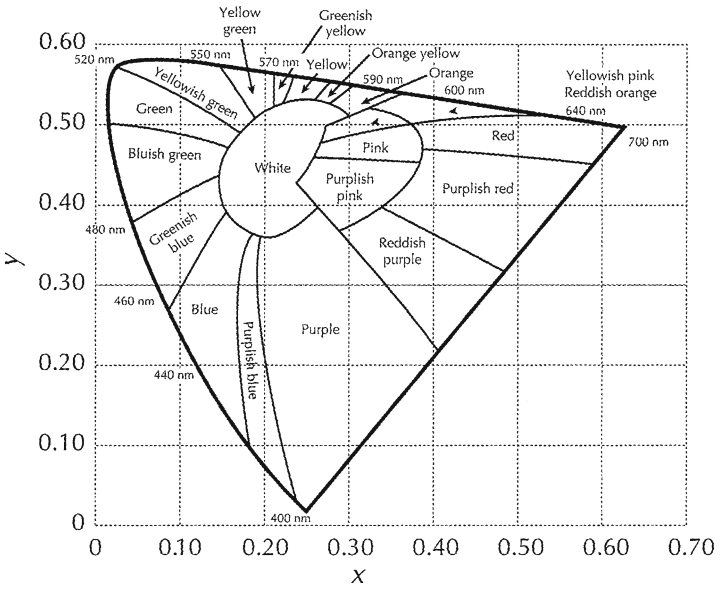


Fig. 11.59 CIE chromaticity curve from 1975. It is less biased to the *green shades* than the 1931 version is, so the change in *color* hue between any two points in this diagram is now roughly equal to the distance between the two points in the diagram. Nevertheless, the 1931 CIE is still predominantly used (From [12])

retina (the *tapetum*) that reflects transmitted light back through the retinal cells, to increase retinal sensitivity; this produces *eye-shine* (bright eyes in the dark). They have pupils that close to slits, which can protect their retinas from the bright daylight sun much better than do round pupils (which cannot close as much). (Some truly nocturnal animals, such as owls, function well with round pupils because they do not come out during the day).

2. *Air versus water:* Humans cannot see well under water (aside from turbulence and waves in the water), because we have only about 1/3 of our imaging power under water, and that is mostly the power due to the crystalline lens. The reason for this is the different refractive indices of air and water, as we saw earlier. Vision can be much better if we wear corrective goggles. Fish and other animals living in the water have more spherical crystalline lenses than humans to provide the extra optical power needed to image in water. In species that live in water the crystalline lens is the main source of refractive power. In fish the crystalline lens is indeed spherical. Spherical aberration is minimized by a decrease in refractive index from the crystalline lens center to the periphery (i.e., away from the optic axis). Some animals can see well both in air and water. The "four-eyed" fish Anableps swims along the surface with the upper half of their eyes in air and the lower half in water. This fish has an elliptically shaped crystalline lens (when viewed from the side). Light transmitted through the upper (air) pupil impinges on the

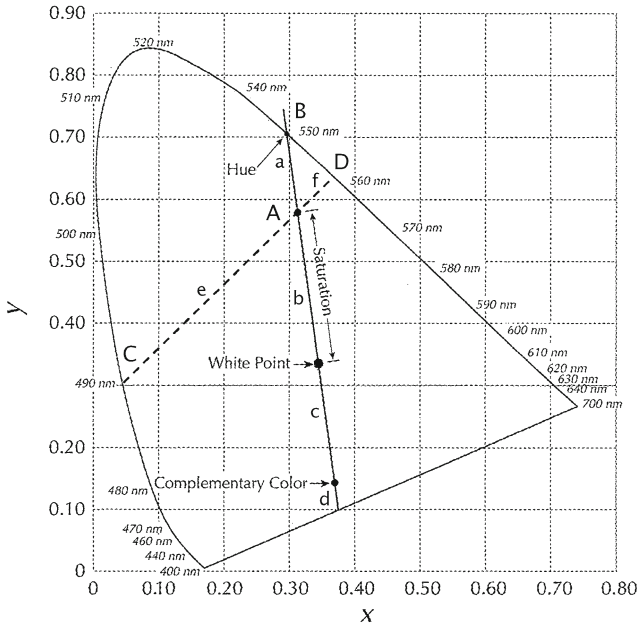


Fig. 11.60 CIE chromaticity curve (1931 version) illustrates several things about a particular *color* in the diagram, represented by point A. The line through A and the point representing *white* (White Point) intersects the CIE boundary at point B, showing that its hue (apparent color) is 550 nm. The saturation of this color (which denotes how close it is to being the pure monochromatic hue) is the ratio of the distance from the white point to the *color* (at A) to that to the hue (at B). This is $b/(a+b)$, where a, b, c, and d are the four line segment distances shown. The *color* complementary to A is the *color* on this line on the other side of the *white point*, which has the same relative distances to the CIE boundary as A, so $c/(c+d) = b/(a+b)$. This figure also shows how *colors* mix. When you mix 490 nm light (point C) with 557 nm light (point D), you end up with light perceived to be a *color* that is a point on the CD *dashed line*, and the location of the point on the line depends on how much light of each is used. If the relative intensities are $f/(e+f)$ and $e/(e+f)$, for the line segment distances e and f as shown, the *color* is A, which is seen to have a hue of 550 nm (point B). (In this example, when ~20% of 490 nm is mixed with ~80% of 557 nm, you see 550 nm) (From [12])

flattened surfaces of the crystalline lens (and so has less refractive power), while that transmitted through the lower (water) pupil impinges on the more curved surfaces of the crystalline lens (and so has more refractive optical power).

3. *Accommodation* occurs in fish, as well as in snakes and adult amphibians, by translating the crystalline lens forward and backward. In many birds and reptiles the ciliary muscles compress the crystalline lens at its equator, moving the anterior surface forward and increasing its curvature. The front of the crystalline lens can actually touch the iris, which is made rigid by sphincter muscles, and this can also increase the curvature of the crystalline lens. In this way diving birds can

increase the refractive power of their eyes by 70–80 D, so they can see well in both air and water.

4. *Eye placement:* In some animals the placement of the two eyes is frontal (including cats and humans), while in others it is lateral—i.e., on the sides of their heads (such as in rabbits, birds, lizards, rodents, etc.). Frontal eyes have overlapping views, which provide good depth perception, while lateral eyes give a better panoramic view—sometimes with independent motion of each eye.

11.7 Summary

Geometric optics, using Snell's Law, can explain much of how the cornea and crystalline lens image light on the retina. This includes developing models of multiple lens systems and optical models of the eye. Image acuity can be lessened by lens aberrations and the effects of diffraction. Imperfect vision can be corrected by using corrective lenses designed by optics models. Models of image formation and the properties of the eye can be linked to vision and visual perception, including the physics of the perception of color. Molecular processes in the rod and cone cells in the retina, which detect the light and send electrical signals to the brain, can be characterized and modeled to further the understanding of vision.

Problems

Basic Imaging

11.1 A thin lens placed 40 cm after an object forms an image 20 cm after it.
(a) Find its focal length by using (11.3).
(b) Calculate the object and image vergences and find the lens power and focal length by using vergences.

11.2 A thin lens with refractive power −15 D is placed 40 cm after an object.
(a) Find the location and type of image by using (11.3).
(b) Repeat (a) by using vergences.

11.3 A thin lens with focal length 20 cm forms an image when the object is at infinity (so parallel rays hit the lens). This lens is replaced by a lens with focal length 50 cm so that the image location is unchanged.
(a) Find the location of the lens (relative to that of the initial lens) by using (11.3).
(b) Repeat (a) by using vergences.
(c) Repeat (a) by using effectivity.

11.4 A thin lens with focal length 30 cm is placed 20 cm after an object, and a second thin lens is placed 10 cm after the first lens. Find the location of the image relative to the second lens, and state the type of image.

11.5 (a) Derive Newton's relation for a thin lens in air that forms a real image, $xx' = f^2$, where $x = d_1 - f$ (the distance the source is in front of the focal point to the left of the lens) and $x' = d_2 - f$ (the distance the image is after the focal point). (Hint: Draw a diagram from a source with a ray that is parallel to the optic axis before the lens and one that is parallel to it after the lens, and consider similar triangles.)
(b) If a source is 20 cm in front of (i.e., to the left of) the primary focal point and the image is 5 cm to the right of the secondary focal point, find the focal length and power of the lens and also confirm that (11.3) is satisfied.

11.6 A thin lens in air is composed of material with an index of refraction of 1.062. For each posterior radius of curvature R_{23} given in (a−c), find the anterior radius of curvature R_{12} needed so this lens has a focal length of 50 mm. Also, for each case sketch the shape of the lens and describe it as being either biconvex, planoconvex, or positive meniscus.
(a) −0.0060 m
(b) ∞
(c) +0.0060 m.
(The parameters in this problem are equivalent to those for the eye crystalline lens, with the ratio of refractive indices of the lens to the humors being 1.42/1.337 = 1.062.)

11.7 Show the equations for a thick lens in air, (11.43) and (11.44), reduce to the Lensmaker's equation for a thin lens, (11.18), under appropriate conditions.

11.8 Show that the thick lens equations for arbitrary media, (11.41) and (11.42), reduce to results for a thick lens in air, (11.43) and (11.44), when $n_1 = n_3 = 1$ and $n_2 = n$.

11.9 A glass lens in air has $n = 1.5$, $R_{12} = 200$ cm, and $R_{23} = -200$ cm.
(a) Assume it is a thin lens and find its focal length.
(b) Now assume that the lens has a thickness $t = 5$ mm. Find its effective and back focal lengths.
(c) Will the rays cross the axis first, i.e., focus first, for the lens described in (a) or (b)? (Compare the focal length for the lens in (a) to the distances in the lens in (b) that incident parallel rays focus beyond the (i) back surface, (ii) middle plane, and (iii) front surface of the lens.)

The Eye and Imaging

11.10 Draw the visual and optical axes on a sketch of the eye.

11.11 Let us explore our blind spot a bit further.
(a) Why does the experiment demonstrating the blind spot in your left eye (using Fig. 11.4) prove it is "nasal?"
(b) Assume the eye acts as a lens in air with a 17 mm focal length (Standard eye model) and determine how far your blind spot is from your fovea (in mm and degrees)? Do you confirm the blind spot is 13–18° away from the fovea?

(c) Repeat the experiment described in Fig. 11.4 using your right eye, with your left eye closed. How do you explain your findings? How can you modify the experiment so you can determine the blind spot in that eye? Does this prove that the blind spot in your right eye is also nasal and located 13–18° away from the fovea. Why?

11.12 Snell's Law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$ (11.6), predicts that rays will undergo total internal reflection (and not be transmitted) when $n_1 > n_2$ for incident angles $\theta_1 > \arcsin(n_2/n_1)$ (at the local planar interface). Show that light rays inside the eye that hit the cornea at angles exceeding 49° cannot leave the eye (assuming the medium in the eye has refractive index 1.33) (Fig. 11.61a). (This total internal reflection can affect the visual inspection of eyes requiring these large angles, except if a *goniolens* is used (Fig. 11.61b). These large-angle rays are transmitted through the eye into a saline/contact lens assembly in this goniolens and then they are reflected to more shallow angles for inspection.)

11.13 By calculating its refractive power, show that the isolated cornea is a negative meniscus lens with negative refractive power, but in the eye it has positive refractive power.

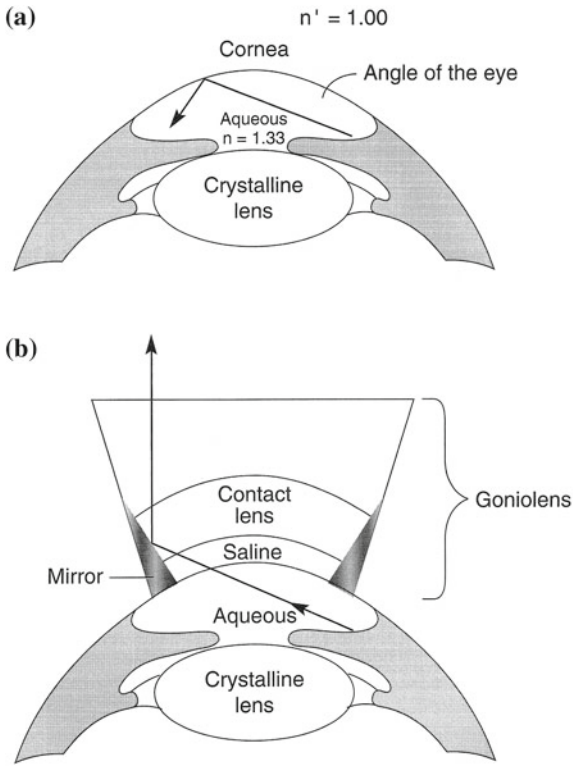


Fig. 11.61 **a** Total internal reflection in the eye and a **b** diagram describing how a goniolens works to inspect the eye, overcoming such total internal reflection (From [52]. Reprinted with permission of McGraw-Hill). For Problem 11.12

11.14 (a) Estimate the loss of light at each of the four ocular interfaces. Assume that each interface is flat and that light hits each at normal incidence. (Use the optical data presented in Table 11.1 for Schematic eye 1.)
(b) Does this change significantly with wavelength?

11.15 Use the parameters in the text for Schematic eye 1 to determine how far past the anterior surface of an optical element incident parallel rays converge to the optic axis to form a focus when the element is assumed to have a finite thickness and compare this to that assuming the thin lens approximation. Do this for the:
(a) Cornea.
(b) Crystalline lens.
(c) Does this suggest that the correction for a thick lens is significant?

11.16 Model the tears on the cornea as a 7 μm thick layer with refractive index 1.33, with anterior and posterior radii of curvature the same as that of the anterior surface of the cornea for Schematic eye 1.
(a) What is the optical power of this tear layer?
(b) What does it correspond to in D?
(c) Do tears affect imaging significantly?

11.17 Thomas Young, famous for seminal experiments and interpretations in optics almost 200 years ago, showed that ocular accommodation was due to the crystalline lens and not the cornea because he could not focus on nearby objects when he immersed his eye in water. We know that his conclusion was right, but was his reasoning faulty? Use (11.49) and (11.57), and what you (should) know about accommodation.

11.18 Equation 11.64 shows that we need 4 D of accommodation for our eyes to image at all distances $\geq$25 cm. Can the similar expression, $1/f_{\text{eye,NP}} - 1/f_{\text{eye,FP}}$, which uses the focal lengths of a person's own near and far points, be used to determine that person's own accommodation? (Hint: Consider a very young person. Also, consider a person with more than 4 D accommodation who still needs corrective lenses and perhaps also bifocals.)

11.19 Estimate the change in the radii of curvature of the crystalline lens surfaces needed to achieve 4 D of accommodation (so a person viewing a source at her far point can then view a source at her near point). Use (11.51), (11.52), and (11.54), and alternately assume (a) that the curvatures of the anterior and posterior surfaces change by the same percentage or (b) that only the anterior surface changes. (Because accommodation is usually cited for an effective imaging element in air, the amount of accommodation required within the eye is larger by the refractive index of the humors, so $\simeq 4\,\text{D} \times 1.337 = 5.35\,\text{D}$ is needed.)

11.20 In a model of the crystalline lens, the lens has a refractive index of 1.42 and the humors before and after it have a refractive index of 1.337, and the magnitudes of the radii of curvature of the two lens surfaces are the same (but they have different signs). Find the radii of curvatures of the anterior and posterior surfaces if the total refractive

power of the crystalline lens is the same as in (11.54) (ignoring the thickness of the lens).

11.21 A crystalline lens is modeled here as in Fig. 11.19, except it is biconvex with radii of curvature that always have equal magnitudes. This magnitude is 7.54 mm when a person views a source at her far point. (The crystalline lens still has an overall refractive power of 22.10 D, ignoring the thickness of the lens.) Find the radius of curvature magnitude when the lens adapts to give 4 D of accommodation (to then allow the person to see her near point), which is equivalent to 5.35 D in the eye (as is explained in Problem 11.19).

11.22 The argument is made that extremely little ultraviolet light with wavelengths from 300 to 400 nm reaches the retina, so it does not cause retinal damage and consequently this light does not damage eyes. Discuss the merits of this argument. Also, where is this light absorbed in the eye?

11.23 The light from a red laser pointer has a wavelength near 650 nm. Where in the eye is most of this wavelength absorbed?

11.24 How well is the transmission spectrum of the eye matched to the spectral responses of rods and cones?

11.25 How well are the spectral responses of the rods and cones matched to the spectrum of solar light? How well are they matched to the spectrum of solar light that actually reaches sea level?

11.26 A nanowatt of light, equally spread across the wavelengths 400–700 nm, is incident on your eye. Approximately what power impinges on your retina?

11.27 Which can cause more retinal damage, light that is highly spatially coherent that can focus according to (11.68) or light that is less coherent that cannot be focused as well. Why?

11.28 A laser enters your eye. Is more damage done if you turn your eye to look at the laser source (which is a reflex) or if you do not stare into the beam? Why?

Visual Acuity and Aberrations

11.29 In a Snellen chart a letter is 8.8 mm high in the 20/20 line. How high is it in the 20/200 line?

11.30 In the text it was said that a 8.8 mm letter in the 20/20 line of the Snellen chart subtended a 5 min arc at the eye of a person standing 20 ft from the chart. This 20/20 (i.e., 20/20 ft) line is also known as the 6/6 (i.e., 6/6 m) line. Does this 5 min arc angle more nearly refer to a person standing 20 ft or 6 m from the chart?

11.31 (a) The letters in the 20/20 line of the Snellen chart are 8.8 mm high and wide. Show that this letter is about 25 μm high and wide on the retina.
(b) Explain why this is considered excellent vision.

11.32 If you use an illuminance much greater than 480 lux in conducting a Snellen eye test, your pupil diameter begins to decrease. Discuss how this could affect the visual acuity you are trying to measure in this test, given the effect of pupil diameter on diffraction and lens aberrations. (Most people think you see better in bright light.)

11.33 (a) Show that the 8.8 mm high letters in the 20/20 (or 6/6) line of the Snellen chart subtend an angle of 1.47 mrad when you are 6 m away (where 1 mrad = 1 milliradian = 1×10^{-3} rad).
(b) The Rayleigh criterion for barely resolving objects separated by an angle θ is that θ needs to be at least $\theta_R = 1.22\lambda/d$, where d is the diameter of the lens (or that of the aperture limiting the lens). Show how this criterion is related to the Airy diameter given by (11.68).
(c) This Rayleigh criterion is a strict limit imposed by diffraction. Show that for 500 nm light, this angle is 0.122 mrad for an aperture diameter of 0.5 cm. How many times the Rayleigh criterion limit is the arc angle needed for 20/20 vision?
(d) The resolution limit for most people is 0.5 mrad, and for the most acute vision under optimum conditions it is 0.2 mrad. How many times the Rayleigh criterion limit are these?

11.34 How small would the pupil diameter need to be to affect the ultimate diffraction limit of the eye lensing system? Given the sizes of cones, would this change make a practical difference?

11.35 (a) Assume for the moment that the rods and cones are 0.1 μm in diameter, that they are tightly packed (with no space between them), and that each is individually connected to the brain by a single neuron. Ignoring optical aberrations, would you see images sharper in the blue or the red? Why?
(b) Ignore the assumptions in part (a), return to the normal conditions, and address the same questions again.

11.36 (a) Reference [64] (as cited in [63]) used experimental data measured using a human eye with 3.0 mm pupil diameter to obtain an analytic fit to the line spread function: $I(i) = 0.47\exp(-3.3i^2) + 0.53\exp(-0.93|i|)$. The distance on the retina from the fovea is i, in arc min. Show that Fig. 11.31 accurately plots this function.
(b) Plot on this same set of axes this same function after each point has been broadened by 1 arc min, and compare them. (One relatively crude, way to do this is divide the x-axis into bins that are 0.1 arc min wide, so you have a set of rectangles that have heights $I(i)$ and widths 0.1 arc min centered at $i = 0.0,\ \pm 0.1,\ \pm 0.2,\ \pm 0.3, \ldots$ arc min. Replace these by rectangles with widths of 1.0 arc min. Sum the contributions at each i and then normalize all points by the value at $i = 0$. This can be made less crude by using a smaller width or a broadening function that is smoother than a rectangle.)

11.37 In Schematic eye 1 and other models of the eye the radius of curvature of the anterior surface of the (biconvex) crystalline lens is larger in magnitude than that

of the posterior surface. Use Fig. 11.28 to explain qualitatively why this asymmetry helps lessen spherical aberration in the eye.

11.38 Dispersion in glass is usually presented by the refractive index at the center of the yellow line doublet (D) from a sodium lamp at 589.3 nm, $n(D)$, and the dispersion constant (or Abbe number or V-number or constringence) $V = (n(D) - 1)/\Delta n$, where $\Delta n = n(F) - n(C)$ is the difference in the indices in the blue (F, 486.1 nm) and red (C, 656.3 nm) lines from a hydrogen lamp. The refractive index is assumed to vary linearly between the red and the blue. For borosilicate crown glass BSC-2: $n(D) = 1.517$ and $V = 64.5$ and for dense flint glass DF-2: $n(D) = 1.617$ and $V = 36.6$. (Chromatic aberration can be minimized in a doublet lens (and achromat), with the two component lens made of two glasses, such as these, with very different dispersion.)
(a) If the focal length of a simple single-component lens composed of either type of glass is designed to be 17 mm in the yellow, what is its focal length in the red and the blue?
(b) If the screen for the image is 17 mm away from the lens (as for the retina), what is the size of the image size (blur) for each range of color? (Ignore the limitations of diffraction. Assume the source is at infinity.)

11.39 The dispersion constants for ocular media in Table 11.3 use a revised definition of dispersion that differs slightly from the earlier definition used in Problem 11.38. Now $V = (n(d) - 1)/\Delta n$, where $\Delta n = n(F') - n(C')$ is the difference in the indices in the blue (F', 480.0 nm) and red (C, 643.8 nm) lines from a cadmium lamp and $n(d)$ is that at the center of the yellow line doublet (d) from a sodium lamp at 587.6 nm. Find the refractive powers at each of the four interfaces in the eye, and the sum of these refractive powers, at each of these three wavelengths. (Assume the refractive index of the cornea is at 587.6 nm and that dispersion constant of the cornea is 51.5.)

11.40 Use (11.69) (describing chromatic aberration in the eye in diopters) to estimate the change in index of refraction for the eye crystalline lens in the red (630 nm) and blue (470 nm) relative to the yellow (578 nm). Compare this to that of glass in Problem 11.39

11.41 Spectral dispersion in water is characterized by $n(D) = 1.333$, $V = 55$ (see Problem 11.38 for definitions). Model the air/anterior cornea interface instead as an air/water interface. Evaluate the chromatic aberration by determining the change in focal length across the visible (in D). How does this compare to the ~2 D of chromatic aberration in the eye?

Schematic Eyes

11.42 Apply simple theory (and ignore the distance between refractive surfaces) to the schematic eyes to find the total refractive power for:
(a) Schematic eye 2
(b) Schematic eye 2′ (accommodated version of part (a)).

(c) Schematic eye 3.
(d) How much refractive power is added in the accommodated version of the Schematic eye 2? Is this enough so these two models can accommodate for near and far vision?

11.43 (advanced problem) Repeat Problem 11.42 using the more exact treatment of the schematic models (i.e., account for the distances between refractive surfaces).

11.44 Find the refractive power of the cornea, crystalline lens, and the entire eye for the Schematic exact eye, ignoring the distances between refractive surfaces. How do these compare to the values with more exact treatment, 43.05 D, 19.11 D, and 58.64 D, respectively?

11.45 (advanced problem) Show that the refractive powers of the cornea and crystalline lens combine to give the overall refractive power of the eye for the Schematic exact eye. The first and second principal points of the cornea are at −0.0496 mm and −0.0506 mm, respectively, and those of the crystalline lens are at 5.678 mm and 5.808 mm, respectively. Also use the data provided in Problem 11.44 and Table 11.1. As part of this problem, show that the positions of the principal points of the overall system agree with those given in Table 11.1.

11.46 (advanced problem) Show that $PP' = NN'$ and $FP = N'F'$ for each model in Table 11.1.

11.47 Model an effective eye lens as a cornea thin lens with focal length 25 mm followed by a "crystalline lens" thin lens with focal length 50 mm, with both reduced lenses in air:
(a) What is their combined effective focal length if their separation is zero?
(b) What is their combined effective focal length if their separation is 9 mm (as in the eye)?
(c) What is this difference in diopters and is it significant?

11.48 Another published set of refractive indices for the eye components has $n_{\text{cornea}} = 1.376$, $n_{\text{aqueous}} = 1.336$, $n_{\text{lens}} = 1.40$, $n_{\text{vitreous}} = 1.337$, and the four radii of curvature as (from anterior to posterior) 7.8, 6.4, 10.1, and −6.1 nm. How do the refractive power and focal length differ from those calculated earlier, using (11.55)–(11.56).

11.49 Let us explore accommodation using the simplified schematic eye (such as Schematic eye 2) for two people, subjects H and M, both having about 9 D of accommodation, by using the data in Table 11.6 [28].
(a) Estimate the ages of the subjects.
(b) Qualitatively, how do the subjects differ in how their eyes accommodate?
(c) Find the refractive power of each interface in each case.

Table 11.6 More data for simplified schematic eyes for unaccommodated (for distant vision) and accommodated (for near vision) conditions

	Subject H	Subject H	Subject M	Subject M
	Unaccomm.	Accomm.	Unaccomm.	Accomm.
Radii of surfaces (mm)				
Cornea	7.74	7.74	7.6	7.6
Anterior lens	11.62	6.90	12.0	5.5
Posterior lens	–5.18	–5.05	–5.74	–4.87
Distance from anterior cornea (mm)				
Anterior lens	3.68	3.34	3.33	3.06
Posterior lens	7.34	7.58	7.17	7.26
First principal point P	1.61	1.96	1.42	1.69
Second principal point P′	1.88	2.33	1.70	2.01
Refractive indices				
Humors	1.336	1.336	1.336	1.336
Crystalline lens	1.413	1.432	1.413	1.415
Power of the complete eye (D)	60.44	69.62	60.04	68.95

Using data from [28]. For Problem 11.49

(d) Use simple theory (and ignore the distance between the refractive surfaces) to find the refractive power of the eye in each case and the amount of accommodation in both subjects.
(e) (advanced problem) Accounting for the distance between the refractive surfaces, confirm the total refractive powers and the principal point locations of each eye, as listed in Table 11.6.

More Vision Models

11.50 Use simple theory (and ignore the distance between the refractive surfaces) to determine how the total refractive power of the eye would change if the refractive index of only one of the following optical elements were increased by 1%?
(a) Cornea.
(b) Aqueous humor.
(c) Crystalline lens.
(d) Vitreous humor.
Start with the values for Schematic eye 1.

11.51 People from Planet X have eyes that are constructed much like ours. They have the same components and the refractive indices of each medium are the same as ours. One difference is that their eyeballs are twice as long as ours. Assume geometric optics theory and ignore the distances between eye components to explore in what other ways their eyes are different. Assume that the radii of curvature of their four refractive interfaces (anterior and posterior surfaces of the cornea and crystalline lens) are in the same proportion as ours.

11.52 Ophthalmologists and optometrists treat 20 ft (or 6 m) as the "optical infinity." For the human eye, what is the difference in refractive power (in D) needed to see at this optical infinity versus the real infinity?

11.53 In a Goldmann tonometer, the force needed to make a circular region of the cornea (with diameter 3.06 mm) flat is measured. The resulting pressure needed to make it flat is equal to the intraocular pressure, and therefore this is a good test for glaucoma. Show that for an intraocular pressure of 15 mmHg, this applied force corresponds to the gravitational force of 1.5 g applied over this area.

Correcting Vision

11.54 A person with myopia has a far point of 530 cm. Show that the prescription for both eyeglasses and contact lenses is $\simeq -0.19$ D. Why are they the same?

11.55 In the text it was shown that the refractive power needed to correct hyperopia with a near point of 190 cm is +3.67 D for eyeglasses placed 1.2 cm in front of the eyes and +3.47 D for contact lenses. Find the prescription for eyeglasses worn 1.5 cm in front of the eyes.

11.56 A person with myopia has a far point 60 cm from her eye. What are the refractive powers of the eyeglasses and contact lenses should she wear? (Give your answer in D.)

11.57 A person with hyperopia has a near point 500 cm from his eye. What are the refractive powers of the corrective eyeglasses and contact lenses should he wear? (Give your answer in D.)

11.58 Aside from corrections for aberrations, what are the potential advantages of using eyeglasses with positive or negative meniscus lenses, rather than biconvex/biconcave or planoconvex/planoconcave lenses?

11.59 Must contact lenses always be meniscus lenses?

11.60 Prove that you can make positive or negative focal length contact lenses even though R_{23} is always >0.

11.61 A contact lens design with refractive power of –4.0 D is needed when it is on the eye. Assume the refractive indices of the contact lens and the cornea are equal and their respective posterior and anterior radii of curvature are equal. Use the data in Fig. 11.19.
(a) Find the radius of curvature of the anterior surface of the contact lens.
(b) Find the refractive power of the contact lens when it is not on the eye.

11.62 Redo Problem 11.61 if the contact lens has a refractive index of 1.46.

11.63 For each case depicted in Fig. 11.62, describe the status of the patient's vision and how it could be corrected, if necessary.

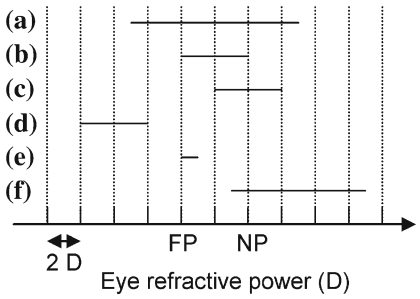


Fig. 11.62 Ranges of eye refractive powers for patients (a–f). The designated NP and FP are the refractive powers that are required to have good vision at the desired near and far points. For Problem 11.63

11.64 Find the prescription in diopters (D) to correct the eyesight for:
(a) A myopic person with a far point of 2 m (using contact lenses).
(b) A hyperopic person with a near point of 1 m who wants to read material 25 cm away and who has very good crystalline lens accommodation (using eyeglasses).
(c) A person with perfect vision for far points who, because of poor accommodation (presbyopia), has a near point of 1 m, and who wants to read material 25 cm away (using eyeglasses).
(d) How do the people in parts (b) and (c) differ? (Could they both use their eyeglasses while attending a baseball game and working at a computer terminal?)
(e) What is the accommodation of the person in part (c) (assuming a standard 17 mm long eyeball for the Standard eye model, with air replacing the humors)?
(f) If the near point for the person in part (a) is 15 cm without contact lenses, what is it when the prescribed contact lenses are worn?

11.65 An ophthalmologist estimates the prescription for a patient based on the smallest lines she can read in the Snellen chart: for 20/20: 0 D to −0.25 D, for 20/30: −0.50 D, for 20/40: −0.75 D, for 20/50: −1.00 D to −1.25 D, for 20/100: −1.75 D to −2.00 D, and for 20/200: −2.00 D to −2.50 D. Find the far point for patients who are able to read each given Snellen chart line (and no better without eyeglasses), using these average prescriptions for which ranges are given. (Note that the optical infinity we are using in our analysis is not the 20 ft value assumed in the Snellen chart. See Problem 11.52.)

11.66 A myopic person has an eyeglass prescription for −3.00 D. What is the appropriate prescription for contact lenses?

11.67 Explain the optical corrections in the following prescription for eyeglasses.

$$\begin{array}{lll} \text{O.D.} & -4.00 + 1.50 \times 90 + 2.3 \\ \text{O.S.} & -3.00 \qquad\quad + 2.3 \end{array}$$

11.68 Approximately how old is the person needing the glasses in Problem 11.67?

11.69 A contact lens made of material with refractive index 1.47 has a posterior radius of curvature set to match the anterior of the cornea (0.0078 m). Find the needed anterior radius of curvature to correct the vision of a:

(a) Myopic person needing −2.00 D correction.
(b) Hyperopic person needing +2.00 D correction.

11.70 A person wears glasses that provide −3.00 D of correction for myopia and wants to have eye surgery (by RK, PRK, or LASIK) to be able to see perfectly without glasses. The person has a cornea with anterior radius of curvature of 0.0078 m. What should the radius of curvature be after surgery? (Remember that −3.00 D refers to the Standard eye in air.)

11.71 A person with astigmatism has corrective lenses with +2 D spherical correction and +1 D cylindrical correction in what we will call the x direction. How could we rephrase spherical and cylindrical corrections for this same prescription if the cylindrical correction were that along the y direction (which is perpendicular to the x direction)?

11.72 We know that a person needs 4 D of accommodation to be able to see both near and far, but does that mean that someone with even more accommodation does not need corrective lenses? Consider as an example someone with a 22-mm-long eyeball in the Reduced eye approximation. Let us say that both persons A and B have 8 D of accommodation. The refractive power of person A can change from 58 to 66 D, while that of person B can change from 52 to 60 D.

11.73 The keratometer (Fig. 11.42) determines the radius of curvature of the anterior (outer) surface of the cornea by tracking the reflection from that surface. Keratometers convert this radius into a K reading, which is the refractive power of the air/cornea surface. However, keratometers sometimes assume slightly different refractive indices for the cornea, often 1.3375, but sometimes other values such as 1.336.
(a) Using $n = 1.3375$, determine the radius of curvature for K readings of 38, 42, 46, and 50.
(b) If the K reading is 44 and you are not sure which n was used, what is the uncertainty in the cornea radius of curvature?

11.74 The stage of keratoconus is characterized by the K reading, as measured by a keratometer (Fig. 11.42). It is mild for K readings <45 D, moderate up to 52 D, advanced up to 60 D, and severe above 60 D. What is the radius of curvature in each case? (The keratometer measurement is sensitive to the central region of the cornea and so these K readings are averaged over the apical (apex) region of the cornea.)

11.75 What happens if the contact lens base curve and the anterior surface of the cornea do not match?

11.76 A contact lens has a density of 0.9 g/cm^3. It has a thickness as given in the chapter and a surface area that can be calculated from the lens diameters given in the chapter. (In calculating this area, assume the lens is flat.) Assume tears have the same surface tension as water.
(a) Estimate the forces needed to pry a typical soft and hard contacts lens loose off your eye.
(b) When you tilt your head down so the contact lenses on your eyes face down, they do not fall off. Why? (Give a numerical answer.)

11.77 If the apex angle of a prism, here the acute angle α relative to the horizontal in Fig. 11.50, is very small, show that the prism deflects light by an angle $(n-1)\alpha$, where n is the refractive index of the prism.

11.78 A prism with power of one prism diopter, 1Δ, deflects light by 1 cm over a distance of 1 m. For prism material with refractive index $n = 1.5$, find α of a prism (as defined in Problem 11.77) with power 10Δ.

11.79 You are designing prisms for eyeglasses that correct diplopia. Say you want to move the apparent image of the left eye to right by the same angle as you want to move the apparent image of the right eye to the left. Explain how you would do this.

Visual Perception and Vision in Animals

11.80 The normalized spectral luminous efficiency is 0.0040 at 420 nm, 0.060 at 460 nm, 0.323 at 500 nm, 1.00 at 555 nm, 0.6310 at 600 nm, 0.1070 at 650 nm, and 0.0041 at 700 nm. How many watts are needed to provide 1,000 lumens at each wavelength?

11.81 In a laboratory, a photometer measures 25 μW due to room lighting over its 0.5 cm^2 area. Find the illuminance in the room in lux. Assume a spectrally-averaged luminous efficiency. Does this value make sense? Why?

11.82 A person with sensitive corneas is comfortable with normal room lighting but needs to wear fairly dark sunglasses—that transmit about 1/8 of the incident light—under normal conditions outside. Less dark sunglasses are not sufficient. Is this all consistent and does it make sense? Why?

11.83 (a) A filter that transmits a fraction T of incident light is said to have an optical density $\mathrm{OD} = -\lg(T)$. Express T in terms of the OD.
(b) Find the optical density of the sunglasses in Problem 11.82.

11.84 Light entering a pinhole camera with a pinhole aperture of diameter d forms an image on the back surface, a distance L_2 away, as in Fig. 11.63:
(a) By using geometrical arguments, show that a point source a distance L_1 from the pinhole, forms an image of dimension $\Delta = [(L_1 + L_2)/L_1]d$, which is $\simeq d$ for the usual case of $L_1 \gg L_2$.
(b) What is the improvement in the resolution of the human eye over a pinhole eye with $d = 1$ mm?

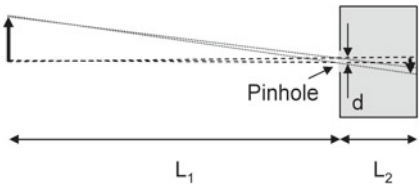


Fig. 11.63 Pinhole camera model of a pinhole eye. Imaging of an extended source is shown. For purposes of illustration a very large pinhole is shown. For Problem 11.84

(c) Let us say d was made small enough so the resolution of the human and pinhole eye were the same. How much more light would be transmitted by the human lensing system? (If the pinhole were really that small, the transmitted light would diffract much and the imaging would, in fact, be very poor. Still, this illustrates that the human lensing system improves both resolution and light throughput over the pinhole analog.)

11.85 Let us model the eye as a lens with 17 mm focal length in air that forms a sharp image on the retina when it is 17 mm away and the source is at infinity (Standard eye model). In some animals accommodation occurs by changing the separation of the crystalline lens and retina, rather than changing the focal length of the crystalline lens. (Many fish can move their lenses using intraocular muscles.) How far would the crystalline lens need to move to form a sharp image on the retina when the object is only 30 cm away, and in which direction would the crystalline lens need to be moved?

References

1. M. Bass (ed.), *Handbook of Optics, Volume III: Classical Optics, Vision, X-Ray Optics*, 2nd edn. (McGraw-Hill, New York, 2001)
2. R.E. Bedford, G. Wyszecki, Axial chromatic aberration of the human eye. J. Opt. Soc. Am. **47**, 564–565 (1957)
3. A.G. Bennett, R.B. Rabbetts, *Clinical Visual Optics*, 2nd edn. (Butterworth, London, 1989)
4. E.A. Boettner, J.R. Wolter, Transmission of the ocular media. Invest. Ophthalmol. Vis. Sci. **1**, 776–783 (1962)
5. P.K. Brown, G. Wald, Visual pigments in single rods and cones of the human retina. Science **144**, 45–51 (1964)
6. P. Buser, M. Imbert, translated by R.H. Kay, *Vision* (MIT, Cambridge, 1992)
7. F.W. Campbell, R.W. Gubisch, Optical quality of the human eye. J. Physiol. **186**, 558–578 (1966)
8. W.N. Charman, Optics of the eye. in *Handbook of Optics, Volume I: Fundamentals, Techniques and Design*, 2nd edn, ed. by M. Bass (McGraw-Hill, New York, 1995), Chap. 24
9. H.H. Emsley, *Visual Optics* (Hatton, London, 1936)
10. G.L. Fain, *Sensory Transduction* (Sinauer, Sunderland, 2003)
11. R.F. Fisher, Presbyopia and the changes with age in the human crystalline lens. J. Physiol. **228**, 765–779 (1973)
12. B. Fortner, T.E. Meyer, *Number by Colors: A Guide to Using Color to Understand Technical Data* (Springer/TELOS, Berlin, 1997)
13. E.B. Goldstein, *Sensation and Perception*, 4th edn. (Brooks/Cole, Pacific Grove, 1996)
14. P.R. Greene, Mechanical considerations in myopia, in *Refractive Anomalies: Research and Clinical Applications*, ed. by T. Grosvenor, M.C. Flom (Butterworth-Heinemann, Boston, 1991), pp. 287–300
15. A. Gullstrand, The optical system of the eye, in *Physiological Optics*, vol. 1, ed. by H. von Helmholtz, pp. 350–358 (1909) Appendix II.3. English tranlsation: J.P.C. Southall (ed.) (Optical Society of America) New York. Reprinted (Dover, 1962)
16. K.B. Gunton, A. Brown, Prism use in adult diplopia. Curr. Opin. Ophthalmol. **23**, 400–404 (2012)
17. A.C. Guyton, J.E. Hall, *Human Physiology and Mechanisms of Disease*, 6th edn. (Saunders, Philadelphia, 1997)

18. R.W. Hart, R.A. Farrell, Light scattering in the cornea. J. Opt. Soc. Am. **59**, 766–774 (1969)
19. W.M. Hart Jr (ed.), *Adler's Physiology of the Eye: Clincal Application*, 9th edn. (Mosby, St. Louis, 1992)
20. J. Hartstein, *Questions and Answers on Contact Lens Practice*, 2nd edn. (Mosby, St. Louis, 1973)
21. E. Hecht, *Optics*, 4th edn. (Addison Wesley, Reading, 2001)
22. D.H. Hubel, *Eye, Brain and Vision* (Scientific American Library, New York, 1987)
23. D.H. Hubel, Exploration of the primary visual cortex, 1955–1978 (Nobel Lecture). Nature **299**, 515–524 (1982)
24. D.H. Hubel, T.N. Wiesel, Brain mechanisms of vision. Sci. Am. **241**, 130 (1979)
25. D.B. Judd, G. Wyszecki, *Color in Business, Science, and Industry*, 3rd edn. (Wiley, New York, 1975)
26. J. Keener, J. Sneyd (eds.), *Mathematical Physiology II: Systems Physiology*, 2nd edn. (Springer, Berlin, 2009)
27. M.V. Klein, T.E. Furtak, *Optics*, 2nd edn. (Wiley, New York, 1986)
28. Y. Le Grand, S.G. El Hage, *Physiological Optics* (Springer, Berlin, 1980)
29. E.R. Lewis, Y.Y. Zeevi, F.S. Werblin, Scanning electron microscopy of vertebrate visual receptors. Brain Res. **15**, 559–562 (1969)
30. A. Lewis, L.V. Del Priore, The biophysics of visual photoreception. Phys. Today, 38–46 (1988)
31. D.E. Lieberman, *The Story of the Human Body: Evolution, Health, and Disease* (Pantheon, New York, 2013)
32. W.F. Long, R. Garzia, J.L. Weaver, Assessment of refraction and refractive errors, in *Handbook of Optics, Volume III: Classical Optics, Vision, X-Ray Optics*, 2nd edn, ed. by M. Bass (McGraw-Hill, New York, 2001), Chap. 11
33. I.A. Mackie, *Medical Contact Lens Practice: A Systematic Approach* (Butterworth Heinemann, Oxford, 1993)
34. W.B. Marks, W.H. Dobelle, E.F. MacNichol Jr., Visual pigments of single primate cones. Science **143**, 1181–1183 (1964)
35. M.J. Mannis, K. Zadnik, C. Coral-Ghanem, N. Kara-Jose, *Contact Lenses in Ophthalmic Practice* (Springer, Berlin, 2004)
36. D.M. Maurice, The structure and transparency of the cornea. J. Physiol. (London) **136**, 263–286 (1957)
37. N.A. McBrien, D.W. Adams, A longitudinal investigation of adult-onset and adult-progression of myopia in an occupational group. Refractive and biometric findings. Invest. Ophthalmol. Vis. Sci. **38**, 321–333 (1997)
38. N.A. McBrien, A. Jobling, A. Gentle, Biomechanics of the sclera in myopia: extracellular and cellular factors. Optom. Vis. Sci. **86**, E23–E30 (2009)
39. J.T. McIlwain, *An Introduction to the Biology of Vision* (Cambridge University Press, Cambridge, 1996)
40. D.D. Michaels, *Visual Optics and Refraction: A Clinical Approach*, 3rd edn. (Mosby, St. Louis, 1985)
41. L. Mlodinow, *Subliminal: How Your Unconscious Mind Rules Your Behavior* (Pantheon, New York, 2012)
42. P. Mouroulis, J. Macdonald, *Geometrical Optics and Optical Design* (Oxford, New York, 1997)
43. National Eye Institute, National Institutes of Health. http://www.nei.nih.gov/photo/charts/index.asp
44. Y. Ohno, Radiometry and photometry review for vision optics, in *Handbook of Optics, Volume III: Classical Optics, Vision, X-Ray Optics*, 2nd edn, ed. by M. Bass (McGraw-Hill, New York, 2001), Chap. 14
45. N. Ohta, A.R. Robertson, *Colorimetery: Fundamentals and Applications* (Wiley, New York, 2005)
46. J.M. Palmer, Radiometry and photometry: units and conversions, in *Handbook of Optics, Volume III: Classical Optics, Vision, X-Ray Optics*, 2nd edn, ed. by M. Bass (McGraw-Hill, New York, 2001), Chap. 7

47. M.S. Rea (ed.), *The IESNA Lighting Handbook*, 9th edn. (Illuminating Engnieering Society of North America, New York, 2000)
48. P. Riordan-Eva, J.P. Whitcher, *Vaughan and Asbury's General Ophthalmology*, 16th edn. (Lange Medical Books/McGraw-Hill, New York, 2004)
49. S.-M. Saw, W.-H. Chua, C.-Y. Hong, H.-Min Wu, W.-Y. Chan, K.-S. Chia, R.A. Stone, D. Tan, Nearwork in early-onset myopia, Invest. Ophthalmol. Vis. Sci. **43**, 332–339 (2002)
50. R.F. Schmidt, F. Lang, G. Thews, *Physiologie des Menschen*, 29th edn. (Springer, Berlin, 2005)
51. S.H. Schwartz, *Visual Perception: A Clinical Orientation*, 2nd edn. (Appleton and Lange, Stamford, 1999)
52. S.H. Schwartz, *Geometrical and Visual Optics: A Clinical Introduction* (McGraw-Hill, New York, 2002)
53. D.H. Sliney, Ocular radiation hazards. in *Handbook of Optics, Volume III: Classical Optics, Vision, X-Ray Optics*, 2nd edn, ed. by M. Bass (McGraw-Hill, New York, 2001), Chap. 15
54. W.J. Smith, *Modern Optical Engineering: The Design of Optical Systems*, 2nd edn. (McGraw-Hill, New York, 1990)
55. G.S. Smith, Human color vision and the unsaturated blue color of the daytime sky. Am. J. Phys. **73**, 590–597 (2005)
56. T.B. Smith, Modeling corneal transparency. Am. J. Phys. **75**, 588–596 (2007)
57. H.A. Stein, B.J. Slatt, R.M. Stein, M.I. Freeman, *Fitting Guide for Rigid and Soft Contact Lenses: A Practical Approach*, 4th edn. (Mosby, St. Louis, 2002)
58. L.N. Thibos, Y. Ming, Z. Xiaoxiao, A. Bradley, The chromatic eye: a new reduced-eye model of ocular chromatic aberration in humans. Appl. Opt. **31**, 3594–3600 (1992)
59. B.N. Tillmann, *Atlas der Anatomie des Menschen* (Springer, Berlin, 2005)
60. M. Tscherning, *Studier Over Myopiers Aetiologi* (C. Myhre, Copenhagen, 1882)
61. D. Vaughan, T. Asbury, *General Ophthalmology*, 10th edn. (Lange Medical, Los Altos, 1983)
62. S. Veronneau-Troutman, Fresnel prisms and their effects on visual acuity and binocularity. Trans. Am. Ophthalmol. Soc. **76**, 610–653 (1978)
63. B.A. Wandell, *Foundations of Vision* (Sinauer, Sunderland, 1995)
64. G. Westheimer, The eye as an optical instrument, in *Handbook of Perception* ed. by J. Thomas et al. (Wiley, New York, 1986), pp. 4.1–4.20
65. T.N. Wiesel, Postnatal development of the visual cortex and the influence of environment (Nobel Lecture). Nature **299**, 583–591 (1982)
66. M.E. Wilkinson, *Essential Optics Review for the Boards*, 2nd edn. (F.E.P. Int. Inc, Coralville, 2009)
67. D.R. Williams, D.H. Brainard, M.J. MacMahon, R. Navarro, Double-pass and interferometric measures of the optical quality of the eye. J. Opt. Soc. Am. A **11**, 3123–3135 (1994)
68. E.C. Woodman, S.A. Read, M.J. Collins, K.J. Hegarty, S.B. Priddle, J.M. Smith, J.V. Perro, Axial elongation following prolonged near work in myopes and emmetropes. Brit. J. Ophthalmol. **95**, 652–656 (2011)
69. G. Wyszecki, W.S. Stiles, *Color Science: Concepts and Methods, Quantitative Data and Formulae*, 2nd edn. (Wiley, New York, 1982)
70. F.A. Young, G.A. Leary, W.R. Baldwin, D.C. West, R.A. Box, E. Harris, C. Johnson, The transmission of refractive errors within Eskimo families. Am. J. Optom. Arch. Am. Acad. Optom. **46**, 676–685 (1969)